

Financial Conditions Targeting

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September 28, 2026

Abstract

Non-fundamental inflows into the stock market loosen financial conditions and raise output, while the policy rate responds gradually. Such financial noise explains up to 20% of the variance of financial conditions and output. These facts motivate a macroeconomic model with noise traders, risk-averse arbitrageurs, and gradual interest rate adjustment. In this model, Financial Conditions Index (FCI) targeting can improve macroeconomic stabilization: the central bank announces its expected FCI as a soft near-term target and adjusts the policy rate to keep conditions near it. This commitment reduces FCI volatility, thereby “recruiting” arbitrageurs to insulate the FCI and aggregate demand from noise.

JEL Codes: E12, E32, E44, E52, G10

Keywords: Financial conditions, monetary policy, financial noise shocks, noise traders, arbitrageurs, inelastic markets, endogenous volatility, forward guidance, policy gradualism, external instruments, SVAR-IV, proxy SVAR

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1. Introduction

“Financial conditions have tightened significantly in recent months... We remain attentive to these developments because persistent changes in financial conditions can have implications for the path of monetary policy.” (Chair Jerome Powell, Economic Club of New York Luncheon, October 19, 2023)

Monetary policy transmission occurs primarily through financial market conditions. Businesses and consumers react to market-based interest rates, stock prices, house prices, and currency exchange rates rather than to the policy rate alone. Financial Conditions Indices (FCIs), which aggregate these asset classes based on their impact on aggregate spending, identify volatile asset prices, especially stock prices, as their main driver in the U.S. and most major economies (see, e.g., Hatzius et al., 2017). These asset prices exhibit excess volatility that cannot be explained by rationally expected cashflows alone (see, e.g., Campbell, 2014). This excess volatility is partly driven by *noise shocks*—changes in asset demand or supply that are unrelated to economic fundamentals. Such noise shocks move asset prices in practice because sophisticated investors face constraints or risks that limit their ability to trade against mispricing (see De Long et al., 1990; Gabaix and Koijen, 2021). We find that such noise shocks explain up to 20% of the variance of financial conditions and of the output gap in the U.S., comparable to the high end of standard structural shocks (see Section 2 and Table 13 in Ramey, 2016). Given this large macroeconomic impact, how should monetary policy react to financial noise?

Bernanke and Gertler (1999, 2001) examine this question within a New Keynesian model with exogenous asset bubbles, arguing that central banks should not target asset prices directly but should instead focus on stabilizing the inflation and output gaps that arise from asset price fluctuations. In contrast, we propose a model with *endogenous* financial volatility in which it is optimal for the central bank to target and partially stabilize financial conditions as an independent objective, *beyond* simply managing their effects on output and inflation.

We start with the evidence. We construct a proxy for financial noise using idiosyncratic flows into the stock market by large investor sectors, following the granular instrumental variable approach of Gabaix and Koijen (2024). We use this noise proxy as an external instrument in a mid-size VAR. An expansionary noise shock loosens financial conditions and raises the output gap, with a delayed and incomplete interest rate response: it resembles a textbook demand shock that monetary policy only imperfectly stabilizes. The largest shocks coincide with major market events such as the Lehman bankruptcy and the dot-com bust. A Campbell-Shiller decomposition shows that the stock price response to the shock reflects expected excess returns rather than expected cash flows or interest rates. The price impact of the flows is large, in line with estimates of inelastic markets, and the output response is consistent with the marginal propensities to consume out of stock market wealth estimated in the literature.

We then develop a model consistent with these facts. The model builds on the “risk-centric” New Keynesian framework developed in Caballero and Simsek (2020, 2023), in which monetary

policy transmits to macroeconomic activity primarily through financial market conditions rather than traditional interest rate channels. The key feature of this framework is that aggregate demand responds directly to the aggregate asset price (which serves as our Financial Conditions Index), capturing wealth effects (which stand in for broader mechanisms through which financial conditions influence spending decisions). Monetary policy operates by adjusting the policy interest rate to influence this aggregate asset price, thereby steering aggregate demand and closing the output gap.

There are two key differences from Caballero and Simsek (2023). First, the aggregate asset price is influenced not only by the policy rate and standard financial forces but also by noise shocks. Specifically, households delegate their portfolio decisions to managers of three types: noise traders, inelastic funds, and risk-averse arbitrageurs. Noise traders create random market flows that must be absorbed by other investors. Inelastic funds cannot absorb these flows, since they passively invest according to the average optimal portfolio benchmark. Risk-averse arbitrageurs can absorb the noisy flows, but their limited capacity leaves substantial scope for noise to impact aggregate asset prices. Second, monetary policy is gradual: the central bank dislikes deviating from the interest rate path that markets anticipate, so it offsets shocks only partially on impact. This gradualism prevents it from fully stabilizing the impact of noise shocks on financial conditions and aggregate demand.

The impact of noise is amplified by an endogenous volatility feedback loop. Noise has a greater impact on aggregate asset prices when return volatility is higher, since higher volatility makes arbitrageurs more reluctant to trade against noise. The larger price impact of noise raises return volatility, which further amplifies the price impact of noise, and so on. This noise-driven volatility leads to “excessive” fluctuations in the output gap.

Our main result is that FCI targeting breaks this feedback loop and significantly improves welfare. Under this framework, the central bank announces its expected FCI as a target for the near term and commits to adjusting the policy rate to keep the actual FCI close to this target. This commitment is soft in the sense that the central bank penalizes deviations from the target but weighs them against its output and interest rate smoothing objectives. Importantly, the central bank does not need to identify noise in real time: it responds to any deviation of the FCI from its expected level regardless of its source. This commitment stabilizes future financial conditions beyond what the central bank’s standard objectives imply. In particular, the central bank effectively resists the asset price movements warranted by macroeconomic shocks, at the cost of temporary output gaps. It also offsets noise shocks faster, at the cost of larger interest rate surprises. Both changes make the return less volatile than under discretionary policy.

This lower return volatility is the key mechanism: it “recruits” arbitrageurs for monetary policy. With lower volatility, arbitrageurs trade more aggressively to counter noisy financial flows and the market becomes more elastic with respect to these flows. This counterforce reverses the volatility feedback loop and reduces the impact of noise on the FCI and output. Perhaps surprisingly, the reduction in the price impact of noise is large enough that, in our calibration,

the policy also reduces the size of interest rate surprises.

We show that starting from a discretionary benchmark, implementing some FCI targeting is always optimal. This is because departures from discretionary policy result in only a second-order loss, while the reduction in the impact of noise generates first-order gains for the central bank’s objectives. The intuition is that FCI targeting improves welfare by addressing a key time inconsistency in monetary policy: committing to stabilize *future* returns helps achieve *today’s* policy objectives by reducing return volatility and recruiting arbitrageurs to trade more aggressively against noise.

FCI targeting is akin to providing *forward guidance* about the future path of the FCI, provided that such guidance is viewed as entailing some degree of commitment rather than mere communication. We contrast it with interest rate forward guidance interpreted as a commitment to the expected policy rate. The two frameworks look similar on the surface, since both penalize deviations from an announced path, but they have very different effects on policy. Interest rate guidance makes the response to noise even more gradual, which amplifies the impact of noise and discourages arbitrageurs from absorbing it. In contrast, FCI targeting accelerates the policy response to noise, while dampening the FCI response to macroeconomic shocks. Consequently, some degree of FCI targeting always reduces FCI volatility and improves the central bank’s policy objectives, whereas interest rate guidance typically has the opposite effect.

We then examine the robustness of FCI targeting in three extensions: an inflation-output trade-off driven by cost-push shocks, belief-driven fluctuations in asset prices, and endogenous arbitrageur entry. Cost-push shocks create a conflict between stabilizing inflation and the output gap. FCI targeting still improves the central bank’s objectives, by mitigating the impact of noise on both output and inflation as in the baseline model. In fact, cost-push shocks add to FCI variance, which raises the impact of noise and makes recruiting arbitrageurs even more valuable. Belief shocks likewise add to FCI variance and make FCI targeting more beneficial. FCI targeting dampens belief-driven fluctuations in the FCI, whether or not the beliefs are justified, which makes arbitrageurs more willing to absorb noise. Finally, FCI targeting remains valuable when arbitrageur entry is increasing in expected returns. FCI targeting reduces FCI variance and expected returns, and therefore the mass of arbitrageurs. But this reduction is a response to the decline in variance and cannot overturn it.

Literature review. Our paper connects two main literatures: one in macroeconomics and one in finance. On the macroeconomics side, our paper is part of an emerging literature on New Keynesian models with risk and asset prices (e.g., Caballero and Farhi, 2018; Caballero and Simsek, 2020, 2021, 2023, 2024b; Fontanier and Simsek, 2026; Pflueger et al., 2020; Pflueger and Rinaldi, 2022; Kekre and Lenel, 2022, 2024; Kekre et al., 2024; Beaudry et al., 2024; Adrian and Duarte, 2018; Adrian et al., 2020). Our main innovation is incorporating financial noise that interferes with monetary policy transmission, demonstrating the benefits of FCI targeting in such an environment. Caballero and Simsek (2026a) extend the analysis to a multi-asset open economy, where the FCI also includes the exchange rate, and compare FCI targeting with

exchange rate targeting.

On the finance side, our paper is related to a large literature that emphasizes asset price fluctuations driven by noise and limits to arbitrage (see Black, 1986; Shleifer and Summers, 1990; De Long et al., 1990 for early contributions). Noise refers to *nonfundamental* demand or supply by market participants that emerges from behavioral biases, institutional frictions, and segmented markets (see Gromb and Vayanos, 2010). Limits to arbitrage refers to the constraints faced by sophisticated investors in trading against such noise (see Shleifer and Vishny, 1997). This framework has been applied to explain asset price fluctuations in markets affecting financial conditions, including treasury bonds (Greenwood and Vayanos, 2014; Vayanos and Vila, 2021), exchange rates (Gabaix and Maggiori, 2015; Gourinchas et al., 2025; Greenwood et al., 2023), and the aggregate stock market (Gabaix and Koijen, 2021). Our main contribution to this literature is to embed noise and limits-to-arbitrage into a macroeconomic model and show that these ingredients create a natural rationale for FCI targeting. We also provide empirical evidence that financial noise drives not only financial conditions but also macroeconomic activity.

Our model shares similarities with Jeanne and Rose (2002); Itskhoki and Mukhin (2025, 2023), who show that exchange rate stabilization can reduce volatility without significantly affecting other macroeconomic variables. Jeanne and Rose (2002) demonstrate that stabilization deters noise trader entry, creating a virtuous cycle, while Itskhoki and Mukhin (2025) show it enables more aggressive trading against noise. While our mechanism resembles that of Itskhoki and Mukhin (2025), our framework differs crucially: noise-driven financial fluctuations have large *macroeconomic* effects (confirmed in Section 2), whereas exchange rate fluctuations have minimal aggregate impact in their model. This large macroeconomic impact is what justifies FCI targeting in our framework.

Our paper is part of a literature that highlights time inconsistency in monetary policy and studies commitment as its solution (Kydland and Prescott, 1977; Barro and Gordon, 1983; Rogoff, 1985; Clarida et al., 1999; Eggertsson and Woodford, 2003). The closest paper is Woodford (2003), who shows that interest rate smoothing might be desirable even though it is not a social objective per se. Similarly, we show that FCI targeting can be welfare improving even though the FCI is not a social objective per se. The distinction from Woodford, as well as the rest of this literature, is the channel. Commitment there works through private expectations of the future *level* of policy rates, inflation, or output. FCI targeting works through expectations of *future FCI volatility*—reducing this volatility recruits arbitrageurs to stabilize financial conditions.

Our paper also connects to the forward guidance literature (Campbell et al., 2012; Woodford, 2013; Svensson, 2015; Bassetto, 2019). We show that forward guidance about the FCI—viewed as a soft commitment to an FCI target—can stabilize financial conditions and output gaps, whereas interest rate guidance interpreted as a commitment cannot. Caballero and Simsek (2026b) show that even without commitment, providing forward guidance about the central bank’s desired FCI level can activate the recruitment effect when markets are uncertain about the central bank’s macroeconomic view. Their delphic guidance (information provision) complements the

odyssean guidance (commitment) analyzed here.

On the empirical side, our paper contributes to a large literature on the interactions between asset prices and monetary policy (see Caballero and Simsek, 2024a for a review). This literature documents that asset prices affect aggregate demand, through credit spreads (Gilchrist and Zakrajšek, 2012), house prices (Mian et al., 2013), and stock market wealth (Chodorow-Reich et al., 2021), that monetary policy transmits through financial markets (Gertler and Karadi, 2015; Caldara and Herbst, 2019; Bauer et al., 2026), and that monetary policy in turn responds to asset prices, viewing them as a driver of economic activity (Cieslak and Vissing-Jorgensen, 2021). A growing literature uses FCIs as a summary statistic for financial market transmission and studies their relationship with monetary policy (Hatzius et al., 2017; Hatzius and Stehn, 2018; Ajello et al., 2023). We take a similar perspective, but we document that financial noise drives sizeable fluctuations in financial conditions and economic activity, and we show that FCI targeting can mitigate these fluctuations. This work complements Caballero et al. (2026), who estimate an FCI analog of the natural rate of interest (FCI-star) using a parsimonious two-equation macroeconomic model following Laubach and Williams (2003).

The rest of the paper is organized as follows. Section 2 presents facts on the macroeconomic effects of financial noise that motivate our theoretical analysis. Section 3 describes the model, characterizes the equilibrium with discretionary policy, and analyzes the destabilizing effects of noise. Section 4 presents our main theoretical results, demonstrating that FCI targeting can reduce financial volatility and improve macroeconomic stability. This section also compares the welfare effects of FCI targeting with interest rate forward guidance and discusses the robustness of FCI targeting in various model extensions. Section 5 provides concluding remarks. The theory appendix A contains the proofs, derivations, and various model extensions. The data appendix B presents the details of the empirical analysis and additional results.

2. The macroeconomic impact of financial noise

We study the effects of stock market noise on financial conditions and macroeconomic activity. We focus on the stock market because it is the main driver of the U.S. FCI-G index over our sample. Three findings emerge. First, financial noise shocks generate responses consistent with aggregate demand shocks. Second, they account for up to 20% of FCI forecast error variance on impact and up to 20% of output gap forecast error variance at a two-year horizon, although the exact contributions are imprecisely estimated. Third, a Campbell–Shiller decomposition attributes most of the stock price response to changes in expected excess returns, with small estimated contributions from expected cash flows and the risk-free rate.

2.1. Data and methodology

We measure financial conditions using the Financial Conditions Impulse on Growth index (FCI-G) constructed by Ajello et al. (2023). The index uses the Fed’s macroeconomic models to

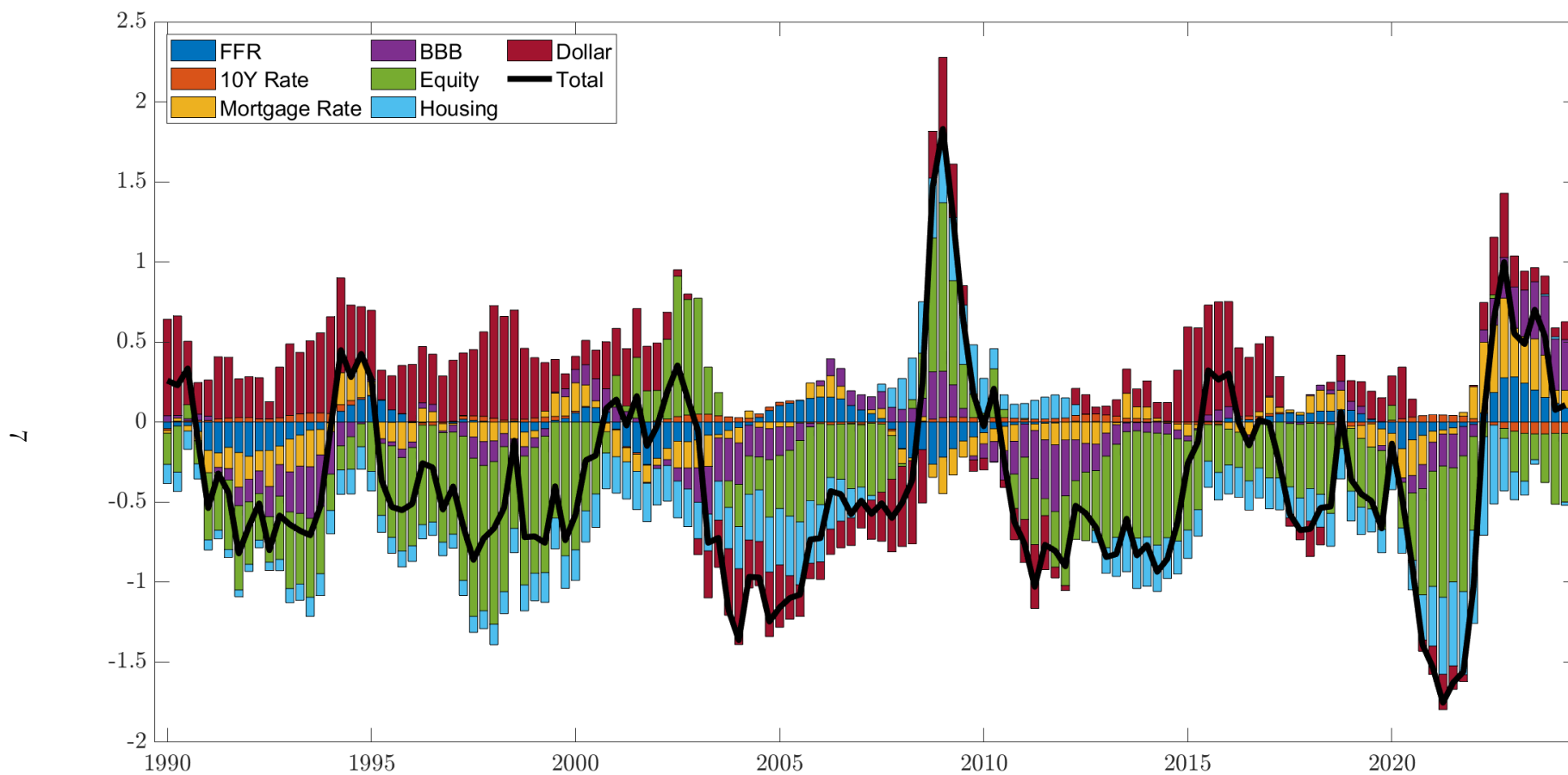


Figure 1: The FCI-G index (with a three-year lookback) and its drivers over 1990Q1–2024Q2. Positive values imply a decrease in GDP growth in the next year. Source: Ajello et al. (2023).

estimate the contribution of changes in seven financial variables over the preceding three years to GDP growth over the next year. These variables are the Dow Jones stock price index, the Zillow house price index, the broad dollar index, the federal funds rate, the 10-year Treasury yield, the 30-year fixed mortgage rate, and the triple-B corporate bond yield. A positive value indicates tighter financial conditions: an index value of 1 implies a one-percentage-point drag on GDP growth over the next year. Figure 1 shows the index and each variable’s contribution over 1990Q1–2024Q2. Stock prices are the main driver, followed by the exchange rate. House prices contribute substantially during the buildup to the global financial crisis and the recovery from the COVID-19 pandemic.

We study the relationship between financial conditions and macroeconomic activity using a vector autoregression (VAR). Our main sample runs from 1990Q1, when the FCI-G becomes available, through 2019Q4, excluding the extreme observations associated with the COVID-19 pandemic. Extending the sample through 2024Q2 leaves the estimated impulse responses and variance decompositions essentially unchanged (see Appendix B.3.2).

The VAR contains 12 variables: the FCI-G, GDP, the output gap measured as the log difference between GDP and CBO potential output, investment, consumption, annualized PCE inflation, the Excess Bond Premium of Gilchrist and Zakrajšek (2012), the three-month nominal interest rate, hours worked, the labor share, labor productivity, and TFP. These variables are standard in monetary and financial VAR specifications (Gilchrist and Zakrajšek, 2012; Gertler and Karadi, 2015). We include a constant and two lags and estimate a Bayesian VAR with a Minnesota prior, choosing its tightness following Giannone et al. (2015). Appendix B.1 provides additional details on data construction, Appendix B.2 contains estimation details, and Appendix B.3.1 reports additional results.

We construct a proxy for financial noise shocks using the granular-IV approach (Gabaix and Koijen, 2024), following Gabaix and Koijen (2021). Using Flow of Funds data, we measure proportional changes in each sector’s equity holdings, Δq_{it} . We residualize these flows using fixed effects, sector-specific trends, macroeconomic observables, and principal components. The resulting residuals, $\Delta \tilde{q}_{it}$, capture sector-specific variation in equity flows after controlling for these factors. Our proxy is their equity-share-weighted average:

$$Z_t^\mu = \sum_{i=1}^I S_{i,t-1} \Delta \tilde{q}_{it}. \quad (1)$$

Here, $S_{i,t-1}$ is sector i ’s share of total equity holdings at $t - 1$. Households account for a large share of these holdings, so their residual flows are the main driver of the proxy. Following Gabaix and Koijen (2021), we interpret the proxy as a measure of shifts in demand for equities. Residualization serves two purposes. First, it controls for common factors that may jointly affect equity flows and valuations. Second, it addresses the mechanical constraint that aggregate flows sum to zero. The principal components absorb the sectors’ common responses to equilibrium

price changes, allowing the residual flows to capture shifts in equity demand whose weighted sum need not be zero. Appendix B.1.2 provides the construction details and shows that the results are robust to alternative residualization choices.

Finally, we remove the component of the proxy predictable from its own lags and those of the VAR variables:

$$\tilde{Z}_t^\mu = Z_t^\mu - L[Z_t^\mu \mid \{Z_\tau^\mu, Y_\tau\}_{\tau < t}],$$

where $L(x \mid y)$ denotes the linear projection of x onto y . The red line in Figure 2 shows the residualized proxy. Substantial variation occurs throughout the sample, including a large negative realization during the global financial crisis and realizations of comparable magnitude in the early 1990s and mid-2000s.

We describe the demeaned observables and the residualized proxy using the following system:

$$Y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell} \quad (2)$$

$$\tilde{Z}_t^\mu = \alpha \varepsilon_t^\mu + v_t, \quad (3)$$

$$Y_t = \sum_{\ell=1}^M B_\ell Y_{t-\ell} + u_t, \quad (4)$$

where Y_t collects the macroeconomic and financial variables, ε_t is the vector of structural shocks, and ε_t^μ is its financial noise component. The residualized proxy $\tilde{Z}_t^\mu$ measures this component with loading α and measurement error v_t . The vector u_t contains the VAR innovations. We assume that ε_t^μ and v_t are mutually independent white noise processes.

Equation (2) gives the structural moving-average representation, with Θ_ℓ describing the responses to structural shocks at horizon ℓ . Equation (3) relates the residualized proxy $\tilde{Z}_t^\mu$ to the financial noise shock, while equation (4) gives the autoregressive representation of the observables, with u_t denoting their Wold innovations. The following subsections explain the identifying assumptions used to recover impulse responses and forecast variance contributions.

2.2. Causal effects of noise shocks

We estimate the effects of financial noise shocks on macroeconomic and financial variables following Plagborg-Møller and Wolf (2021). We augment the baseline VAR with the proxy, forming $X_t = [Z_t^\mu, Y_t']'$, and estimate the resulting Bayesian VAR. We identify the responses recursively, ordering the proxy first.

Impulse responses. Figure 3 shows the responses to an expansionary financial noise shock—an exogenous inflow into equities. Financial conditions loosen on impact, and a positive output gap develops, while inflation responds little. The interest rate rises with a delay, providing only a partial offset to the shock. These responses resemble those of an aggregate demand shock that

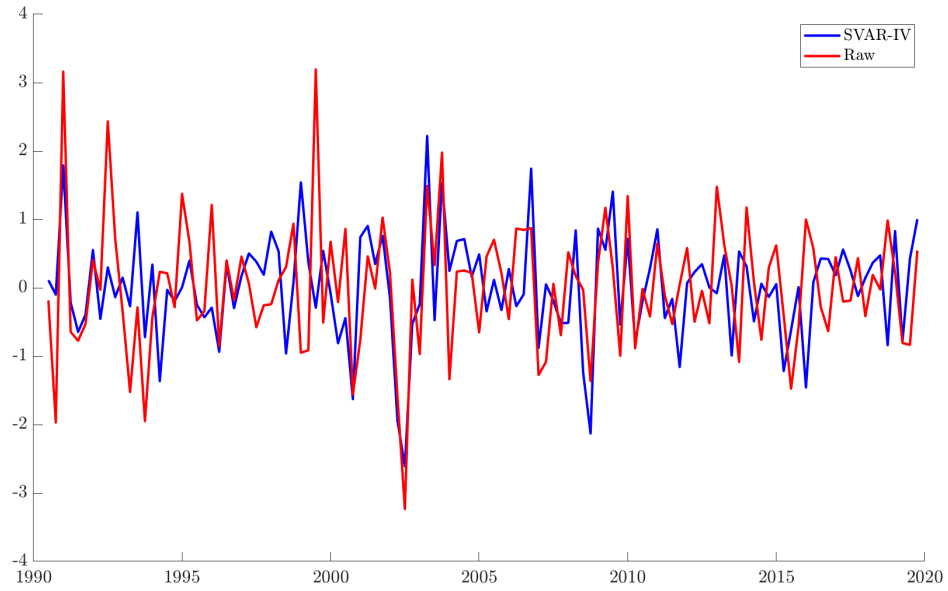


Figure 2: Red: raw shock, $\tilde{Z}_t^\mu$ as in (3). Blue: shock identified using SVAR-IV as in (5).

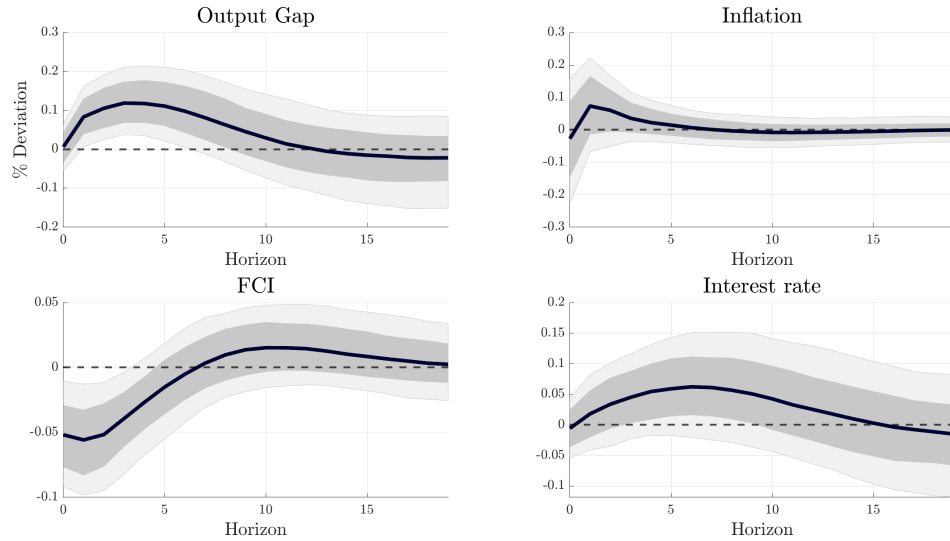


Figure 3: Impulse response to a financial noise shock. Shaded and light shaded grey bands indicate 68% and 90% credible sets, respectively. Appendix B.3.1 reports the responses of all variables in the VAR.

monetary policy incompletely stabilizes. The results are robust to including additional principal components and changes in consumer sentiment when constructing the proxy, and to alternative identification and estimation methods (Appendix B.3.3).

2.3. Importance for macroeconomic fluctuations

We next estimate the contribution of financial noise shocks to output fluctuations. This contribution helps assess the scope for stabilization through FCI targeting, which works by endogenously reducing the impact of noise shocks.

Forecast variance ratios. We quantify this contribution under several alternative identifying assumptions. Following Plagborg-Møller and Wolf (2022), we define the forecast variance ratio (FVR) for variable i at horizon h as

$$FVR_{i,h} = 1 - \frac{\text{Var}(Y_{i,t+h} | \{Y_\tau\}_{\tau < t}, \{\varepsilon_\tau^\mu\}_{t \leq \tau < \infty})}{\text{Var}(Y_{i,t+h} | \{Y_\tau\}_{\tau < t})}.$$

The FVR measures the fraction of forecast error variance eliminated by knowing the current and future realizations of the financial noise shock.

Within this framework, alternative assumptions about the relationship between the proxy $\tilde{Z}_t^\mu$, the structural shock ε_t^μ , and the Wold innovations u_t yield different estimates or bounds for the FVR.

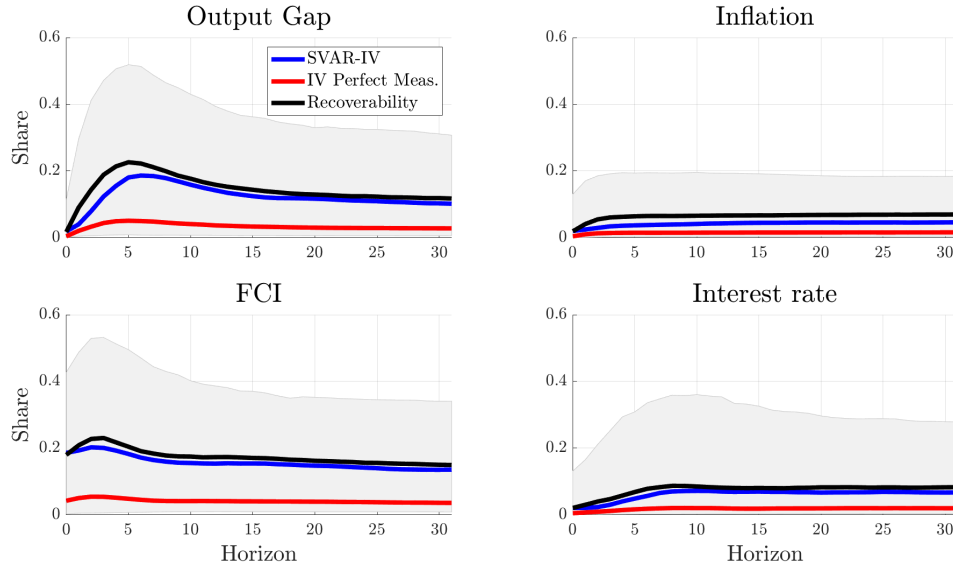


Figure 4: Identified Forecast Variance Ratios for the noise shock. Blue: SVAR-IV, assuming invertibility. Red: lower bound, assumes perfect measurement of the shocks. Black: recoverability-based FVR. The three lines are the medians, horizon by horizon, of each object. Light grey areas are 90% credible regions of the identified set (Plagborg-Møller and Wolf, 2022). Appendix B.3.1 reports the ratios for all variables in the VAR.

The first assumption is *invertibility*: the structural shock can be recovered as a linear combination of contemporaneous Wold innovations,

$$\varepsilon_t^\mu = q' u_t. \quad (5)$$

Under this assumption, we use the proxy in an SVAR-IV procedure (Mertens and Ravn, 2013) to identify q and recover the shock. Intuitively, the proxy identifies the pattern of comovement between the structural shock and the Wold innovations, which determines the weights used to recover the shock from those innovations. Appendix B.2 provides implementation details. The blue line in Figure 2 shows the resulting shock series. If invertibility fails, this procedure can overestimate the shock’s contribution to forecast error variance.

A second assumption is that the proxy measures the shock without error, so $v_t = 0$ for all t . This assumption also point-identifies the FVR. If the proxy contains measurement error, however, imposing $v_t = 0$ understates the shock’s contribution: a weak correlation between the proxy and the observables is attributed entirely to the shock’s limited importance. The resulting FVR therefore provides a lower bound on the shock’s true contribution (Plagborg-Møller and Wolf, 2022). The red line in Figure 2 shows the residualized proxy, which under this assumption is proportional to the structural shock. In practice, the proxy of Gabaix and Koijen (2021) may capture only part of the financial noise affecting markets. The lower bound measures the contribution of the directly measured noise, whereas the invertibility-based estimate uses additional structure to recover the full contribution of financial noise.

The third assumption is *recoverability*: the structural shock can be recovered from current and future Wold innovations (Chahrour and Jurado, 2022; Plagborg-Møller and Wolf, 2022; Forni et al., 2023),

$$\varepsilon_t^\mu = q'(L^{-1})u_t,$$

where $q(L^{-1})$ is a lead polynomial and L^{-1} denotes the lead operator. Recoverability is weaker than invertibility because it allows future observations to reveal the shock. Plagborg-Møller and Wolf (2022) show how to compute an upper bound on the FVR under this assumption.

Figure 4 reports the FVRs under the three identifying assumptions. The invertibility-based SVAR-IV estimates are almost indistinguishable from the recoverability-based upper bounds. Both reach approximately 20% of output gap forecast error variance at a two-year horizon. At the longest horizon, where little predictability remains, the contribution is around 12%, approximating the share of unconditional output gap variance. Financial noise also accounts for up to 20% of FCI forecast error variance on impact, but only a modest share of inflation and interest rate fluctuations. These estimates indicate a substantial contribution to output and financial conditions, although the wide credible regions imply considerable uncertainty about its magnitude.

Assessing the magnitude of the explained variance. To assess the magnitude of this contribution, we compare it with a variant of the Main Business Cycle shock of Angeletos et al. (2020). Following their procedure in Angeletos et al. (2020, Supplementary Appendix K), we construct a shock that maximizes the output gap variance explained at business cycle frequencies, subject to being orthogonal to the identified financial noise shock. This accounting exercise provides a benchmark: how much of the remaining output gap variance can a *single* shock explain when chosen specifically for that purpose?

In the SVAR-IV specification, this constrained Main Business Cycle shock explains around 33% of output gap forecast error variance at an eight-quarter horizon, compared with approximately 20% for the financial noise shock. The contribution of financial noise is therefore substantial relative to that of a shock chosen specifically to maximize the remaining variation at business cycle frequencies.

2.4. What are these shocks really capturing?

Given the identification strategy and the estimated macroeconomic importance of these shocks, we examine whether they capture other sources of financial market fluctuations, such as investors rationally anticipating higher future cash flows. We first relate the largest shock realizations to market events, then assess whether changes in cash flows and risk-free rates can account for the stock price response.

Narrative account. We examine the five largest negative and positive realizations of the SVAR-IV shock. The negative realizations occur in 2008Q4, 2002Q3, 2000Q4, 2016Q1, and 1994Q2, all quarters of tightening financial conditions. These episodes coincide with the acute phase of the global financial crisis following Lehman’s bankruptcy; the corporate governance crisis surrounding WorldCom’s accounting fraud and bankruptcy; the collapse of the dot-com boom; global spillovers from stock market turbulence in China; and the 1994 bond market selloff, respectively.

The largest positive realizations occur in 2003Q2, 1991Q1, 2003Q4, 2006Q4, and 1999Q1. The first two coincide with the end of major combat operations in Iraq and the resolution of the Gulf War, respectively. The late-2003 and late-2006 observations coincide with equity gains amid strong economic activity. The early-1999 observation accompanies the technology-stock rally and recovery from the financial turmoil surrounding Russia’s default and LTCM’s collapse. The largest realizations thus correspond to recognizable financial market episodes.

These episodes may also involve news about economic fundamentals. We therefore examine whether the responses of cash flows and the risk-free rate are large enough to account for the stock price response to the identified shock.

Decomposition of stock price movements. We extend the VAR to include real stock prices and dividends, the VIX, and long-term earnings growth forecasts, measured by the aggregate

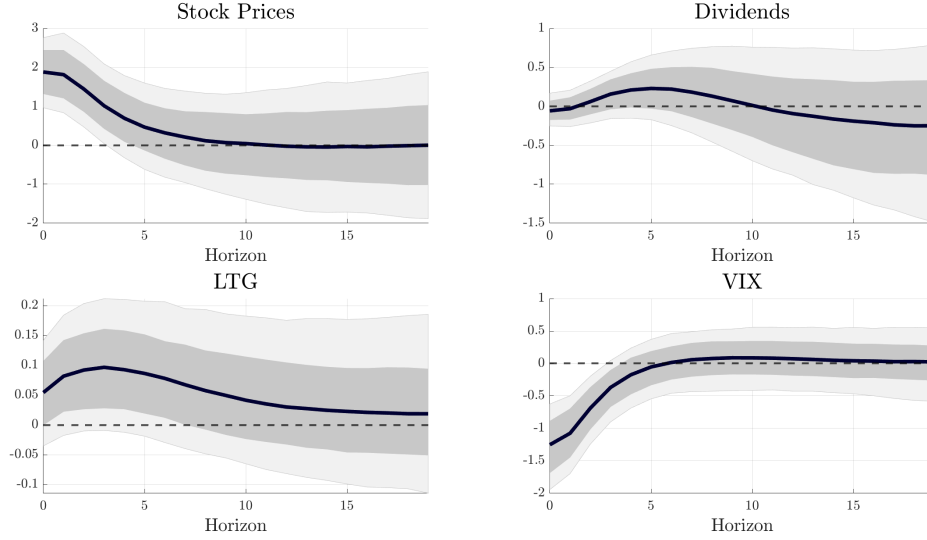


Figure 5: Impulse responses of selected financial variables to a financial noise shock. Shaded and light shaded grey bands indicate 68% and 90% credible sets, respectively. Stock prices and dividends are measured in log points times 100. LTG is measured in percentage points, so 0.1 corresponds to a change in forecasts of 0.1 p.p.

Long Term Growth (LTG) variable from Bordalo et al. (2024). The macroeconomic responses are essentially unchanged. Figure 5 shows that a positive shock raises stock prices by about 1.9% on impact, with prices returning to their pre-shock level after a few quarters. The VIX falls sharply, while dividends and expected earnings growth rise temporarily. The latter two responses are imprecisely estimated, and the dividend response is not statistically significant. Local projections yield similar patterns. Appendix B.3 reports similar results for earnings, an additional variable related to cash flows in standard macroeconomic models.

To assess whether observed changes in fundamentals can account for the stock price response, we use a Campbell–Shiller decomposition *conditional* on the identified shock. Let p_t denote the log real stock price, d_t log real dividends, and r_{t+1} the log real equity return. Up to a log-linear approximation error, suppressing the constant and imposing the usual terminal condition, we have:

$$\begin{aligned}
 p_t &= \sum_{j=0}^{\infty} \rho^j [(1 - \rho)d_{t+1+j} - r_{t+j+1}] \\
 &= \sum_{j=0}^{\infty} \rho^j [(1 - \rho)d_{t+1+j} - r_{t+j+1}^f - rp_{t+j+1}].
 \end{aligned}$$

The first line expresses the stock price in terms of future dividends and returns. The second separates returns into the real risk-free-rate component, $r_{t+1}^f = i_t - \pi_{t+1}$, and the excess return, $rp_{t+1} = r_{t+1} - r_{t+1}^f$.

Let $\mathcal{I}_{t-1}$ denote the information available before the shock, and define the response of a variable x to the financial noise shock as

$$\Delta^\mu \mathbb{E}(x) \equiv \mathbb{E}[x \mid \mathcal{I}_{t-1}, \varepsilon_t^\mu] - \mathbb{E}[x \mid \mathcal{I}_{t-1}].$$

Applying this operator to the present-value relation gives the impact response of stock prices:

$$\Delta^\mu \mathbb{E}(p_t) = N_{d,t}^\mu + N_{rf,t}^\mu + N_{rp,t}^\mu,$$

where

$$\begin{aligned} N_{d,t}^\mu &= \Delta^\mu \mathbb{E} \left[(1 - \rho) \sum_{j=0}^{\infty} \rho^j d_{t+1+j} \right], \\ N_{rf,t}^\mu &= \Delta^\mu \mathbb{E} \left[- \sum_{j=0}^{\infty} \rho^j r_{t+j+1}^f \right], \\ N_{rp,t}^\mu &= \Delta^\mu \mathbb{E} \left[- \sum_{j=0}^{\infty} \rho^j r p_{t+j+1} \right]. \end{aligned}$$

These terms measure the contributions of shock-induced revisions in expected dividends, risk-free rates, and excess returns to the stock price response.

We set $\rho = 0.99$, consistent with an annual dividend yield of approximately 4%. We use the estimated impulse responses of dividends, nominal interest rates, and inflation to compute the cash-flow and risk-free-rate contributions, $N_{d,t}^\mu$ and $N_{rf,t}^\mu$, where we truncate the sums at 25 years. We then obtain the excess-return contribution, $N_{rp,t}^\mu$, as the residual required to match the estimated stock price response.

Variable	Median [P10, P90]
Impact Price Reaction	1.88 [1.15, 2.61]
Cash-Flow	-0.02 [-0.48, 0.51]
Risk-Free	-0.09 [-1.44, 0.58]
CF + RF	-0.10 [-1.44, 0.68]
Flow Size (%)	0.24 [0.22, 0.26]
M_{GK}	7.79 [4.85, 10.68]

Table 1: Median with [10th, 90th] percentiles for selected magnitudes. The second through fourth rows correspond to the price movement explained by each term (Cash-Flows, Risk-Free rates, and their sum). “Flow size” refers to the average inflow into the stock market, in percentage of the total market size. M_{GK} is the ratio between the price impact and the flow, which corresponds to the inverse of the elasticity estimated in Gabaix & Koijen.

Table 1 reports the decomposition. At the posterior medians, stock prices rise by 1.88% on impact, while dividends and real risk-free rates contribute -0.02 and -0.09 percentage points,

respectively. These two components would therefore imply a small price decline. The increase is accounted for by changes in expected excess returns. Although the estimates are imprecise, the 90th percentile of the combined cash-flow and risk-free-rate contribution is 0.68 percentage points, less than half the median stock price response. The decomposition thus provides little support for anticipated changes in cash flows or risk-free rates as the main explanation for the price increase. Our estimates reveal a conditional version of the excess volatility puzzle: the stock price response is large relative to the response implied by measured cash flows.

Magnitudes of flows and market elasticity. The bottom rows of Table 1 report the flow size and the Gabaix and Koijen (2021) multiplier, calculated as the ratio of the impact price response to the flow. The one-standard-deviation shock in the extended VAR (Figure 5) corresponds to an equity inflow of about 0.24% of stock market wealth. The impact price response of 1.9% implies a multiplier of approximately 8, somewhat above the range of 4 to 6 estimated in Gabaix and Koijen (2021). Interpreted causally, this estimate implies an inelastic stock market: small flows produce large price changes. This interpretation comes with a caveat. The flows do not predict cash flows, but they may coincide with excess optimism or pessimism shared by most investors. Part of the price response may then reflect these belief shifts rather than the price impact of the flows themselves. Section 4.4.2 shows that our mechanism remains valid when asset prices are partly driven by these types of aggregate belief fluctuations.

Plausibility of macroeconomic responses. We assess the plausibility of the estimated macroeconomic responses through back-of-the-envelope calculations using the stock market’s weight in the FCI-G and estimates of spending out of stock market wealth. We focus on point estimates to compare the magnitudes implied by these channels with the estimated responses.

The contemporaneous weight of stock prices in the FCI-G is around -0.02 . Holding the other components constant, a stock price increase of 1.9% therefore lowers the index by approximately 0.038. The estimated FCI response is about -0.05 , with the remaining decline reflecting changes in the other components. Thus, stock prices mechanically account for most of the estimated loosening of financial conditions.

Stock market wealth averages approximately 120% of annual GDP in our sample.¹ The estimated shock therefore represents an equity inflow of about 0.29% of GDP and raises stock market wealth by approximately 2.3% of GDP. Chodorow-Reich et al. (2021) estimate that a dollar increase in stock market wealth raises annual consumption expenditure by 3.2 cents at a two-year horizon (see also Di Maggio et al., 2020). Applied to the estimated wealth increase, this implies additional annual consumption of approximately 0.073% of GDP, comparable to our estimated output response of about 0.08% at the same horizon (Figure 3). Although the propensity to consume out of stock market wealth is modest, the stock of wealth is large relative

¹We arrive at this number by aggregating the corporate equity holdings for all sectors in the construction of the shock.

to GDP, so even a small percentage increase in stock prices can generate a sizable spending response. We also note that in recent years, stock market wealth over GDP roughly doubled, which suggests that this channel may be even stronger than what historical averages would suggest.

Several considerations affect this comparison in opposite directions. The wealth shock in Chodorow-Reich et al. (2021) is essentially permanent, whereas ours reverses within about two years, which tends to dampen the consumption response. In the other direction, a Keynesian multiplier amplifies the effect on output, and the loosening of other components of financial conditions can further stimulate spending. Assessing these forces jointly requires a quantitative model, but the calculation supports a substantial role for stock market wealth effects in the observed output response.

3. A macroeconomic model with financial noise

We develop a risk-centric macroeconomic model with three key features: (i) financial noise drives the aggregate asset price (FCI) away from the central bank’s desired level because monetary policy adjusts gradually; (ii) sophisticated investors (arbitrageurs) trade against this noise; and (iii) uncertainty about future financial conditions limits their willingness to do so. Financial noise consequently affects both financial conditions and macroeconomic activity, consistent with the evidence in Section 2. In Section 4, we study how FCI targeting changes this interaction between monetary policy and arbitrageurs.

3.1. Environment

We summarize the real side of the economy and describe financial markets in greater detail. Appendix A.2 provides the full environment.

Real economy. Log potential output follows

$$y_t^* = y_{t-1}^* + \varepsilon_{z,t}, \quad \text{where } \sigma_z^2 \equiv \text{var}(\varepsilon_{z,t}), \quad (6)$$

and $\varepsilon_{z,t}$ is an i.i.d. *aggregate supply shock*. In the baseline model, firms’ prices are fully sticky, so log output, y_t , is determined by aggregate demand and can differ from potential output. Appendix A.5 and Section 4.4.1 extend the analysis to partially flexible prices and a trade-off between inflation and output stabilization.

There are two types of households: hand-to-mouth households and asset-holding households. Hand-to-mouth households supply all labor and consume their entire income. Their spending responds to output, generating a Keynesian multiplier. This household structure separates labor supply decisions from the consumption and portfolio decisions of asset-holding households, whose spending drives aggregate demand.

Asset-holding households own the aggregate risky asset, or market portfolio, which is a claim on firms' share of output, νY_t . They have expected log utility and delegate portfolio allocation to the managers described below. Their consumption follows the optimal log-utility rule, modified by an *aggregate demand shifter*, δ_t . This shifter captures factors that affect aggregate spending, such as consumer sentiment, fiscal policy, or changes in discount rates.

These assumptions imply the *output–asset price relation*

$$y_t = m + p_t + \delta_t, \quad (7)$$

where p_t is the log price of the market portfolio and m combines households' marginal propensity to consume and the Keynesian multiplier. Higher asset prices raise spending and output through a consumption wealth effect. In practice, the prices of stocks, bonds, and real estate affect spending through several channels. We therefore interpret p_t as the model counterpart to an FCI and Eq. (7) as capturing the broader relationship between asset prices and aggregate demand.²

A higher demand shifter δ_t raises spending and output at a given asset price. We assume it follows an AR(1) process,

$$\delta_t = \varphi_\delta \delta_{t-1} + \varepsilon_{\delta,t}, \quad \text{where } \sigma_\delta^2 \equiv \text{var}(\varepsilon_{\delta,t}), \quad (8)$$

and $\varepsilon_{\delta,t}$ is an i.i.d. *aggregate demand shock*, independent of supply shocks.

Remark 1 (Policy transmits via FCI). *In the output–asset price relation, the interest rate has no direct effect on output. It operates entirely through the price of the market portfolio. This feature follows from our assumptions, including log utility, but it also has empirical support when p_t is interpreted as a broad FCI. In Appendix B.3.6, we combine our estimates from Section 2 with the counterfactual methods of McKay and Wolf (2023) to ask how a monetary policy shock would propagate if financial conditions did not respond to it. In this counterfactual, the short-term interest rate rises, financial conditions remain unchanged, and the output gap and inflation barely respond. Bauer et al. (2026) reach a similar conclusion with a different approach: separately identifying the effects of monetary policy shocks on risk appetite and on the expected rate path, they find that transmission operates primarily through risk appetite and risk asset prices.*

Financial markets. Households delegate portfolio allocation to managers, who invest in the market portfolio and a risk-free asset in zero net supply.³ The central bank sets the log risk-free

²We caution that p_t is stated in price, rather than the usual rates convention of FCIs. Thus, it has the opposite sign convention of the standard FCIs, where an increase in the index typically means tightening (see Section 2).

³Managers do not consume themselves. They simply make a portfolio choice for households. For simplicity, each household invests with a very large random sample of managers. This ensures there is no portfolio and wealth heterogeneity across households, even though there is heterogeneity across managers.

return, $r_t^f = \log R_t^f$, as described below. The log return on the market portfolio, $r_{t+1} = \log R_{t+1}$, is approximately

$$r_{t+1} = \rho - (1 - \beta)m + (1 - \beta)y_{t+1} + \beta p_{t+1} - p_t,$$

where β is households' discount factor and $\rho \equiv -\log \beta$ is the corresponding discount rate. Substituting the output–asset price relation (7) gives

$$r_{t+1} = \rho + p_{t+1} - p_t + (1 - \beta)\delta_{t+1}. \quad (9)$$

The return therefore depends on the change in the log asset price and the demand shifter through its effect on cash flows. Appendix A.2 provides the derivation.

In each period, a fraction η of managers are “noise traders,” whose portfolio weight on the market portfolio is $\omega_t^N = 1 + \mu_t/\eta$. Their deviation from the benchmark weight of one generates aggregate noise μ_t , the flow that other investors must absorb (see Remark 2). We assume aggregate noise follows an AR(1) process,

$$\mu_t = \varphi_\mu \mu_{t-1} + \varepsilon_{\mu,t}, \quad \text{where } \sigma_\mu^2 \equiv \text{var}(\varepsilon_{\mu,t}), \quad (10)$$

and $\varepsilon_{\mu,t}$ is an i.i.d. *financial noise shock*, independent of supply and demand shocks.

A fraction $1 - \eta - \alpha$ of managers operate “inelastic funds,” maintaining the benchmark portfolio weight $\omega_t^I = 1$. The remaining fraction α are “arbitrageurs,” or elastic funds, who choose their portfolio weights optimally as described below. Combining these positions gives the market-clearing condition

$$\alpha \omega_t^A + \eta \left(1 + \frac{\mu_t}{\eta}\right) + (1 - \eta - \alpha) = 1 \implies \omega_t^A = 1 - \frac{\mu_t}{\alpha}. \quad (11)$$

Arbitrageurs therefore absorb aggregate noise by adjusting their portfolio weight away from one, with a larger adjustment required when their share α is smaller.

Arbitrageurs observe the risk-free rate and current noise μ_t , then choose their portfolio weight to maximize the expected log of assets under management:

$$\max_{\omega_t^A} E_t \left[\log \left(\alpha W_t \left(R_t^f + \omega_t^A (R_{t+1} - R_t^f) \right) \right) \right],$$

where W_t is aggregate household wealth.

The first-order condition is $E_t[M_{t+1}(R_{t+1} - R_t^f)] = 0$, where

$$M_{t+1} = \frac{1}{R_t^f + \omega_t^A (R_{t+1} - R_t^f)}.$$

Using a log-normal approximation for market and portfolio returns, we obtain

$$\omega_t^A \sigma_{t,r_{t+1}} = \frac{E_t[r_{t+1}] + \frac{(\sigma_{t,r_{t+1}})^2}{2} - r_t^f}{\sigma_{t,r_{t+1}}}, \quad (12)$$

where $\sigma_{t,r_{t+1}}$ is the conditional standard deviation of the market return. Arbitrageurs' exposure to risk equals the equilibrium Sharpe ratio.

Combining the optimality condition with market clearing, (11), gives the *financial market equilibrium condition*:

$$E_t[r_{t+1}] = r_t^f + \frac{1}{2}(\sigma_{t,r_{t+1}})^2 - \frac{(\sigma_{t,r_{t+1}})^2}{\alpha} \mu_t. \quad (13)$$

Substituting the return equation (9) yields the equilibrium asset-pricing relation:

$$p_t = \rho + E_t[p_{t+1}] + (1 - \beta)E_t[\delta_{t+1}] - \left(r_t^f + \frac{1}{2}(\sigma_{t,r_{t+1}})^2 \right) + \frac{(\sigma_{t,r_{t+1}})^2}{\alpha} \mu_t. \quad (14)$$

Equation (14) shows that, all else equal, the price impact of noise *increases with return variance*. Greater return risk makes arbitrageurs less willing to absorb noise traders' flows, so market clearing requires a larger adjustment in prices and expected returns. The price impact is also larger when the arbitrageur share α is smaller and aggregate asset demand is therefore less elastic. Through the output–asset price relation (7), this sensitivity carries over to aggregate activity: all else equal, the effect of financial noise on output is proportional to return variance. This dependence is the distinguishing feature of our model.⁴

Remark 2 (Interpreting noise shocks). *Noisy flows can arise from investor sentiment, client inflows and redemptions, or mechanical portfolio rebalancing. Our mechanism requires these flows to be less sensitive to return variance than arbitrageurs' demand. For simplicity, we assume they are entirely insensitive to variance. The limits-to-arbitrage literature supports this emphasis on risk as a constraint on arbitrage (see, e.g., Shleifer and Vishny, 1997).*

3.2. Equilibrium with discretionary monetary policy

We next characterize equilibrium under *discretionary* monetary policy. The central bank sets the nominal interest rate i_t^f , which equals the real interest rate r_t^f under fully sticky prices. We therefore treat r_t^f as its policy instrument. The central bank seeks to close the output gap, $\tilde{y}_t \equiv y_t - y_t^*$.

⁴In standard New Keynesian models with log utility, a log-normal Euler equation approximation yields $y_t = \rho + E_t[y_{t+1}] - \left(r_t^f + \frac{1}{2}(\sigma_{t,y_{t+1}})^2 \right)$. Comparing this with Eqs. (7) and (14) shows that financial noise creates a wedge in the Euler equation in proportion to the return variance.

3.2.1. Benchmark without policy gradualism

As a benchmark, suppose the central bank can adjust the interest rate freely. It can then close the output gap in every period and state. Combining the output–asset price relation (7), the return equation (9), and the asset-pricing relation (14) with the shock processes (6) and (8) gives the equilibrium:

$$\begin{aligned} p_t &= p_t^* \equiv y_t^* - m - \delta_t, \\ r_t^f &= \rho - \frac{1}{2}\sigma^2 + (1 - \beta\varphi_\delta)\delta_t + \frac{\sigma^2}{\alpha}\mu_t, \\ r_{t+1} &= \rho + \delta_t - \beta\delta_{t+1} + \varepsilon_{z,t+1}, \\ \text{where } \sigma^2 &\equiv \text{var}_t(r_{t+1}) = \sigma_z^2 + \beta^2\sigma_\delta^2. \end{aligned} \tag{15}$$

The first line defines p_t^* , or “pstar,” the aggregate asset price consistent with output at potential. The second gives the interest rate that achieves this price, or “rstar.” The central bank fully offsets financial noise through interest rate adjustments, insulating asset prices and output from it. The third line gives the return when output remains at potential in every period. The last line gives the conditional variance of this return and shows that it is driven by supply and demand shocks. In this benchmark, noise does not create variance because it is offset by policy.

3.2.2. Equilibrium with gradual interest rate adjustment

Financial markets respond to noise over short horizons, while central banks typically adjust the policy rate gradually toward the level required by their objectives (see, e.g., Clarida et al., 1999; Woodford, 2003; Stein and Sunderam, 2018). This gradualism limits the immediate policy response to noise, consistent with the delayed interest rate response documented in Section 2. We capture it tractably by assuming that the central bank penalizes deviations from the interest rate path previously anticipated by markets.⁵

The central bank observes all current-period shocks and sets the risk-free interest rate under discretion to solve

$$G_t = \min_{r_t^f} \tilde{y}_t^2 + \frac{1}{\theta} \left(r_t^f - E_{t-1} [r_t^f] \right)^2 + \beta E_t [G_{t+1}]. \tag{16}$$

The first term captures output gap stabilization. The second introduces interest rate smoothing around the path anticipated by markets in the previous period, $E_{t-1}[r_t^f]$. The parameter $\theta > 0$ governs the degree of gradualism: a higher θ makes the central bank more willing to adjust the rate in response to news. As $\theta \rightarrow \infty$, the smoothing term vanishes and we recover the benchmark without gradualism. As $\theta \rightarrow 0$, the policy rate converges to its previously anticipated level.

The optimality condition for problem (16) yields an interest rate rule reminiscent of the

⁵Penalizing deviations from the previously anticipated rate path is a modeling device that captures gradualism while retaining analytical tractability. In practice, gradual adjustment unfolds over a longer horizon.

Taylor rule:

$$r_t^f = E_{t-1}[r_t^f] + \theta \tilde{y}_t. \quad (17)$$

Here, we use $\frac{d\tilde{y}_t}{dr_t^f} = -1$. The central bank adjusts the rate relative to its anticipated path in proportion to the current output gap. Interest rate smoothing leads it to accommodate part of the shock rather than fully stabilize output as in (15). The following proposition characterizes the resulting equilibrium under discretion.

Proposition 1 (Equilibrium with Discretionary Policy). *Suppose the central bank sets policy according to (16) and the parameters satisfy $(1 + \theta)^2 \alpha^2 > 4\sigma_\mu^2 \sigma_{macro}^2$, where σ_{macro}^2 is defined in (22). Then there is a locally stable equilibrium in which the asset price, output, and interest rate are*

$$p_t = p_t^* + \frac{1}{1 + \theta} \left[(1 - \beta\varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad \text{where } p_t^* \equiv y_t^* - m - \delta_t, \quad (18)$$

$$y_t = y_t^* + \frac{1}{1 + \theta} \left[(1 - \beta\varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad (19)$$

$$r_t^f = r_t^{f*} - \frac{1}{1 + \theta} \left[(1 - \beta\varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad (20)$$

$$\text{where } r_t^{f*} \equiv \rho - \frac{1}{2}\sigma^2 + (1 - \beta\varphi_\delta) \delta_t + \frac{\sigma^2}{\alpha} \mu_t.$$

The return is

$$r_{t+1} = E_t[r_{t+1}] + \varepsilon_{z,t+1} - \left(\beta - \frac{1 - \beta\varphi_\delta}{1 + \theta} \right) \varepsilon_{\delta,t+1} + \frac{1}{1 + \theta} \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t+1}, \quad (21)$$

and its variance $\sigma^2 = \text{var}_t(r_{t+1})$ is the smaller positive solution to the fixed point problem:

$$\sigma^2 = \sigma_{macro}^2 + \frac{1}{(1 + \theta)^2} \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2, \quad \text{where } \sigma_{macro}^2 = \sigma_z^2 + \left(\beta - \frac{1 - \beta\varphi_\delta}{1 + \theta} \right)^2 \sigma_\delta^2. \quad (22)$$

Greater noise variance σ_μ^2 increases the return variance σ^2 as well as the expected policy loss

$$G^e \equiv E[G_t] = \frac{1}{1 - \beta} \frac{1}{1 + \theta} \left[(1 - \beta\varphi_\delta)^2 \sigma_\delta^2 + \left(\frac{\sigma^2}{\alpha} \right)^2 \sigma_\mu^2 \right]. \quad (23)$$

Under gradualism, demand and noise innovations generate deviations of the asset price from p^* and corresponding output gaps, as shown in Eqs. (18)–(19). The central bank offsets a fraction $\frac{\theta}{1 + \theta}$ of their impact on the output gap immediately, leaving the fraction $\frac{1}{1 + \theta}$ uncorrected. Supply shocks, however, generate no output gaps. A permanent increase in potential output raises p^* and the equilibrium asset price one-for-one without requiring an interest rate adjustment [see (15)]. Gradualism therefore does not impede the adjustment to supply shocks.

Equation (20) expresses the policy rate as r^* minus the remaining output gap.⁶ Equivalently,

$$r_t^f = \rho - \frac{1}{2}\sigma^2 + (1 - \beta\varphi_\delta) \left[\varphi_\delta \delta_{t-1} + \frac{\theta}{1 + \theta} \varepsilon_{\delta,t} \right] + \frac{\sigma^2}{\alpha} \left[\varphi_\mu \mu_{t-1} + \frac{\theta}{1 + \theta} \varepsilon_{\mu,t} \right].$$

The anticipated rate path fully accommodates the predictable components of demand and noise, insulating output from them. Gradualism limits the response to the current demand and noise shocks, which the central bank offsets only partially on impact.

Equations (21)–(22) characterize the equilibrium return and its variance. Gradualism limits the response to noise shocks, allowing them to contribute to return variance. Equation (22) accordingly decomposes the return variance into a macro-induced component, σ_{macro}^2 , generated by supply and demand shocks, and a noise-induced component.

Crucially, the *noise-induced component of return variance is endogenous*. Greater return variance makes arbitrageurs less willing to absorb noise, increasing its price impact and generating further variance. Equation (22) captures this feedback through a quadratic fixed point problem. Under the proposition’s parameter restriction, it has two positive solutions. We select the smaller, locally stable solution. The larger solution is locally unstable.⁷ The stable solution is increasing in the noise variance σ_μ^2 .

Return variance feeds back into the real economy. The price impact of noise, σ^2/α , grows with return variance, so noise shocks generate larger output gaps and larger interest rate adjustments when returns are more volatile. The expected policy loss in (23) reflects this: its noise component, $(\sigma^2/\alpha)^2 \sigma_\mu^2$, increases with the return variance as well as with the noise variance.

In sum, financial noise generates output gaps and interest rate adjustments that increase with return variance. It thus worsens the trade-off between output stabilization and interest rate smoothing. This motivates a policy framework that stabilizes financial conditions in order to reduce return variance and strengthen arbitrageurs’ willingness to absorb noise.

4. FCI targeting

We show that adopting a soft FCI target for the next period can improve the central bank’s ability to meet its conventional objectives. The gain arises from addressing a time inconsistency in monetary policy. Under discretion, the central bank stabilizes financial conditions according to its current objectives, taking arbitrageurs’ exposure to future return risk as given. Committing to greater future stabilization reduces this risk and enables arbitrageurs to trade more aggressively against noise today. These benefits are first order, while the cost of reduced policy flexibility is

⁶Under gradualism, r^* has the same form as in the benchmark but incorporates the equilibrium return variance σ^2 , which differs from its benchmark value.

⁷The larger solution is locally unstable in the sense that a small increase (resp. decrease) in volatility would further increase (resp. decrease) the price impact, which would further increase (resp. decrease) the variance, and so on. In contrast, the smaller solution is robust to small fluctuations in volatility.

second order for a small degree of targeting. FCI targeting provides the commitment device.

Throughout, we impose the restriction that the central bank announces its expected FCI as the target. The central bank therefore hits the target on average, and the announcement only restricts the speed at which it responds to news. This restriction isolates the benefits of commitment that operate through lower return volatility, which is our recruitment mechanism. Remarks 4 and 5 discuss a flexible target and the unconstrained commitment solution.

With this target, the policy dampens the FCI response to macroeconomic shocks and accelerates the interest rate response to noise. Over the relevant range of targeting intensities, both changes reduce return volatility and recruit arbitrageurs to absorb noise, lessening its impact on economic activity. We first characterize the equilibrium and establish the improvement in the central bank's objective. We then compare FCI targeting with interest rate forward guidance and examine model extensions.

4.1. Equilibrium with FCI targeting

In each period, the central bank announces expected p^* , $E_t[p_{t+1}^*]$, as its target for the next period's FCI. This is the asset price it expects to be consistent with output at potential, where $p_{t+1}^* = y_{t+1}^* - m - \delta_{t+1}$. In this baseline model, it is also the expected FCI, $E_t[p_{t+1}]$, since the central bank expects to close next period's output gap. The central bank sets the current interest rate using a modified objective that includes deviations from the target announced in the previous period:

$$G_t^{FCI} = \min_{r_t^f} \tilde{y}_t^2 + \frac{1}{\theta} \left(r_t^f - E_{t-1} [r_t^f] \right)^2 + \frac{\psi}{\theta} (p_t - E_{t-1} [p_t^*])^2 + \beta E_t [G_{t+1}^{FCI}]. \quad (24)$$

Alongside output gap stabilization and interest rate smoothing, the central bank seeks to keep the aggregate asset price close to its pre-announced target. The parameter $\psi \geq 0$ governs the strength of FCI targeting. Setting $\psi = 0$ recovers the discretionary policy in Section 3.2.2.

FCI targeting changes the central bank's *operational* objective, while its *true* objective remains output gap stabilization and interest rate smoothing, as specified in (16). We evaluate the policy using this original loss function. The gains from FCI targeting must therefore come from improving these conventional objectives.

The optimality condition for problem (24) yields

$$r_t^f = E_{t-1}[r_t^f] + \theta \tilde{y}_t + \psi (p_t - E_{t-1}[p_t^*]), \quad (25)$$

using $\frac{d\tilde{y}_t}{dr_t^f} = \frac{dp_t}{dr_t^f} = -1$, as before. Relative to the discretionary rule (17), the central bank also responds to the gap between the FCI and its pre-announced target. The normalization by θ in (24) makes ψ the coefficient on this gap. The following proposition characterizes the equilibrium and shows that the FCI stabilization term does what it is designed to do: it stabilizes the FCI and reduces return variance.

Proposition 2 (FCI Targeting Stabilizes Financial Conditions). *Suppose the central bank follows the FCI targeting policy in (24) with $\psi \geq 0$, and the parameters satisfy $(1 + \theta)^2 \alpha^2 > 4\sigma_\mu^2 \sigma_{macro}^2(0)$ and $\beta(1 + \theta + \varphi_\delta) > 1$.*

(i) *There is a locally stable equilibrium in which the asset price, output, and interest rate are*

$$p_t = E_{t-1}[p_t^*] + \frac{1}{1 + \theta + \psi} \left[(1 + \theta) \varepsilon_{z,t} - (\theta + \beta \varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad (26)$$

$$\text{where } E_{t-1}[p_t^*] = y_{t-1}^* - m - \varphi_\delta \delta_{t-1},$$

$$y_t = y_t^* + \frac{1}{1 + \theta + \psi} \left[-\psi \varepsilon_{z,t} + (1 + \psi - \beta \varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad (27)$$

$$r_t^f = r_t^{f*} - \frac{1}{1 + \theta + \psi} \left[-\psi \varepsilon_{z,t} + (1 + \psi - \beta \varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad (28)$$

$$\text{where } r_t^{f*} \equiv \rho - \frac{1}{2} \sigma^2 + (1 - \beta \varphi_\delta) \delta_t + \frac{\sigma^2}{\alpha} \mu_t.$$

The equilibrium return is

$$r_{t+1} = E_t[r_{t+1}] + \frac{1 + \theta}{1 + \theta + \psi} \varepsilon_{z,t+1} - \left(\beta - \frac{1 + \psi - \beta \varphi_\delta}{1 + \theta + \psi} \right) \varepsilon_{\delta,t+1} + \frac{1}{1 + \theta + \psi} \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t+1}. \quad (29)$$

The return variance $\sigma^2 = \text{var}_t(r_{t+1})$ is the smaller positive solution to the fixed point problem

$$\sigma^2 = \sigma_{macro}^2(\psi) + \frac{1}{(1 + \theta + \psi)^2} \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2, \quad (30)$$

$$\text{where } \sigma_{macro}^2(\psi) = \left(\frac{1 + \theta}{1 + \theta + \psi} \right)^2 \sigma_z^2 + \left(\beta - \frac{1 + \psi - \beta \varphi_\delta}{1 + \theta + \psi} \right)^2 \sigma_\delta^2.$$

(ii) Let $\bar{\psi} = \arg \min_{\psi \geq 0} \sigma_{macro}^2(\psi) > 0$. For $\psi \in [0, \bar{\psi})$, increasing ψ strictly reduces the return variance σ^2 as well as both of its components, $\sigma_{macro}^2(\psi)$ and $\frac{1}{(1 + \theta + \psi)^2} \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2$.

Equation (26) shows that, under discretion ($\psi = 0$), positive supply and noise shocks raise the asset price, while positive demand shocks lower it. For a given return variance σ^2 , FCI targeting dampens *all* these responses through the factor $\frac{1}{1 + \theta + \psi}$. Each shock therefore moves financial conditions less from the pre-announced target than under discretion.

Equations (27) and (28) show the policy adjustments behind this stabilization. For a given return variance, the central bank responds *faster* to noise: the fraction of its price impact that passes through to asset prices and output falls from $\frac{1}{1 + \theta}$ under discretion to $\frac{1}{1 + \theta + \psi}$. For noise shocks, FCI targeting effectively raises the speed of adjustment from θ to $\theta + \psi$.

Stabilizing the FCI against macroeconomic shocks instead requires *resisting* asset price changes warranted by fundamentals. A positive supply shock raises potential output and p^* . Under discretion, the interest rate remains unchanged and output rises with potential. Under FCI targeting, the central bank raises the rate to moderate the asset price increase, creating a

negative output gap of $-\frac{\psi}{1+\theta+\psi}\varepsilon_{z,t}$. A positive demand shock lowers p^* . The central bank then raises the rate by less than under discretion to moderate the asset price decline, producing a larger positive output gap: the coefficient $\frac{1+\psi-\beta\varphi_\delta}{1+\theta+\psi}$ increases with ψ .

Equations (29)–(30) describe returns and their conditional variance. Since FCI targeting dampens the asset price response to every shock, for a given σ^2 stronger targeting reduces both macro-induced and noise-induced return variance as long as $\psi < \bar{\psi}$.⁸ These direct effects are reinforced by the endogenous variance feedback: lower return risk encourages arbitrageurs to absorb more noise, reducing its price impact, $\frac{\sigma^2}{\alpha}$, and further lowering noise-induced variance.

FCI targeting changes both monetary policy and the market's response to noise. The central bank resists asset price movements induced by macroeconomic shocks and responds faster to noise shocks, keeping financial conditions closer to the announced target. The resulting reduction in return risk recruits arbitrageurs to absorb more noise, further reducing its price impact and volatility. FCI targeting thus reverses the adverse feedback between risk and noise: stabilization encourages arbitrageurs to trade against noise, and their trading delivers further stabilization.

Remark 3 (FCI targeting does not condition on the source of the shock). *The policy rule (25) responds to the gap between the FCI and its target regardless of what moves the FCI. The central bank therefore does not need to distinguish noise from fundamental movements, a common objection to policies that respond to asset prices (see Bernanke and Gertler, 2001). The flip side is that FCI targeting also dampens asset price movements that are warranted by fundamentals. Most starkly, it leans against a price increase driven by higher productivity, which the discretionary policy would accommodate. This opens a negative output gap of $-\frac{\psi}{1+\theta+\psi}\varepsilon_{z,t}$. Whether the gains from recruiting arbitrageurs outweigh this cost is the question we turn to next.*

4.2. Macro-stabilization effects of FCI targeting

We now show that a small degree of FCI targeting improves the central bank's objectives. We evaluate FCI targeting using the original loss function G_t in (16), which captures output gap stabilization and interest rate smoothing. Averaging across shock realizations gives

$$G^e(\psi) = E[G_t(\psi)] = \frac{1}{1-\beta} E \left[\tilde{y}_t(\psi)^2 + \frac{1}{\theta} \left(r_t^f(\psi) - E_{t-1} \left[r_t^f(\psi) \right] \right)^2 \right], \quad (31)$$

where $E[\cdot]$ denotes the unconditional expectation. The second equality follows from stationarity of the per-period loss.

FCI targeting affects expected loss through two channels: for a given return variance, it changes the central bank's responses to shocks, and it changes the return variance itself. We

⁸To understand this threshold $\bar{\psi}$, note that FCI targeting always reduces the return response to supply shocks but not necessarily to demand shocks. By (9), the return response to a positive demand shock is the sum of a price decline and a smaller cash-flow increase, $(1-\beta)\varepsilon_{\delta,t+1}$. FCI targeting shrinks the price decline but not the cash-flow effect. The net response therefore falls with ψ until the price decline drops below the cash-flow effect and rises thereafter. Macro-induced variance is minimized at $\bar{\psi}$, where this increase begins to outweigh the decline in supply-induced variance. Since the cash-flow effect is small when β is close to one, $\bar{\psi}$ is large.

separate these effects by writing

$$G^e(\psi) = \hat{G}(\psi, \sigma^2(\psi)). \quad (32)$$

Here, $\hat{G}(\psi, \sigma^2)$ denotes the expected loss (31) when the output gap and the interest rate surprise follow (27)–(28) with the return variance σ^2 held fixed (Appendix A.3 gives the closed form).

Proposition 3 (Macro-Stabilization Effects of FCI Targeting). *Consider the equilibrium in Proposition 2. Holding return variance fixed, expected loss $\hat{G}(\psi, \sigma^2)$ is increasing in targeting intensity, with zero derivative at $\psi = 0$. Holding targeting intensity fixed, expected loss is increasing in return variance. Since a small degree of FCI targeting reduces return variance, it strictly reduces expected loss:*

$$\left. \frac{dG^e(\psi)}{d\psi} \right|_{\psi=0} = \left. \frac{\partial \hat{G}}{\partial \sigma^2} \frac{d\sigma^2}{d\psi} \right|_{\psi=0} < 0.$$

The loss-minimizing targeting intensity is therefore positive, $\psi^* = \arg \min_{\psi \geq 0} G^e(\psi) > 0$.

FCI targeting distorts the central bank’s responses to shocks, as described in Section 4.1. Supply shocks open output gaps that discretion would avoid. Demand shocks generate larger output gaps. Noise shocks require larger interest rate adjustments. Nonetheless, these costs are second order near $\psi = 0$. For a given return variance, discretion minimizes the central bank’s loss, so small departures from its responses have only a second-order effect on the loss. FCI targeting also reduces return variance, and this effect is first order. Lower return variance recruits arbitrageurs and reduces the price impact of noise, so noise shocks generate smaller output gaps and require smaller interest rate adjustments. This eases the trade-off between output stabilization and interest rate smoothing.

FCI targeting achieves these gains by addressing a time inconsistency in policy. A *future* discretionary central bank takes return risk as given and minimizes future output gaps and interest rate surprises. In doing so, it ignores how its actions affect the return risk that *current* arbitrageurs anticipate. FCI targeting supplies the commitment that addresses this time inconsistency: the central bank deliberately deviates from future discretion to reduce current arbitrageurs’ return risk and recruit them in support of its policy objectives.

Numerical illustration. Figure 6 illustrates these gains in a numerical example. We set noise’s contribution to return variance under discretion to 20 percent, using its estimated contribution to FCI forecast error variance in Section 2 as a guide. We set $\theta = 2$, so the central bank adjusts the interest rate by two percentage points for each percentage point of output gap. Appendix A.1 provides the remaining parameters.

In the left panel, total return variance falls by more than half at the loss-minimizing targeting intensity, ψ^* . Macro-induced variance falls by about half as policy dampens asset price responses

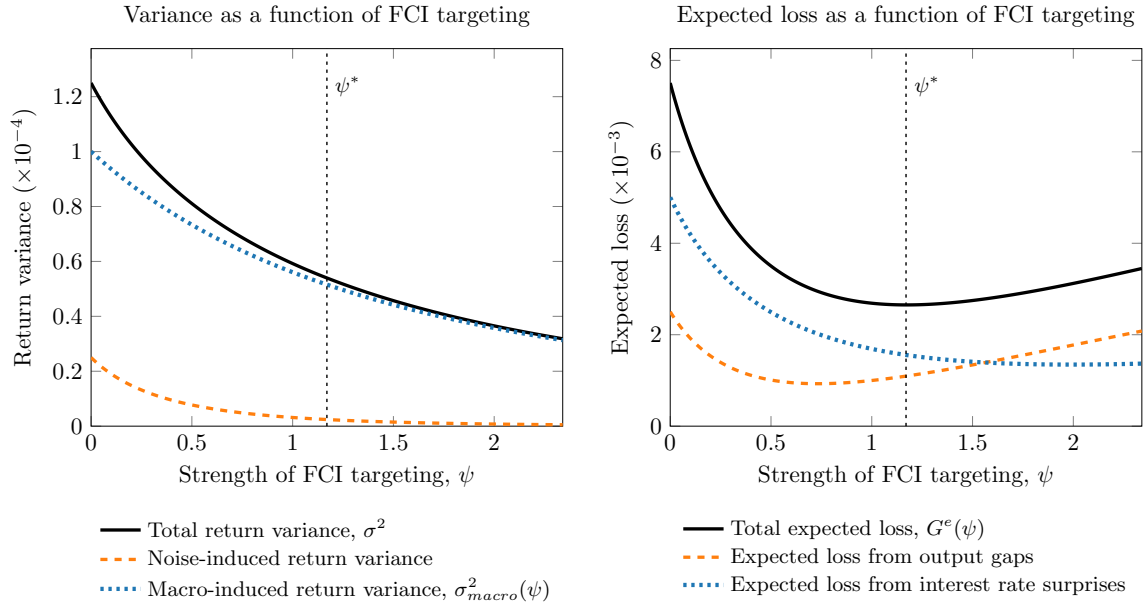


Figure 6: The left panel shows the effect of FCI targeting ψ on return variance (solid black line) and its components induced by noise shocks (dashed orange line) and by macroeconomic shocks (dotted blue line). The right panel shows the effect on the expected loss (solid black line) and its components from output gaps (dashed orange line) and from interest rate surprises (dotted blue line). The vertical lines illustrate the loss-minimizing level of FCI targeting. We use the parameters in Appendix A.1.

to supply and demand shocks. Noise-induced variance falls by about 90 percent. This is partly because policy responds faster to noise and, more importantly, because arbitrageurs trade more aggressively against noise in view of lower return risk. In fact, the price impact of noise falls by more than half, and this accounts for most of the decline in noise-induced variance.

The right panel shows expected loss and its two components, from output gaps and from interest rate surprises. At ψ^* , expected loss falls by about 65 percent, and both components fall by more than half. The decline in the interest rate component is notable, since one might expect that keeping financial conditions near a target requires large policy rate adjustments. For a given return variance, FCI targeting does require the interest rate to respond more strongly to noise shocks as well as to supply shocks. However, the decline in variance reduces the price impact of noise and therefore the required interest rate response to it. In our calibration, the second effect dominates and the variance of interest rate surprises falls by about 70 percent. Intuitively, as FCI targeting recruits arbitrageurs to do more of the stabilization, the central bank needs to do less.

As targeting becomes stronger, the costs of dampening asset price responses to macroeconomic shocks eventually outweigh the gains. The loss-minimizing intensity, $\psi^* \approx 1.2$, implies a coefficient of about 1.2 on the FCI gap in the policy rule (25), compared with 2 on the output gap.

Remark 4 (Fixed target vs. discretionary target announcements). *Appendix A.8 allows the central bank to choose its announced target freely each period. The chosen target departs from expected p_{star} . Following a shock that calls for an interest rate increase, the central bank promises tighter financial conditions next period by setting its target below expected p_{star} . This lowers expected future asset prices and, through the asset-pricing relation, current asset prices, reducing the interest rate adjustment needed today. The target thus introduces history dependence: future policy helps stabilize the economy in response to current shocks. The benefits of this mechanism are well understood in models with interest rate smoothing (see Woodford, 2003). We fix the target at the expected FCI to focus on commitment to lower volatility and isolate the recruitment mechanism. Moreover, this target reflects the central bank's macroeconomic outlook, and the realized FCI is centered around it. In contrast, a freely chosen target departs from the outlook by design, and the realized FCI systematically differs from it. This might create communication challenges. In our numerical examples, the fixed target captures most of the gains available with a freely chosen target (a 65 versus 70 percent reduction in expected loss).*

Remark 5 (FCI targeting vs the full commitment solution). *Appendix A.8 also characterizes the full linear commitment solution. Like FCI targeting, full commitment dampens asset price responses to shocks, reducing return risk and recruiting arbitrageurs to absorb noise. It also exploits history dependence, as the freely chosen target in Remark 4 does, without the restrictions of a particular framework. In our numerical examples, FCI targeting with a fixed target nonetheless captures most of the gains from full commitment (65 versus 72 percent).*

4.3. FCI targeting vs interest rate forward guidance

Central banks routinely communicate their expected policy rate path. Suppose markets interpret this forward guidance as a commitment, so that the central bank finds it costly to deviate from the announced path, for instance to preserve its credibility. This has a resemblance to FCI targeting, which makes deviations from the announced FCI target costly. Nonetheless, we show that the two policies are very different: unlike FCI targeting, interest rate forward guidance typically worsens the central bank's policy objectives. Noise affects output because the central bank offsets it only gradually, and guidance makes this response even more gradual. This amplifies the impact of noise and discourages arbitrageurs from absorbing it.

Formally, suppose that in each period the central bank announces its expected policy rate, $E_t[r_{t+1}^f]$, and penalizes deviations from the announced rate:

$$G_t^r = \min_{r_t^f} \tilde{y}_t^2 + \frac{1 + \psi_r}{\theta} \left(r_t^f - E_{t-1} [r_t^f] \right)^2 + \beta E_t [G_{t+1}^r]. \quad (33)$$

The parameter $\psi_r \geq 0$ governs the strength of the commitment, and the central bank's true objective remains (16). Since the central bank announces its own expectation, which markets share, the announced rate is $E_{t-1}[r_t^f]$ and the guidance term simply adds to the smoothing term. The optimality condition yields

$$r_t^f = E_{t-1} [r_t^f] + \theta_r \tilde{y}_t, \quad \text{where } \theta_r \equiv \frac{\theta}{1 + \psi_r}. \quad (34)$$

Forward guidance lowers the speed of adjustment from θ to θ_r . The following result characterizes the equilibrium.

Proposition 4 (Interest Rate Forward Guidance). *Suppose the central bank follows (33) with $\psi_r \geq 0$ and the parameters satisfy the condition of Proposition 1 with θ replaced by θ_r .*

(i) *There is a locally stable equilibrium given by Proposition 1 with θ replaced by θ_r . Expected loss under the true objective is*

$$G^e(\psi_r) = \hat{G}^r(\psi_r, \sigma^2) \equiv \frac{1}{1 - \beta} \frac{\theta + \theta_r^2}{\theta(1 + \theta_r)^2} \left[(1 - \beta\varphi_\delta)^2 \sigma_\delta^2 + \left(\frac{\sigma^2}{\alpha} \right)^2 \sigma_\mu^2 \right]. \quad (35)$$

(ii) *A small degree of guidance raises return variance and expected loss,*

$$\left. \frac{d\sigma^2}{d\psi_r} \right|_{\psi_r=0} > 0 \quad \text{and} \quad \left. \frac{dG^e}{d\psi_r} \right|_{\psi_r=0} = \frac{\partial \hat{G}^r}{\partial \sigma^2} \frac{d\sigma^2}{d\psi_r} \Big|_{\psi_r=0} > 0,$$

as long as, under discretion,

$$\frac{1}{1 + \theta} \left(\frac{\sigma^2}{\alpha} \right)^2 \sigma_\mu^2 > (1 - \beta\varphi_\delta) \left(\beta - \frac{1 - \beta\varphi_\delta}{1 + \theta} \right) \sigma_\delta^2. \quad (36)$$

Guidance slows the central bank’s response to every shock, and in particular to noise. With a slower response, noise moves asset prices by more. This raises noise-induced return variance, so arbitrageurs absorb less noise and its price impact rises. Guidance thus runs the recruitment mechanism in reverse. Its only offsetting effect is that asset prices move less with demand shocks, which lowers demand-induced variance. Condition (36) says that the noise effect dominates whenever noise-induced variance exceeds a multiple of demand-induced variance. This multiple is small when demand shocks are persistent, as in our calibration, so the condition is easily satisfied. Intuitively, a persistent demand shock requires little policy rate response to begin with, since anticipated future rate increases already tighten current financial conditions. Slowing the policy rate response further therefore does little to reduce demand-induced variance, while it raises noise-induced variance.

The effect on the central bank’s objective follows from the envelope argument of Proposition 3. For a given return variance, the cost of the slower response is second order at $\psi_r = 0$. However, guidance raises return variance, and this effect is first order. Committing to the expected interest rate therefore worsens the central bank’s objective under plausible parameters. In contrast, committing to the expected FCI improves the objective (Proposition 3). FCI targeting speeds up the response to noise while dampening the asset price response to macroeconomic shocks, so it lowers both noise-induced and macro-induced variance, as well as total return variance.

4.4. Robustness of FCI targeting

Our baseline model is stylized. In this section, we show that the rationale for the benefits of FCI targeting holds in more complex economic environments—ranging from policy-side extensions like inflation-output trade-offs to market-side extensions like time-varying arbitrageur beliefs and endogenous arbitrageur supply.

4.4.1. FCI targeting with inflation and output trade-off

In Appendix A.5, we extend the analysis to partially flexible prices, so that the central bank faces a trade-off between stabilizing inflation and output. Our main results continue to hold in this setting.

We endogenize inflation via the standard New Keynesian Phillips curve

$$\pi_t = \kappa \tilde{y}_t + \beta E_t [\pi_{t+1}] + u_t, \quad (37)$$

where π_t denotes nominal price inflation and u_t is a cost-push shock that follows an AR(1) process with persistence φ_u . The central bank’s true objective adds inflation gaps to the loss:

$$G_t = \min_{r_t^f} \tilde{y}_t^2 + \zeta \pi_t^2 + \frac{1}{\theta} \left(r_t^f - E_{t-1} [r_t^f] \right)^2 + \beta E_t [G_{t+1}]. \quad (38)$$

The parameter ζ is the relative weight on inflation gaps, and we normalize the inflation target

to zero. As in the baseline model, the policy instrument is the real interest rate r_t^f , which the central bank now controls through the nominal rate.⁹

Benchmark without gradualism. As in Section 3.2, we start with the benchmark in which the central bank adjusts the interest rate freely. Cost-push shocks create a trade-off: the central bank cannot close the output gap and the inflation gap at the same time. Under discretion, it resolves the trade-off as in the textbook New Keynesian model (Clarida et al., 1999):

$$\tilde{y}_t = -Y_u u_t \quad \text{and} \quad \pi_t = \Pi_u u_t, \quad \text{where } \Pi_u = \frac{1}{1 + \kappa^2 \zeta - \beta \varphi_u} \quad \text{and} \quad Y_u = \kappa \zeta \Pi_u. \quad (39)$$

A positive cost-push shock raises inflation, and the central bank contracts output to contain it. Supply, demand, and noise shocks do not affect output or inflation, as before. The output–asset price relation implies that the asset price is

$$p_t = p_t^o \equiv p_t^* - Y_u u_t. \quad (40)$$

We call p_t^o the *optimal asset price*. It plays the role of p^* in the baseline model.

Discretion. With gradualism, the central bank sets the interest rate under discretion to minimize the loss in (38). The optimality condition yields

$$r_t^f = E_{t-1} \left[r_t^f \right] + \theta (\tilde{y}_t + \kappa \zeta \pi_t). \quad (41)$$

Compared to the rule (17) in the baseline model, the central bank also responds to inflation, as in a Taylor rule. Proposition 5 in the appendix (with $\psi = 0$) shows that the asset price, output, and inflation are the benchmark values plus a common deviation,

$$p_t = p_t^o + w_t, \quad \tilde{y}_t = -Y_u u_t + w_t, \quad \pi_t = \Pi_u u_t + \kappa w_t, \quad (42)$$

where

$$w_t = \frac{1}{1 + \theta_\pi} \left[(1 - \beta \varphi_\delta) \varepsilon_{\delta,t} + (1 - \varphi_u) Y_u \varepsilon_{u,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right] \quad \text{and} \quad \theta_\pi \equiv \theta (1 + \kappa^2 \zeta).$$

As in Proposition 1, gradualism leaves the predictable component of policy unchanged and limits only the response to surprises. Demand and noise surprises pass through to the asset price and output as before. Likewise, cost-push shocks have a larger impact on inflation and a smaller impact on output than in the benchmark without gradualism, because the central bank contracts output only gradually. Noise generates inflation gaps as well as output gaps.

⁹The nominal rate equals the real rate plus expected inflation and an inflation risk premium. Appendix A.5 derives this relation from the pricing of the nominal bond. The central bank can implement any real rate through its choice of the nominal rate, and alternative assumptions about the risk premium leave our results unchanged.

Note also that cost-push shocks add to return variance. The return variance solves a fixed point problem as in Proposition 1, with macro-induced component

$$\sigma_{macro}^2 = \sigma_z^2 + \left(\beta - \frac{1 - \beta\varphi_\delta}{1 + \theta_\pi} \right)^2 \sigma_\delta^2 + \left(\frac{\theta_\pi + \varphi_u}{1 + \theta_\pi} \right)^2 Y_u^2 \sigma_u^2.$$

The last term is new. Cost-push shocks move the optimal asset price, $p_t^o = p_t^* - Y_u u_t$, because the central bank tightens financial conditions to stabilize inflation. Gradualism mitigates this effect, as the factor $\frac{\theta_\pi + \varphi_u}{1 + \theta_\pi} < 1$ shows, but does not overturn it. The higher return variance raises the price impact of noise, which makes recruiting arbitrageurs more valuable.

FCI targeting. With FCI targeting, the central bank announces the expected optimal price, $E_t[p_{t+1}^o]$, as its target for the next period. This is the appropriate target for two reasons. It is the ideal level of the FCI: the asset price that implements the output gap the central bank desires, accounting for cost-push shocks. Targeting expected p^* instead would lead the central bank to resist the output contractions it uses to stabilize inflation. It is also the expected FCI, as Section 4.1 requires, since the deviation w_t has zero mean.

The central bank penalizes deviations from the target announced in the previous period. Its operational objective is

$$G_t^{FCI} = \min_{r_t^f} \tilde{y}_t^2 + \zeta \pi_t^2 + \frac{1}{\theta} \left(r_t^f - E_{t-1}[r_t^f] \right)^2 + \frac{\psi}{\theta} (p_t - E_{t-1}[p_t^o])^2 + \beta E_t[G_{t+1}^{FCI}], \quad (43)$$

and the optimality condition yields

$$r_t^f = E_{t-1}[r_t^f] + \theta(\tilde{y}_t + \kappa\zeta\pi_t) + \psi(p_t - E_{t-1}[p_t^o]). \quad (44)$$

The central bank also responds to the gap between the FCI and its pre-announced target. Proposition 5 shows that the equilibrium asset price is

$$p_t = E_{t-1}[p_t^o] + \frac{1}{1 + \theta_\pi + \psi} \left[(1 + \theta_\pi) \varepsilon_{z,t} - (\theta_\pi + \beta\varphi_\delta) \varepsilon_{\delta,t} - (\theta_\pi + \varphi_u) Y_u \varepsilon_{u,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad (45)$$

and output and inflation are given by (42) with the deviation

$$w_t = \frac{1}{1 + \theta_\pi + \psi} \left[-\psi \varepsilon_{z,t} + (1 + \psi - \beta\varphi_\delta) \varepsilon_{\delta,t} + (1 - \varphi_u + \psi) Y_u \varepsilon_{u,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right].$$

Importantly, FCI targeting reduces return variance as in the baseline model. For a given return variance, it dampens the asset price response to all shocks, including cost-push shocks, through the common factor $\frac{1}{1 + \theta_\pi + \psi}$. This lowers macro-induced variance, including its cost-push component, and the recruitment mechanism then operates as before: lower return variance reduces the price impact of noise, which lowers noise-induced variance further (Proposition 5 in

the appendix). Since noise now generates inflation gaps as well as output gaps, the lower price impact stabilizes both.

Proposition 6 in the appendix shows that, as in the baseline model, a small degree of FCI targeting improves the central bank's true objective. Cost-push shocks induce nonzero gaps on average, but for a given return variance discretionary policy already minimizes the losses they induce. Small deviations from this policy generate second-order losses, while the reduction in return variance generates first-order gains through the same mechanism as in Proposition 3.

4.4.2. FCI targeting with time-varying arbitrageur beliefs

In Appendix A.6, we extend the analysis to asset prices that fluctuate with arbitrageurs' beliefs about future productivity as well as with noise. Our main results continue to hold. Belief shocks create a new source of FCI variance, which strengthens the impact of noise and the importance of recruitment, as cost-push shocks do in the previous section.

We capture belief shocks by allowing arbitrageurs to receive a signal about next period's productivity, $n_{z,t} = \varepsilon_{z,t+1} + e_{z,t}$. Their posterior expectation is $E_t[y_{t+1}^*] = y_t^* + b_t$, where b_t denotes *the belief shock*. It is proportional to the signal and has an ex-ante distribution $b_t \sim N(0, \sigma_b^2)$. Good news, $b_t > 0$, leads arbitrageurs to expect faster productivity growth, and vice versa. We remain agnostic about whether these beliefs are correct, since this does not affect the equilibrium. We assume the central bank shares arbitrageurs' beliefs, and we write E_t for expectations under this common belief.

Discretion. Proposition 7 in the appendix (with $\psi = 0$) shows that the asset price is

$$p_t = p_t^* + \frac{1}{1+\theta} \left[(1 - \beta\varphi_\delta) \varepsilon_{\delta,t} + b_t + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad \text{where } p_t^* = y_t^* - m - \delta_t,$$

and output deviates from potential by the same amount. Belief shocks do not affect p^* , which depends on current productivity, but they raise the asset price by shifting expectations of future prices. Absent gradualism, the central bank would keep the asset price at p^* by raising the interest rate one-for-one with the belief shock. With gradualism, it offsets belief shocks only gradually, so a belief shock acts like a noise shock with a unit price impact. Unlike noise, the price impact of belief shocks does not depend on return variance, since beliefs are common across investors and do not induce trade. Like noise, belief shocks generate output gaps.

Note also that belief shocks add to return variance. The return variance solves a fixed point problem as in Proposition 1, with macro-induced component

$$\sigma_{macro}^2 = \sigma_z^2 + \left(\beta - \frac{1 - \beta\varphi_\delta}{1 + \theta} \right)^2 \sigma_\delta^2 + \frac{\sigma_b^2}{(1 + \theta)^2},$$

where σ_z^2 now denotes the variance of productivity innovations conditional on the news. The

last term is new. It allows the model to attribute part of the variance of financial conditions to belief-driven fluctuations, a natural candidate for the large share of this variance that Section 2 does not attribute to noise. However, as long as some noise remains, this additional variance raises the price impact of noise and makes recruiting arbitrageurs more valuable.

FCI targeting. With FCI targeting, the central bank announces expected p^* , $E_t [p_{t+1}^*] = y_t^* + b_t - m - \varphi_\delta \delta_t$, as its target, which incorporates the news. The operational objective and the interest rate rule are as in Section 4.1. Proposition 7 shows that the equilibrium asset price is

$$p_t = E_{t-1} [p_t^*] + \frac{1}{1 + \theta + \psi} \left[(1 + \theta) (\varepsilon_{z,t} - b_{t-1}) - (\theta + \beta \varphi_\delta) \varepsilon_{\delta,t} + b_t + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \quad (46)$$

where $\varepsilon_{z,t} - b_{t-1}$ is the productivity surprise relative to the previous period's belief. FCI targeting dampens the asset price response to all shocks, including belief shocks, through the factor $\frac{1}{1 + \theta + \psi}$. It responds faster to belief shocks, as to noise, and reduces the return variance they generate. The resulting reduction in return variance lowers the price impact of noise as in the baseline model. Proposition 3 then extends directly: a small degree of FCI targeting recruits arbitrageurs to insulate the FCI from noise and improves the central bank's objectives.

4.4.3. FCI targeting with endogenous arbitrageur supply

In Appendix A.7, we investigate whether our results are robust to endogenous arbitrageur supply. This extension is relevant since FCI targeting affects arbitrageurs' equilibrium returns, potentially influencing their market participation. Our main results still hold. While FCI targeting reduces return variance and arbitrageurs' excess returns, inducing some exit, this exit effect cannot be strong enough to overturn the initial variance decline.

We endogenize the fraction of arbitrageurs α via condition (A.59)

$$\frac{1}{2} \frac{1}{1 - \beta} c(\alpha) = E \left[\sum_{t=0}^{\infty} \beta^t (E_t [\log (R_{t+1}^A)] - E_t [\log (R_{t+1})]) \right].$$

Here, the left side captures the (normalized) cost of entry as an arbitrageur: we assume this supply curve is upward sloping. The right side captures the benefits from entry: the expected excess return generated by the arbitrageurs relative to the inelastic funds (which generate the return on the market portfolio). *In equilibrium*, the excess return in a period satisfies

$$E_t [\log R_{t+1}^A] - E_t [\log R_{t+1}] = \frac{1}{2} \left(\frac{\mu_t}{\alpha} \right)^2 \sigma_{t,r_{t+1}}^2.$$

The excess return is increasing in noisy flows μ_t and in *the return variance*. When variance is higher, arbitrageurs require a higher Sharpe ratio to absorb noisy flows. A higher Sharpe ratio in turn raises the arbitrageurs' expected return.

Since FCI targeting reduces the return variance $\sigma_{t,r_{t+1}}^2$, it might also lower α and induce

some arbitrageur exit. However, since this exit *is caused* by the decline in $\sigma_{t,r_{t+1}}^2$, it cannot be strong enough to undo the decline in $\sigma_{t,r_{t+1}}^2$. This holds even if the supply of arbitrage capital is perfectly elastic. Free entry pins down the noisy flow absorbed by each arbitrageur relative to return variance. Thus, FCI targeting still reduces the variance and thus the noise impact, along with the mass of arbitrageurs. Appendix A.7 verifies this logic and shows that our results qualitatively still hold in this extended model with endogenous arbitrageur entry.

Beyond the specific mechanisms in our model, there are reasons to think that FCI targeting might actually *increase* arbitrageur entry. With discretion, trading against noise in the FCI is arguably difficult and requires a sophisticated understanding of macroeconomic conditions: in our model, arbitrageurs need to forecast $p_{t+1}^* = y_{t+1}^* - m - \delta_{t+1}$. In practice, only a handful of institutional investors have this type of knowledge. When the Fed announces its FCI target, it provides potentially valuable information about the FCI it expects to see, $E_t[p_{t+1}^*]$. This might make trading against noise in the FCI more accessible to a broader set of financial market players: in the above model, it might reduce the cost function $c(\alpha)$. Stabilization can also reduce noise itself: Jeanne and Rose (2002) show that a stabilizing policy framework deters noise trader entry, creating a virtuous cycle of reduced volatility. While these effects are beyond our scope, they illustrate that endogenizing market participation not only fails to eliminate but can even reinforce the beneficial effects of FCI targeting.

5. Final Remarks

This paper theoretically and empirically investigates how monetary policy should respond to macroeconomic fluctuations driven by financial noise.

We first document the quantitative importance of financial noise using idiosyncratic flows into the stock market as an external instrument in a VAR. We find that an expansionary noise shock affects the economy similarly to a demand shock: it loosens financial conditions and raises the output gap, with an incomplete interest rate response. We further find that noise shocks account for up to 20% of the variance of financial conditions and up to 20% of the variance of the output gap in the U.S. The stock price response to these shocks reflects changes in expected excess returns rather than dividends or interest rates, and the output response is consistent with estimated marginal propensities to consume out of stock market wealth.

We then develop a model consistent with these facts. Households delegate their portfolios to noise traders, inelastic funds, and risk-averse arbitrageurs, so noisy flows move the aggregate asset price, which in turn drives aggregate demand. Monetary policy adjusts the interest rate gradually and therefore offsets noise only partially. The impact of noise is amplified by an endogenous volatility feedback loop: higher return volatility makes arbitrageurs less willing to trade against noise, which raises the price impact of noise and volatility further.

In this environment, it is optimal for the central bank to target financial conditions as an independent objective, beyond managing their effects on output and inflation. FCI targeting,

whereby the central bank announces its expected FCI as a target for the near term and adjusts the policy rate to keep the actual FCI close to it, breaks the feedback loop. The central bank does not need to identify noise: it responds to any deviation of the FCI from its expected level regardless of its source. It effectively dampens the FCI response to macroeconomic shocks and accelerates its response to noise, which lowers return volatility. Lower volatility recruits arbitrageurs to trade more aggressively against noise, further reducing the impact of noise on financial conditions and output. The cost of the commitment is second order for a small degree of targeting, while the gains from recruitment are first order. Interest rate forward guidance interpreted as a commitment does the opposite: it slows the response to noise and, under plausible parameters, worsens the central bank's objectives.

Our analysis speaks to a live debate about how central banks should treat financial conditions. Central banks increasingly acknowledge that financial conditions matter for the transmission of policy, as in the remarks by Chair Powell that open this paper, but they stop short of stating where they would like conditions to be. Our results suggest that going one step further and announcing a soft, temporary target for financial conditions would let markets do part of the work. Market participants know not to “fight the Fed.” A central bank that commits to stabilizing financial conditions puts this adage to work: investors who expect the commitment to hold trade toward the target, and the central bank has to do less of the stabilizing itself.

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Online Appendices: Not for Publication

A. Theory Appendix

This appendix contains details related to the theoretical model. Section A.1 describes the parameters we use in the numerical examples that illustrate our findings. Section A.2 provides the microfoundations for the model. Section A.3 contains the proofs omitted from the main text. The remaining Sections A.4-A.8 provide the details of various extensions that we discuss in the main text.

A.1. Details of the numerical exercise

We next describe the parameters we use in the numerical examples that illustrate the results in Sections 4.1-4.2. We consider a quarterly calibration and set $\beta = 0.99$. For the speed of interest rate adjustment, we set $\theta = 2$, so that the central bank changes the interest rate by 2 percentage points in response to a 1 percent output gap (cf. the interest rate rule (17)). We set the persistence of demand and noise shocks to $\varphi_\delta = \varphi_\mu = 0.95$ to match (roughly) the quarterly autocorrelation of the policy interest rate observed in the data.

In our model, the price impact of a unit change in asset demand (as a fraction of supply) at a given interest rate is

$$\mathcal{I} \equiv \frac{\sigma^2}{\alpha}.$$

Recent empirical analyses find that this price impact is large. For instance, Gabaix and Koijen (2021) suggest that for the stock market it could be as large as 5. To be conservative, we set the fraction of elastic funds α to target a price impact equal to one under discretion, $\mathcal{I} = \frac{\sigma^2}{\alpha} = 1$. Since the central bank offsets part of the noise, the equilibrium response of the asset price to a unit noise shock under discretion is $\frac{1}{1+\theta}\mathcal{I}$.

We set the rest of the parameters as follows

$$\begin{aligned} \sigma_{macro}^2(0) &= (0.01)^2, \\ \sigma_\delta^2 &= \sigma_z^2 = \frac{\sigma_{macro}^2(0)}{1 + \left(\beta - \frac{1-\beta\varphi_\delta}{1+\theta}\right)^2}, \\ \sigma_\mu^2 &= \frac{1}{4}(1+\theta)^2 \sigma_{macro}^2(0), \\ \alpha &= \sigma^2(0) = \frac{5}{4}\sigma_{macro}^2(0). \end{aligned} \tag{A.1}$$

Here, $\sigma_{macro}^2(0)$ denotes the macro-induced return variance under discretion [see (22)]. We set the macro-induced standard deviation to 1% to match (roughly) the standard deviation of quarterly output growth in the data, and we set the variances of demand and supply shocks to be identical.¹⁰ We set the noise variance so that, under discretion, the noise-induced return variance $\frac{1}{(1+\theta)^2}\mathcal{I}^2\sigma_\mu^2$ equals one quarter of the macro-induced variance, i.e., noise accounts for 20 percent of the total return variance, in line with the evidence in Section 2. The last line then follows from the price impact normalization. The code that produces the figures is in the replication folder.

¹⁰In this calibration, the level of $\sigma_{macro}^2(0)$ does not change the optimal level of FCI targeting since all other variances scale with $\sigma_{macro}^2(0)$.

A.2. Microfoundations of the model

In this section, we provide the microfoundations of the model that we summarize in Section 3.1 and use throughout the paper. The real side of the economy is the same as the baseline model in Caballero and Simsek (2023). The financial market side is different and allows for noise shocks.

The economy is set in discrete time $t \in \{0, 1, \dots\}$. The model consists of four agent types: asset-holding households (households), hand-to-mouth agents, portfolio managers, and the central bank. The hand-to-mouth agents primarily serve to decouple labor supply decisions from household consumption behavior. The asset-holding households are the main drivers of aggregate demand through their consumption and savings choices. The portfolio managers act on behalf of these households by making portfolio allocation decisions that determine asset prices in financial markets. The central bank conducts monetary policy.

A.2.1. Supply side

Hand-to-mouth agents provide all of the labor supply and spend all of their income (they do not save). Their problem is

$$\begin{aligned} \max_{L_t} \log C_t^{HM} - \chi \frac{L_t^{1+\varphi}}{1+\varphi}, \\ Q_t C_t^{HM} = W_t L_t + T_t. \end{aligned} \quad (\text{A.2})$$

Here, φ denotes the inverse Frisch elasticity of labor supply, Q_t denotes the nominal price for the final good, W_t denotes the nominal wage, and T_t denotes lump-sum transfers from the government (described subsequently). The optimality condition implies a standard labor supply equation

$$\frac{W_t}{Q_t} = \chi L_t^\varphi C_t^{HM}. \quad (\text{A.3})$$

The rest of the supply side is similar to the standard New Keynesian Model. A competitive final goods producer combines the intermediate goods according to the CES technology,

$$Y_t = \left(\int_0^1 Y_t(\nu)^{\frac{\varepsilon_t-1}{\varepsilon_t}} d\nu \right)^{\varepsilon_t/(\varepsilon_t-1)} \quad \text{where } Y_t(\nu) = Z_t L_t(\nu)^{1-\nu}. \quad (\text{A.4})$$

Here, $\varepsilon_t > 1$ denotes the elasticity of substitution that determines the firm markups in equilibrium. We assume it is stochastic around a steady-state level $\varepsilon^* > 1$, which allows us to accommodate cost-push shocks. With these technologies, the demand for intermediate good firms satisfies,

$$Y_t(\nu) \leq \left(\frac{Q_t(\nu)}{Q_t} \right)^{-\varepsilon_t} Y_t, \quad (\text{A.5})$$

$$\text{where } Q_t = \left(\int_0^1 Q_t(\nu)^{1-\varepsilon_t} d\nu \right)^{1/(1-\varepsilon_t)}. \quad (\text{A.6})$$

$Q_t(\nu)$ denotes the nominal price set by the intermediate good firm ν and Q_t is the ideal price index.

The goods market clearing condition is:

$$Y_t = C_t^H + C_t^{HM}. \quad (\text{A.7})$$

Here, C_t^H and C_t^{HM} denote consumption by the asset holding households and the hand-to-mouth agents, respectively.

The labor market clearing condition is

$$\int_0^1 L_t(\nu) d\nu = L_t. \quad (\text{A.8})$$

Finally, to simplify the distribution of output across factors, we assume the government taxes part of the firms' profits lump-sum and redistributes to the hand-to-mouth agents to ensure they receive their production share of output. Specifically, each intermediate firm pays lump-sum taxes determined as follows:

$$T_t = (1 - \nu) Q_t Y_t - W_t L_t. \quad (\text{A.9})$$

This ensures that in equilibrium hand-to-mouth agents receive and spend their production share of output, $(1 - \nu) Q_t Y_t$, and consume [see (A.2)]

$$C_t^{HM} = (1 - \nu) Y_t. \quad (\text{A.10})$$

Substituting this into the goods market clearing condition (A.7), we further obtain

$$Y_t = \frac{C_t^H}{\nu}. \quad (\text{A.11})$$

Hand-to-mouth agents create a Keynesian multiplier effect, but output is ultimately determined by (asset-holding) *households'* spending, C_t^H .

Flexible-price equilibrium. Consider a benchmark without nominal rigidities. In this benchmark, the intermediate good firm ν solves

$$\Pi = \max_{Q, L} QY - W_t L - T_t, \quad (\text{A.12})$$

$$\text{where } Y = Z_t L^{1-\nu} = \left(\frac{Q}{Q_t} \right)^{-\varepsilon_t} Y_t.$$

The firm takes as given the aggregate price, wage, and output, Q_t, W_t, Y_t , and chooses its price, labor input, and output Q, L, Y .

The optimal price is

$$Q = \frac{\varepsilon_t}{\varepsilon_t - 1} W_t \frac{1}{(1 - \nu) Z_t L^{-\nu}}. \quad (\text{A.13})$$

The firm sets an optimal markup over the marginal cost, where the markup depends (inversely) on the elasticity of substitution and the marginal cost depends on the wage and (inversely) on the marginal product of labor.

In equilibrium, all firms choose the same prices and allocations, $Q_t = Q$ and $L_t = L$. Substituting this into (A.13), we obtain a labor demand equation,

$$\frac{W_t}{Q_t} = \frac{\varepsilon_t - 1}{\varepsilon_t} (1 - \nu) Z_t L_t^{-\nu}. \quad (\text{A.14})$$

Combining this with the labor supply equation (A.3), and substituting the hand-to-mouth consumption

(A.10), we obtain the equilibrium labor as the solution to,

$$\chi (L_t^*)^\varphi (1 - \nu) Y_t^* = \frac{\varepsilon_t - 1}{\varepsilon_t} (1 - \nu) Z_t (L_t^*)^{-\nu}.$$

In equilibrium, output is $Y_t^* = Z_t (L_t^*)^{1-\nu}$. Therefore, the flexible-price equilibrium conditions are:

$$\begin{aligned} \chi (L_t^*)^{1+\varphi} &= \frac{\varepsilon_t - 1}{\varepsilon_t}, \\ Y_t^* &= Z_t (L_t^*)^{1-\nu}. \end{aligned} \tag{A.15}$$

Potential output. Consider the flexible-price allocation in which the firms' markups are at their steady-state level, $\varepsilon_t = \varepsilon^*$, that is:

$$\begin{aligned} \chi (L^*)^{1+\varphi} &= \frac{\varepsilon^* - 1}{\varepsilon^*}, \\ Y_t^* &= Z_t (L^*)^{1-\nu}. \end{aligned} \tag{A.16}$$

We refer to L^* as the potential labor supply and $Y^* = Z_t (L^*)^{1-\nu}$ as the potential output. In the main text, we assume the central bank attempts to keep labor and output demand at these levels. In particular, the central bank attempts to stabilize the output fluctuations driven by shocks to ε_t (or markups), because these shocks are distortionary. This enables us to accommodate cost-push shocks that create a trade-off for the central bank for stabilizing inflation and output.¹¹

Sticky prices and demand-driven output. We next describe the equilibrium with nominal rigidities. We start with the special case with full price stickiness and then extend the analysis to partially flexible prices. With fully sticky prices, intermediate good firms have a preset nominal price that remains fixed over time, $Q_t(\nu) = Q^*$. This implies the nominal price for the final good is also fixed and given by $Q_t = Q^*$. Then, each intermediate good firm ν at time t solves the following version of problem (A.12),

$$\begin{aligned} \Pi &= \max_L Q^* Y - W_t L - T_t \\ \text{where } Y &= Z_t L^{1-\nu} \leq Y_t. \end{aligned} \tag{A.17}$$

Since the firm operates with a markup, for small aggregate demand shocks (which we assume) it optimally chooses to meet the demand for its goods, $Y = Z L^{1-\nu} = Y_t$. Therefore, each firm's output is determined by aggregate demand, which is driven by households' spending C_t^H according to (A.11).

Partially flexible prices and the New Keynesian Phillips curve. We next allow for partially flexible prices. With partially flexible prices, each firm still optimally serves the demand and output is still determined by aggregate demand. However, inflation is also endogenous and reacts to output gaps as well as other (cost-push) shocks. We derive a Phillips curve that describes inflation.

We consider the setup in the textbook New Keynesian model in which in each period a randomly

¹¹The central bank does not attempt to stabilize the distortions generated by the *average* markup, because this would induce an average inflationary bias. In practice, these average distortions should ideally be corrected by other policy tools rather than monetary policy.

selected fraction, $1 - \theta$, of firms reset their nominal prices. The firms that do not adjust their price in period t set their labor input to meet the demand for their goods.

Consider the firms that adjust their price in period t . Let Q_t^{adj} denote the optimal price set by these firms. We assume Q_t^{adj} solves the following version of problem (A.12)

$$\begin{aligned} \max_{Q_t^{adj}} \sum_{h=0}^{\infty} \theta^h E_t \left\{ M_{t,t+h} \left(Y_{t+h|t} Q_t^{adj} - W_{t+h} L_{t+h|t} - T_t \right) \right\}, \quad (A.18) \\ \text{where } Y_{t+h|t} = Z_{t+h} L_{t+h|t}^{1-\nu} = \left(\frac{Q_t^{adj}}{Q_{t+h}} \right)^{-\varepsilon_{t+h}} Y_{t+h} \\ \text{and } M_{t,t+h} = \beta^h \frac{1/P_{t+h}}{1/P_t} \frac{Q_t}{Q_{t+h}}. \end{aligned}$$

The terms $L_{t+h|t}$ and $Y_{t+h|t}$ denote the input and the output of the firm (that resets its price in period t) in a future period $t+h$. The term $M_{t,t+h}$ is the stochastic discount factor (SDF) between periods t and $t+h$. Here, P_t denotes the end-of-period price of the market portfolio which we describe later in the appendix.¹²

The optimality condition for problem (A.18) is

$$\begin{aligned} \sum_{h=0}^{\infty} \theta^h E_t \left\{ M_{t,t+h} Q_{t+h}^{\varepsilon_{t+h}} Y_{t+h} \left(Q_t^{adj} - \frac{\varepsilon_{t+h}}{\varepsilon_{t+h} - 1} \frac{W_{t+h}}{(1-\nu) Z_{t+h} L_{t+h|t}^{-\nu}} \right) \right\} = 0, \quad (A.19) \\ \text{where } L_{t+h|t} = \left(\frac{Q_t^{adj}}{Q_{t+h}} \right)^{\frac{-\varepsilon_{t+h}}{1-\nu}} \left(\frac{Y_{t+h}}{Z_{t+h}} \right)^{\frac{1}{1-\nu}}. \end{aligned}$$

We next combine Eq. (A.19) with the remaining equilibrium conditions to derive the New Keynesian Phillips curve. Specifically, we log-linearize the equilibrium around the allocation that features real potential outcomes (with constant markups) and zero inflation, that is, $L_t = L^*, Y_t = Y_t^*, \frac{\varepsilon_t}{\varepsilon_t - 1} = \frac{\varepsilon^*}{\varepsilon^* - 1}, Q_t = Q^*$ for each t , where L^* is given by (A.16) and $Y_t^* = Z_t (L^*)^{1-\nu}$. Throughout, we use the notation $\tilde{x}_t = \log(X_t/X_t^*)$ to denote the log-linearized version of the corresponding variable X_t and we use $\tilde{\mu}_t = \frac{\varepsilon_t}{\varepsilon_t - 1} - \frac{\varepsilon^*}{\varepsilon^* - 1}$ to denote the deviation of the desired markup from its steady-state level. We also let $W_t^{norm} = \frac{W_t}{Z_t Q_t}$ denote the normalized (productivity-adjusted) real wage.

We first log-linearize the labor-supply equilibrium condition (A.3) and use $C_t^{HM} = (1-\nu) Y_t$ to obtain

$$\tilde{w}_t^{norm} = \varphi \tilde{l}_t + \tilde{y}_t. \quad (A.20)$$

Log-linearizing Eqs. (A.4) and (A.8), we also obtain

$$\tilde{y}_t = (1-\nu) \tilde{l}_t. \quad (A.21)$$

¹²Consistent with the financial market side of our model, we assume the SDF is determined by asset-holding households' wealth rather than their consumption. In equilibrium, asset-holding households' wealth is equal to the value of the market portfolio. The exact specification does not affect our analysis because we log-linearize the equation and the interaction of the SDF and prices, $M_{t,t+h} Q_t^{adj}$, generates second-order terms that drop out of the log linearization.

Finally, we log-linearize Eq. (A.19) (and linearize for $\tilde{\mu}_t$) to obtain

$$\sum_{h=0}^{\infty} (\theta\beta)^h E_t \left\{ \tilde{q}_t^{adj} - \left(\tilde{w}_{t+h}^{norm} + \nu \tilde{l}_{t+h|t} + \tilde{q}_{t+h} \right) - \tilde{\mu}_{t+h} \right\} = 0, \quad (\text{A.22})$$

$$\text{where } \tilde{l}_{t+h|t} = \frac{-\varepsilon^* (\tilde{q}_t^{adj} - \tilde{q}_{t+h})}{1 - \nu} + \tilde{l}_{t+h}.$$

The second line uses $\tilde{y}_t = (1 - \nu) \tilde{l}_t$.

We next combine Eqs. (A.20 – A.22) and rearrange terms to obtain a closed-form solution for the price set by adjusting firms

$$\tilde{q}_t^{adj} = (1 - \theta\beta) \sum_{h=0}^{\infty} (\theta\beta)^h E_t [\Theta \tilde{y}_{t+h} + \tilde{q}_{t+h} + \tilde{\mu}_{t+h}],$$

$$\text{where } \Theta = \frac{1 + \varphi}{1 - \nu + \nu\varepsilon^*}$$

Since the expression is recursive, we can also write it as a difference equation

$$\tilde{q}_t^{adj} = (1 - \theta\beta) (\Theta \tilde{y}_t + \tilde{q}_t + \tilde{\mu}_t) + \theta\beta E_t [\tilde{q}_{t+1}^{adj}]. \quad (\text{A.23})$$

Here, we have used the law of iterated expectations, $E_t[\cdot] = E_t[E_{t+1}[\cdot]]$.

Next, we consider the aggregate price index (A.6)

$$Q_t = \left((1 - \theta) (Q_t^{adj})^{1-\varepsilon} + \int_{S_t} (Q_{t-1}(\nu))^{1-\varepsilon} d\nu \right)^{1/(1-\varepsilon)}$$

$$= \left((1 - \theta) (Q_t^{adj})^{1-\varepsilon} + \theta Q_{t-1}^{1-\varepsilon} \right)^{1/(1-\varepsilon)},$$

where we have used the observation that a fraction θ of prices are the same as in the last period. The term, S_t , denotes the set of sticky firms in period t , and the second line follows from the assumption that adjusting firms are randomly selected. Log-linearizing the equation, we further obtain $\tilde{q}_t = (1 - \theta) \tilde{q}_t^{adj} + \theta \tilde{q}_{t-1}$. After substituting inflation, $\pi_t = \tilde{q}_t - \tilde{q}_{t-1}$, this implies

$$\pi_t = (1 - \theta) (\tilde{q}_t^{adj} - \tilde{q}_{t-1}). \quad (\text{A.24})$$

Hence, inflation is proportional to the price change by adjusting firms.

Finally, note that Eq. (A.23) can be written in terms of the price change of adjusting firms as

$$\tilde{q}_t^{adj} - \tilde{q}_{t-1} = (1 - \theta\beta) (\Theta \tilde{y}_t + \tilde{\mu}_t) + \tilde{q}_t - \tilde{q}_{t-1} + \theta\beta E_t [\tilde{q}_{t+1}^{adj} - \tilde{q}_t].$$

Substituting $\pi_t = \tilde{q}_t - \tilde{q}_{t-1}$ and combining with Eq. (A.24), we obtain the New Keynesian Phillips curve

(37) that we use in the main text

$$\begin{aligned}
\pi_t &= \kappa \tilde{y}_t + \beta E_t [\pi_{t+1}] + u_t, \\
\text{where } \kappa &= \frac{1-\theta}{\theta} (1-\theta\beta) \frac{1+\varphi}{1-\nu+\nu\varepsilon^*} \\
\text{and } u_t &= \frac{1-\theta}{\theta} (1-\theta\beta) \tilde{\mu}_t, \text{ where } \tilde{\mu}_t = \frac{\varepsilon_t}{\varepsilon_t-1} - \frac{\varepsilon^*}{\varepsilon^*-1}.
\end{aligned} \tag{A.25}$$

A.2.2. Demand side and financial markets

We next describe households' consumption-savings and portfolio allocation decisions. In equilibrium, together with monetary policy, these decisions determine aggregate demand, asset prices, and output.

Financial assets. There are two assets. There is a market portfolio, which is a claim on firms' profits νY_t (the firms' share of output). We let P_t denote the ex-dividend price of the market portfolio (which we also refer to as "the aggregate asset price" or "aggregate asset prices"). The gross return of the market portfolio is

$$R_{t+1} = \frac{\nu Y_{t+1} + P_{t+1}}{P_t}. \tag{A.26}$$

There is also a risk-free asset in zero net supply. Its gross return R_t^f is set by the central bank, as we describe in the main text.

Households' consumption-savings decisions. Households have standard preferences:

$$E_t \left[\sum_{h=0}^{\infty} \beta^h \log C_{t+h}^H \right], \tag{A.27}$$

along with the budget constraint

$$\begin{aligned}
W_{t+1} + C_{t+1}^H &= W_t \left((1-\omega_t) R_t^f + \omega_t R_{t+1} \right) \\
&= D_{t+1} + K_{t+1}, \\
\text{where } D_{t+1} &= W_t \left[(1-\omega_t) (R_t^f - 1) + \omega_t \frac{\nu Y_{t+1}}{P_t} \right] \\
\text{and } K_{t+1} &= W_t \left[1 - \omega_t + \omega_t \frac{P_{t+1}}{P_t} \right].
\end{aligned} \tag{A.28}$$

W_t denotes the end-of-period wealth and ω_t denotes the market portfolio weight in period t . The term $W_t \left((1-\omega_t) R_t^f + \omega_t R_{t+1} \right)$ is the beginning-of-period wealth in period $t+1$. The second line breaks this term into a component that captures the interest and dividend income (D_{t+1}) and a residual component that captures the capital (K_{t+1}).

Households take their portfolio allocation as given (delegated to the portfolio managers) and make a consumption-savings decision. We assume their consumption follows the rule

$$C_t^H = (1-\beta) (D_t + K_t \exp(\delta_t)). \tag{A.29}$$

When $\delta_t = 0$, this is the optimal rule given the log preferences in (A.27). When $\delta_t > 0$ (resp. $\delta_t < 0$), households spend more (resp. less) than the optimal rule. We refer to δ_t as an *aggregate demand shock*

and view it as a modeling device to capture various factors that affect aggregate spending in practice, e.g., a consumer sentiment shock, a fiscal policy shock, or a discount rate shock. Having the demand shock multiply K_t rather than $D_t + K_t$ does not play an important role beyond simplifying the expressions.¹³

The portfolio managers (the market) and the portfolio allocation. Households delegate their portfolio choice to managers. The portfolio managers are infinitesimal and they do not consume themselves. They simply make a portfolio choice for households. For simplicity, each household invests with a continuum of managers randomly sampled from all managers. This assumption ensures that there is no portfolio heterogeneity across households and thus no individual wealth dynamics.

In each period, a fraction η of portfolio managers are “noise traders” and their portfolio weight is $\omega_t^N = 1 + \frac{1}{\eta}\mu_t$. That is, they deviate from the optimal portfolio benchmark by an amount $\frac{1}{\eta}\mu_t$. We refer to μ_t as the aggregate noise—the total amount of flow that needs to be absorbed by other investors. Among the remaining managers, a mass $1 - \eta - \alpha$ represent “inelastic funds” and their portfolio weight is $\omega_t^I = 1$. Finally, a mass α of managers are “arbitrageurs” (or elastic funds) who choose their portfolio weights to maximize expected log assets-under-management, after observing the risk-free rate $r_t^f = \log R_t^f$ and the current noise μ_t .¹⁴

$$\max_{\omega_t^A} E_t \left[\log \left(\alpha W_t \left(R_t^f + \omega_t^A \left(R_{t+1} - R_t^f \right) \right) \right) \right].$$

As we describe in the main text, the optimality condition is approximately given by (12)

$$\omega_t^A \sigma_{t,r_{t+1}} = \frac{E_t[r_{t+1}] + \frac{(\sigma_{t,r_{t+1}})^2}{2} - r_t^f}{\sigma_{t,r_{t+1}}}.$$

Financial market clearing. Financial markets are in equilibrium when the households in the aggregate hold the market portfolio, both before and after the portfolio allocation:

$$W_t = P_t \quad \text{and} \quad \omega_t = \alpha \omega_t^A + \eta \left(1 + \frac{\mu_t}{\eta} \right) + (1 - \eta - \alpha) = 1. \quad (\text{A.30})$$

Output-asset price relation. We next derive the equilibrium condition (7) that we use in the main text. Combining Eqs. (A.28) and (A.30), we obtain $D_t = \nu Y_t, K_t = P_t$. In equilibrium, dividends are equal to the firms’ share of output. Capital is equal to the (ex-dividend) value of the market portfolio. Substituting these observations into the consumption rule in (A.29), we obtain

$$C_t^H = (1 - \beta) (\nu Y_t + P_t \exp(\delta_t)).$$

¹³We could alternatively capture demand shocks as shocks to households’ discount factor β in a fully optimizing framework. We prefer our approach where we view demand shocks as small consumption “mistakes” because doing so simplifies the analysis and gives us greater flexibility in specifying the process for δ_t .

¹⁴We assume arbitrageurs maximize log-wealth in line with the households’ preferences in (A.27). In the special case where households follow the optimal rule ($\sigma_s^2 = 0$), this problem results in portfolio allocations that maximize the households’ utility. We formulate the portfolio problem in terms of wealth, rather than consumption, because we allow consumption to deviate from the optimal rule. In our setup, wealth is a more accurate representation of welfare, as it captures the *ideal* consumption a household could choose if she followed the optimal rule.

Substituting Eq. (A.11) ($C_t^H = \nu Y_t$) into this expression yields Eq. (7)

$$\begin{aligned} Y_t &= (1 - \beta) \frac{1}{\nu\beta} P_t \exp(\delta_t) \\ \implies y_t &= m + p_t + \delta_t, \quad \text{where } m \equiv \log\left(\frac{1 - \beta}{\nu\beta}\right). \end{aligned} \quad (\text{A.31})$$

We refer to this as *the output-asset price relation*. In equilibrium, output depends on aggregate wealth, P_t , the MPC out of wealth, $1 - \beta$, the demand shock, δ_t , and the Keynesian multiplier, $1/(\nu\beta)$. The second line describes the relation in logs and obtains the derived parameter m .

Financial market equilibrium condition. We next derive the equilibrium condition (13) that we use in the main text. Eq. (A.30) implies $\omega_t^A = 1 - \frac{\mu_t}{\alpha}$. Substituting this into (12), we obtain (13)

$$E_t[r_{t+1}] = r_t^f + \frac{(\sigma_{t,r_{t+1}})^2}{2} - \mu_t \frac{(\sigma_{t,r_{t+1}})^2}{\alpha}.$$

In equilibrium, the expected return on the market portfolio depends on the risk premium, return variance, and noise. The impact of noise is increasing in the return variance and decreasing in the mass of arbitrageurs.

Campbell-Shiller approximation to the equilibrium return. We next derive the Campbell-Shiller approximation in (9). First note that Eq. (A.26) implies

$$\begin{aligned} r_{t+1} &= \log\left(\frac{\nu Y_{t+1}}{P_{t+1}} \frac{P_{t+1}}{P_t} + \frac{P_{t+1}}{P_t}\right) \\ &= \log\left(\frac{\nu Y_{t+1}}{P_{t+1}} + 1\right) + \log\left(\frac{P_{t+1}}{P_t}\right) \\ &= \log(1 + X_{t+1}) + p_{t+1} - p_t. \end{aligned} \quad (\text{A.32})$$

Here, we have defined the dividend price ratio, $X_t = \nu Y_t / P_t$.

Setting the demand shifter to zero ($\delta_t = 0$) and output equal to its potential $Y = Y^*$, Eq. (A.31) implies $Y^* = (1 - \beta) \frac{1}{\nu\beta} P^*$. This implies $X^* = \nu Y_t^* / P_t^* = \frac{1 - \beta}{\beta}$.

Finally, log-linearize (A.32) around $X_{t+1} = X^*$. Let $x_{t+1} = \log(X_{t+1}/X^*)$ denote the log deviation of the dividend price ratio from its steady-state level. Consider the term, $\log(1 + X_{t+1}) = \log(1 + X^* \exp(x_{t+1}))$. Using a Taylor approximation around $x_{t+1} = 0$, we obtain

$$\begin{aligned} \log(1 + X_{t+1}) &\approx \log(1 + X^*) + \frac{X^*}{1 + X^*} x_{t+1} \\ &\approx \log\left(\frac{1}{\beta}\right) + (1 - \beta) \left(\log\left(\frac{\nu Y_{t+1}}{P_{t+1}}\right) - \log\left(\frac{1 - \beta}{\beta}\right) \right). \end{aligned}$$

Substituting this into (A.32) and collecting the constant terms, we obtain Eq. (9)

$$\begin{aligned} r_{t+1} &= \rho - (1 - \beta)m + (1 - \beta)y_{t+1} + \beta p_{t+1} - p_t \\ &= \rho + p_{t+1} + (1 - \beta)\delta_{t+1} - p_t, \end{aligned}$$

where the second line substitutes the output asset price relation (7) to simplify the expression.

Present discounted value relation. Substituting Eq. (9) into the financial market equilibrium condition (13), we also obtain the present discounted value relation (14) that describes the equilibrium asset price

$$p_t = \rho + E_t[p_{t+1}] + (1 - \beta) E_t[\delta_{t+1}] - \left(r_t^f + \frac{1}{2} (\sigma_{t,r_{t+1}})^2 \right) + \mu_t \frac{(\sigma_{t,r_{t+1}})^2}{\alpha}.$$

A.3. Omitted proofs

This section contains the proofs omitted from the main text.

Proof of Propositions 1 and 2. We prove Proposition 2. Proposition 1 is the special case with $\psi = 0$. We conjecture and verify a linear equilibrium in which the return variance σ^2 is constant, the output gap and the interest rate surprise are linear functions of the current shocks $(\varepsilon_{z,t}, \varepsilon_{\delta,t}, \varepsilon_{\mu,t})$, and the asset price and output respond one-for-one to the interest rate, $\frac{dp_t}{dr_t^f} = \frac{dy_t}{dr_t^f} = -1$.

Benchmark without gradualism. Consider first the interest rate that closes the output gap. Setting $p_t = p_t^*$ in the asset price equation (14) and using $E_t[p_{t+1}] = E_t[p_{t+1}^*] = y_t^* - m - \varphi_\delta \delta_t$ (which holds since the expected future output gap is zero in the conjectured equilibrium), we obtain $r_t^{f*} = \rho - \frac{1}{2}\sigma^2 + (1 - \beta\varphi_\delta)\delta_t + \frac{\sigma^2}{\alpha}\mu_t$, as claimed. Since $E_t[p_{t+1}]$ does not depend on r_t^f , Eq. (14) implies $\frac{dp_t}{dr_t^f} = -1$, and Eq. (7) implies $\frac{dy_t}{dr_t^f} = -1$. This verifies the conjecture and implies that, for any interest rate,

$$p_t - p_t^* = y_t - y_t^* = -\left(r_t^f - r_t^{f*}\right). \quad (\text{A.33})$$

Output gap, interest rate, and asset price. Since the continuation value in (24) depends only on future shocks, the first-order condition with respect to r_t^f is the interest rate rule (25). To evaluate it, note that the conjectured equilibrium implies $E_{t-1}[\tilde{y}_t] = 0$ and thus $E_{t-1}[r_t^f] = E_{t-1}[r_t^{f*}]$ by (A.33). Using $\delta_t - E_{t-1}[\delta_t] = \varepsilon_{\delta,t}$ and $\mu_t - E_{t-1}[\mu_t] = \varepsilon_{\mu,t}$, the interest rate surprise is

$$r_t^f - E_{t-1}[r_t^f] = (1 - \beta\varphi_\delta)\varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha}\varepsilon_{\mu,t} - \tilde{y}_t. \quad (\text{A.34})$$

Likewise, since $p_t^* - E_{t-1}[p_t^*] = \varepsilon_{z,t} - \varepsilon_{\delta,t}$, the FCI gap is $p_t - E_{t-1}[p_t^*] = \tilde{y}_t + \varepsilon_{z,t} - \varepsilon_{\delta,t}$. Substituting both expressions into (25) and solving for $\tilde{y}_t$ gives (27). Eq. (28) then follows from (A.33), and Eq. (26) from $p_t = E_{t-1}[p_t^*] + \tilde{y}_t + \varepsilon_{z,t} - \varepsilon_{\delta,t}$. The output gap is a linear function of current shocks with zero mean, which verifies $E_{t-1}[\tilde{y}_t] = 0$.

Return and its variance. The return in (9) implies $r_{t+1} - E_t[r_{t+1}] = p_{t+1} - E_t[p_{t+1}] + (1 - \beta)\varepsilon_{\delta,t+1}$. Substituting (26) gives (29). The expected return follows from (13) and (28):

$$E_t[r_{t+1}] = r_t^f + \frac{1}{2}\sigma^2 - \frac{\sigma^2}{\alpha}\mu_t = \rho + (1 - \beta\varphi_\delta)\delta_t - \tilde{y}_t. \quad (\text{A.35})$$

Calculating the conditional variance of (29) gives the fixed point problem (30). That is, σ^2 is a root of the quadratic

$$P(x; \psi) = a(\psi)x^2 - x + \sigma_{macro}^2(\psi), \quad \text{where } a(\psi) \equiv \frac{\sigma_\mu^2}{(1 + \theta + \psi)^2 \alpha^2}, \quad (\text{A.36})$$

which has two positive roots if and only if $4a(\psi)\sigma_{macro}^2(\psi) < 1$. The smaller root,

$$\sigma^2 = \frac{2\sigma_{macro}^2(\psi)}{1 + \sqrt{1 - 4a(\psi)\sigma_{macro}^2(\psi)}}, \quad (\text{A.37})$$

corresponds to the (locally) stable equilibrium.

Comparative statics. Let $u \equiv \frac{1}{1+\theta+\psi}$, which decreases from $\frac{1}{1+\theta}$ toward zero as ψ increases. The macro-induced variance is a convex quadratic in u ,

$$\sigma_{macro}^2 = (1+\theta)^2 \sigma_z^2 u^2 + (Bu - (1-\beta))^2 \sigma_\delta^2, \quad \text{where } B \equiv \theta + \beta\varphi_\delta,$$

minimized at $u^* = \frac{(1-\beta)B\sigma_\delta^2}{(1+\theta)^2\sigma_z^2 + B^2\sigma_\delta^2}$. The condition $\beta(1+\theta+\varphi_\delta) > 1$ is equivalent to $(1-\beta)(1+\theta) < B$, which implies $u^* < \frac{1}{1+\theta}$. Hence $\sigma_{macro}^2(\psi)$ is decreasing in ψ on $[0, \bar{\psi})$ and increasing thereafter, where $\bar{\psi} = \frac{1}{u^*} - 1 - \theta > 0$. Over $[0, \bar{\psi})$, both $a(\psi)$ and $\sigma_{macro}^2(\psi)$ decrease with ψ . It follows from (A.37) that σ^2 decreases, and so does the noise-induced variance $a(\psi)(\sigma^2)^2$. Moreover, since $a(\psi)\sigma_{macro}^2(\psi)$ decreases over this range, the assumed condition $4a(0)\sigma_{macro}^2(0) < 1$ ensures that the two positive roots exist for every $\psi \in [0, \bar{\psi}]$. For $\psi > \bar{\psi}$, they exist as long as $4a(\psi)\sigma_{macro}^2(\psi) < 1$.

Finally, for Proposition 1, Eq. (A.37) with $\psi = 0$ shows that σ^2 is increasing in σ_μ^2 (through $a(0)$). The expected loss (23) follows from substituting (19) and (20) into (16) and using (A.34). It is increasing in σ_μ^2 both directly and through σ^2 . $\square$

Proof of Proposition 3. Proposition 2 implies that the output gap and the interest rate surprise are

$$\begin{aligned} \tilde{y}_t &= \frac{1}{1+\theta+\psi} \left[-\psi\varepsilon_{z,t} + (\psi+c)\varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha}\varepsilon_{\mu,t} \right], \\ r_t^f - E_{t-1}[r_t^f] &= \frac{1}{1+\theta+\psi} \left[\psi\varepsilon_{z,t} + (\theta c - \psi(1-c))\varepsilon_{\delta,t} + (\theta+\psi)\frac{\sigma^2}{\alpha}\varepsilon_{\mu,t} \right], \end{aligned} \quad (\text{A.38})$$

where $c \equiv 1 - \beta\varphi_\delta \in (0, 1]$. Substituting these into (31) and treating σ^2 as given, we obtain

$$\hat{G}(\psi, \sigma^2) = \frac{1}{1-\beta} \left[\frac{(1+\theta)\psi^2}{\theta(1+\theta+\psi)^2} \sigma_z^2 + \frac{(\psi+c)^2 + \frac{1}{\theta}(\theta c - \psi(1-c))^2}{(1+\theta+\psi)^2} \sigma_\delta^2 + \frac{\theta + (\theta+\psi)^2}{\theta(1+\theta+\psi)^2} \left(\frac{\sigma^2}{\alpha} \right)^2 \sigma_\mu^2 \right]. \quad (\text{A.39})$$

This is increasing in σ^2 since σ^2 enters only the last term. Differentiating each term with respect to ψ and simplifying, we obtain

$$\frac{\partial \hat{G}(\psi, \sigma^2)}{\partial \psi} = \frac{2\psi}{(1-\beta)\theta(1+\theta+\psi)^3} \left[(1+\theta)^2 \sigma_z^2 + (1+\theta-c)^2 \sigma_\delta^2 + \left(\frac{\sigma^2}{\alpha} \right)^2 \sigma_\mu^2 \right],$$

which is positive for $\psi > 0$ and zero at $\psi = 0$. Finally, differentiating (32) and using $\frac{\partial \hat{G}}{\partial \psi}|_{\psi=0} = 0$ gives $\frac{dG^e}{d\psi}|_{\psi=0} = \frac{\partial \hat{G}}{\partial \sigma^2} \frac{d\sigma^2}{d\psi}|_{\psi=0}$, which is negative since Proposition 2 shows that $\frac{d\sigma^2}{d\psi} < 0$ over the range $\psi \in [0, \bar{\psi})$. This completes the proof. $\square$

A.4. FCI targeting vs interest rate forward guidance

This section proves Proposition 4 in Section 4.3. Throughout, $c \equiv 1 - \beta\varphi_\delta$.

Proof of Proposition 4. *Part (i).* Since the continuation value in (33) depends only on future shocks, the first-order condition with respect to r_t^f is (34), using $\frac{d\tilde{y}_t}{dr_t^f} = -1$. This is the discretionary rule (17) with θ_r in place of θ , so the proof of Proposition 1 in Appendix A.3 applies verbatim and yields the equilibrium in (18)–(22) with θ replaced by θ_r . In this equilibrium, $E_{t-1}[\tilde{y}_t] = 0$, so the announced rate $E_t[r_{t+1}^f]$ coincides with the rate path markets anticipate, as conjectured. For the expected loss, let $X_t \equiv c\varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha}\varepsilon_{\mu,t}$. Eq. (A.34) holds for any speed of adjustment, so combining it with (34) gives $\tilde{y}_t = \frac{X_t}{1+\theta_r}$ and $r_t^f - E_{t-1}[r_t^f] = \frac{\theta_r X_t}{1+\theta_r}$. Substituting into (31) gives (35).

Part (ii). Let $h(\theta_r) \equiv \frac{\theta+\theta_r^2}{\theta(1+\theta_r)^2}$. Then $h'(\theta_r) = \frac{2(\theta_r-\theta)}{\theta(1+\theta_r)^3}$, which is zero at $\theta_r = \theta$, so $\frac{\partial \hat{G}^r}{\partial \psi_r}|_{\psi_r=0} = 0$, while $\frac{\partial \hat{G}^r}{\partial \sigma^2} > 0$ since σ^2 enters only the bracketed term. Hence $\frac{dG^e}{d\psi_r}|_{\psi_r=0} = \frac{\partial \hat{G}^r}{\partial \sigma^2} \frac{d\sigma^2}{d\psi_r}|_{\psi_r=0}$ and the two derivatives have the same sign. For the sign of $\frac{d\sigma^2}{d\psi_r}$, note that by part (i), σ^2 solves $P_r(\sigma^2) \equiv a_r(\sigma^2)^2 - \sigma^2 + \sigma_{macro,r}^2 = 0$, where $a_r \equiv \frac{\sigma_\mu^2}{(1+\theta_r)^2\alpha^2}$ and $\sigma_{macro,r}^2 \equiv \sigma_z^2 + \left(\beta - \frac{c}{1+\theta_r}\right)^2 \sigma_\delta^2$ are the coefficients of (22) with θ replaced by θ_r . Moreover, σ^2 is the smaller root of P_r , at which $P_r'(\sigma^2) = 2a_r\sigma^2 - 1 < 0$. Differentiating $P_r(\sigma^2) = 0$ with respect to θ_r gives

$$\frac{d\sigma^2}{d\theta_r} = \frac{1}{1 - 2a_r\sigma^2} \left[(\sigma^2)^2 \frac{\partial a_r}{\partial \theta_r} + \frac{\partial \sigma_{macro,r}^2}{\partial \theta_r} \right],$$

where $\frac{\partial a_r}{\partial \theta_r} = -\frac{2a_r}{1+\theta_r}$ and $\frac{\partial \sigma_{macro,r}^2}{\partial \theta_r} = \frac{2c}{(1+\theta_r)^2} \left(\beta - \frac{c}{1+\theta_r}\right) \sigma_\delta^2$. Hence $\frac{d\sigma^2}{d\theta_r} < 0$ if and only if $(1+\theta_r)a_r(\sigma^2)^2 > c\left(\beta - \frac{c}{1+\theta_r}\right)\sigma_\delta^2$. At $\psi_r = 0$ we have $\theta_r = \theta$ and $a_r(\sigma^2)^2 = \frac{1}{(1+\theta)^2} \left(\frac{\sigma^2}{\alpha}\right)^2 \sigma_\mu^2$, so this is condition (36). Since $\frac{d\theta_r}{d\psi_r} = -\frac{\theta}{(1+\psi_r)^2} < 0$, $\frac{d\sigma^2}{d\psi_r}|_{\psi_r=0} > 0$ if (36) holds and < 0 if it fails. $\square$

A.5. FCI targeting with inflation and output trade-off

This section analyzes the extension in Section 4.4.1. We first characterize the benchmark without gradualism, which determines the FCI target. We then show that the equilibrium with gradualism has the same structure as in the baseline model, with the speed of adjustment θ replaced by $\theta_\pi \equiv \theta(1 + \kappa^2\zeta)$ and cost-push surprises added to the shocks, so that Propositions 2 and 3 extend directly.

Environment with inflation. Consider the baseline model from Section 4 but suppose inflation follows the New Keynesian Phillips curve derived in Appendix A.2 [see (A.25)],

$$\pi_t = \kappa \tilde{y}_t + \beta E_t[\pi_{t+1}] + u_t, \quad \text{where } u_t = \varphi_u u_{t-1} + \varepsilon_{u,t} \text{ and } \sigma_u^2 \equiv \text{var}(\varepsilon_{u,t}).$$

Here, $\pi_t \simeq \log \frac{Q_t}{Q_{t-1}}$ denotes inflation measured as the log change of the nominal price index Q_t . The cost-push shocks are independent of the supply, demand, and noise shocks.

There is a *nominal* risk-free bond with rate $\exp(i_t^f)$ set by the central bank, in addition to the real risk-free bond with rate $\exp(r_t^f)$ and the market portfolio with real return R_{t+1} . Both bonds are in zero net supply. The real bond is traded as in Section 3, so the financial market equilibrium condition (13) is unchanged. For completeness, we specify the mapping between the nominal and the real rate, although it is not central to the analysis. We assume the nominal bond is priced by the inelastic funds, which hold the market portfolio and maximize expected log wealth. Their optimality condition implies

$$E_t \left[M_{t+1}^I \left(\frac{\exp(i_t^f)}{Q_{t+1}/Q_t} - \exp(r_t^f) \right) \right] = 0, \quad \text{where } M_{t+1}^I = \frac{1}{R_{t+1}}.$$

Assuming R_{t+1} and inflation $\frac{Q_{t+1}}{Q_t}$ are (approximately) log-normally distributed, we obtain

$$i_t^f = r_t^f + \left[E_t[\pi_{t+1}] - \frac{1}{2} \text{var}_t(\pi_{t+1}) \right] - \text{cov}_t(\pi_{t+1}, r_{t+1}). \quad (\text{A.40})$$

The nominal rate equals the real rate plus expected inflation (adjusted for a Jensen term) and an inflation risk premium. Equation (A.40) maps the central bank's choice of the nominal rate one-to-one into the real rate. We therefore describe policy in terms of r_t^f and report the implied nominal rate. Alternative assumptions about which investors price the nominal bond change the risk premium term but not the equilibrium allocation, since the policy instrument is the real rate.

The central bank's true objective is (38). With FCI targeting, it solves (43), where the optimal asset price p_t^o is defined in (40) and characterized next. Discretion is the special case $\psi = 0$.

Benchmark without gradualism. Suppose the central bank adjusts the interest rate freely ($\theta \rightarrow \infty$) and sets policy under discretion, taking its future policy as given. We conjecture an equilibrium in which output and inflation depend only on the cost-push shock, $\tilde{y}_t = -Y_u u_t$ and $\pi_t = \Pi_u u_t$, so that $E_t[\pi_{t+1}] = \Pi_u \varphi_u u_t$ does not depend on the current interest rate. The NKPC then implies $\frac{d\pi_t}{dr_t^f} = \kappa \frac{d\tilde{y}_t}{dr_t^f} = -\kappa$, and the optimality condition is

$$\tilde{y}_t = -\kappa \zeta \pi_t. \quad (\text{A.41})$$

With a positive cost-push shock, the central bank targets a negative output gap to contain inflation. The output gap is more negative when it has a greater impact on inflation (higher κ) and when the central

bank puts a greater weight on inflation (higher ζ). Substituting the conjecture into the NKPC gives $\Pi_u = -\kappa Y_u + \beta \varphi_u \Pi_u + 1$, and combining with (A.41) gives the coefficients in (39).

The output–asset price relation (7) implies that the asset price is $p_t = p_t^* + \tilde{y}_t = p_t^o$, the optimal asset price in (40). Substituting $p_t = p_t^o$ and $E_t[p_{t+1}] = E_t[p_{t+1}^o]$ into the asset pricing relation (14), the interest rate that implements the benchmark is

$$r_t^{f,o} = \rho - \frac{1}{2}\sigma^2 + (1 - \beta\varphi_\delta)\delta_t + (1 - \varphi_u)Y_u u_t + \frac{\sigma^2}{\alpha}\mu_t. \quad (\text{A.42})$$

Relative to r^* in (15), the interest rate also rises with the cost-push shock to implement the output contraction. The return (9) is $r_{t+1} = E_t[r_{t+1}] + \varepsilon_{z,t+1} - \beta\varepsilon_{\delta,t+1} - Y_u\varepsilon_{u,t+1}$, so the return variance is $\sigma^2 = \sigma_z^2 + \beta^2\sigma_\delta^2 + Y_u^2\sigma_u^2$. Cost-push shocks add to return variance, whereas noise does not, as in the baseline benchmark.

Equilibrium with gradualism and FCI targeting. With gradualism, the central bank solves (43) and announces the expected optimal price $E_t[p_{t+1}^o]$ as its target. Let

$$\begin{aligned} X_t &\equiv r_t^{f,o} - E_{t-1}[r_t^{f,o}] = (1 - \beta\varphi_\delta)\varepsilon_{\delta,t} + (1 - \varphi_u)Y_u\varepsilon_{u,t} + \frac{\sigma^2}{\alpha}\varepsilon_{\mu,t}, \\ \xi_t &\equiv p_t^o - E_{t-1}[p_t^o] = \varepsilon_{z,t} - \varepsilon_{\delta,t} - Y_u\varepsilon_{u,t}. \end{aligned} \quad (\text{A.43})$$

X_t is the interest rate surprise that implements the benchmark response to current shocks, and ξ_t is the surprise in the target.

Proposition 5 (Equilibrium with Inflation and FCI Targeting). *Suppose the central bank follows the FCI targeting policy in (43) with $\psi \geq 0$, and the parameters satisfy $(1 + \theta_\pi)^2 \alpha^2 > 4\sigma_\mu^2 \sigma_{macro}^2(0)$ and $\beta(1 + \theta_\pi + \varphi_\delta) > 1$, where $\sigma_{macro}^2(\psi)$ is defined in (A.46). Then there is a locally stable equilibrium in which*

$$\tilde{y}_t = -Y_u u_t + w_t, \quad \pi_t = \Pi_u u_t + \kappa w_t, \quad p_t = E_{t-1}[p_t^o] + \xi_t + w_t, \quad r_t^f = r_t^{f,o} - w_t, \quad (\text{A.44})$$

where $r_t^{f,o}$ is the benchmark rate in (A.42), evaluated at the equilibrium return variance, and

$$w_t = \frac{X_t - \psi \xi_t}{1 + \theta_\pi + \psi}.$$

The return is

$$r_{t+1} = E_t[r_{t+1}] + \frac{(1 + \theta_\pi)\varepsilon_{z,t+1} - (\theta_\pi + \varphi_u)Y_u\varepsilon_{u,t+1} + \frac{\sigma^2}{\alpha}\varepsilon_{\mu,t+1}}{1 + \theta_\pi + \psi} - \left(\beta - \frac{1 + \psi - \beta\varphi_\delta}{1 + \theta_\pi + \psi}\right)\varepsilon_{\delta,t+1}, \quad (\text{A.45})$$

and its variance σ^2 is the smaller positive solution to

$$\begin{aligned} \sigma^2 &= \sigma_{macro}^2(\psi) + \frac{1}{(1 + \theta_\pi + \psi)^2} \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2, \\ \text{where } \sigma_{macro}^2(\psi) &= \left(\frac{1 + \theta_\pi}{1 + \theta_\pi + \psi}\right)^2 \sigma_z^2 + \left(\beta - \frac{1 + \psi - \beta\varphi_\delta}{1 + \theta_\pi + \psi}\right)^2 \sigma_\delta^2 + \left(\frac{\theta_\pi + \varphi_u}{1 + \theta_\pi + \psi}\right)^2 Y_u^2 \sigma_u^2. \end{aligned} \quad (\text{A.46})$$

Let $\bar{\psi} = \arg \min_{\psi \geq 0} \sigma_{macro}^2(\psi) > 0$. For $\psi \in [0, \bar{\psi})$, stronger FCI targeting strictly reduces σ^2 and both of

its components. The nominal interest rate is $i_t^f = r_t^f + \Pi_u \varphi_u u_t - \frac{1}{2} \text{var}_t(\pi_{t+1}) - \text{cov}_t(\pi_{t+1}, r_{t+1})$, where the variance and the covariance are constants determined by (A.44) and (A.45).

The equilibrium has the structure of Proposition 2. The predictable component of cost-push shocks generates the benchmark inflation-output trade-off, which gradualism does not affect since it is built into the anticipated rate path. Surprises in all shocks, including cost-push surprises, are split between output gaps and interest rate surprises at the speed $\theta_\pi > \theta$: an interest rate change now stabilizes inflation as well as output, so the central bank is willing to move faster. FCI targeting dampens the asset price response to every shock by the common factor $\frac{1+\theta_\pi}{1+\theta_\pi+\psi}$ for a given return variance, which lowers σ^2 and the price impact of noise. Since noise now generates inflation gaps κw_t as well as output gaps, the reduction in its price impact stabilizes both.

Macro-stabilization effects of FCI targeting. We evaluate the policy with the true loss (38), averaged across shock realizations:

$$G^e(\psi) = E[G_t(\psi)] = \frac{1}{1-\beta} E \left[\tilde{y}_t^2 + \zeta \pi_t^2 + \frac{1}{\theta} \left(r_t^f - E_{t-1} \left[r_t^f \right] \right)^2 \right]. \quad (\text{A.47})$$

Substituting (A.44) and using $Y_u = \kappa \zeta \Pi_u$, the cross terms cancel and

$$\tilde{y}_t^2 + \zeta \pi_t^2 = (Y_u^2 + \zeta \Pi_u^2) u_t^2 + (1 + \kappa^2 \zeta) w_t^2.$$

The first term is the benchmark loss from cost-push shocks and does not depend on policy. Hence

$$G^e(\psi) = \frac{1}{1-\beta} (Y_u^2 + \zeta \Pi_u^2) \frac{\sigma_u^2}{1-\varphi_u^2} + \hat{G}_\pi(\psi, \sigma^2(\psi)), \quad (\text{A.48})$$

where

$$\hat{G}_\pi(\psi, \sigma^2) \equiv \frac{1}{1-\beta} E \left[(1 + \kappa^2 \zeta) w_t^2 + \frac{1}{\theta} \left(r_t^f - E_{t-1} \left[r_t^f \right] \right)^2 \right]$$

is the expected loss when w_t and the interest rate surprise follow (A.44) with the return variance σ^2 held fixed, the analogue of $\hat{G}$ in (32).

Proposition 6 (Macro-Stabilization Effects of FCI Targeting with Inflation). *Consider the equilibrium in Proposition 5. Holding return variance fixed, expected loss $\hat{G}_\pi(\psi, \sigma^2)$ is increasing in targeting intensity, with zero derivative at $\psi = 0$. Holding targeting intensity fixed, expected loss is increasing in return variance. Since a small degree of FCI targeting reduces return variance, it strictly reduces expected loss:*

$$\left. \frac{dG^e(\psi)}{d\psi} \right|_{\psi=0} = \left. \frac{\partial \hat{G}_\pi}{\partial \sigma^2} \frac{d\sigma^2}{d\psi} \right|_{\psi=0} < 0.$$

The loss-minimizing targeting intensity is therefore positive, $\psi^* = \arg \min_{\psi \geq 0} G^e(\psi) > 0$.

Proof of Proposition 5. We conjecture and verify a linear equilibrium in which σ^2 is constant, $\tilde{y}_t = -Y_u u_t + w_t$ and $\pi_t = \Pi_u u_t + \kappa w_t$ with $E_{t-1}[w_t] = 0$, and $\frac{d\tilde{y}_t}{dr_t^f} = \frac{dp_t}{dr_t^f} = -1$. In this equilibrium, $E_t[\pi_{t+1}] = \Pi_u \varphi_u u_t$ and $E_t[p_{t+1}] = E_t[p_{t+1}^o]$ do not depend on r_t^f , so the NKPC implies $\frac{d\pi_t}{dr_t^f} = -\kappa$,

and the continuation value depends only on future shocks. The first-order condition with respect to r_t^f is therefore the rule (44).

Predictable components. Taking E_{t-1} of (44) and using $E_{t-1}[p_t - E_{t-1}[p_t^o]] = E_{t-1}[w_t] = 0$ (verified below) gives $E_{t-1}[\tilde{y}_t + \kappa\zeta\pi_t] = 0$. The conjectured predictable components satisfy this condition and the NKPC by the same calculation as in the benchmark. Note also that $\tilde{y}_t + \kappa\zeta\pi_t = (1 + \kappa^2\zeta)w_t$ since $Y_u = \kappa\zeta\Pi_u$.

Output gap, interest rate, and asset price. As in the benchmark, the rate $r_t^{f,o}$ implements $\tilde{y}_t = -Y_u u_t$, and since $\frac{d\tilde{y}_t}{dr_t^f} = -1$, we have $w_t = \tilde{y}_t + Y_u u_t = -\left(r_t^f - r_t^{f,o}\right)$ for any rate. Since $E_{t-1}[w_t] = 0$, we have $E_{t-1}[r_t^f] = E_{t-1}[r_t^{f,o}]$, and the interest rate surprise is $r_t^f - E_{t-1}[r_t^f] = X_t - w_t$. The output-asset price relation (7) gives $p_t - p_t^o = \tilde{y}_t + Y_u u_t = w_t$, so the FCI gap is $p_t - E_{t-1}[p_t] = w_t + \xi_t$. Substituting into (44),

$$X_t - w_t = \theta(1 + \kappa^2\zeta)w_t + \psi(w_t + \xi_t),$$

which gives $w_t = \frac{X_t - \psi\xi_t}{1 + \theta_\pi + \psi}$. This is a linear function of current shocks with zero mean, verifying $E_{t-1}[w_t] = 0$ and (A.44).

Return and its variance. The return (9) implies $r_{t+1} - E_t[r_{t+1}] = p_{t+1} - E_t[p_{t+1}] + (1 - \beta)\varepsilon_{\delta,t+1} = \xi_{t+1} + w_{t+1} + (1 - \beta)\varepsilon_{\delta,t+1}$, which gives (A.45) and, taking the conditional variance, (A.46). This fixed point problem has the form of (A.36) with θ_π in place of θ . Let $u \equiv \frac{1}{1 + \theta_\pi + \psi}$ and $B \equiv \theta_\pi + \beta\varphi_\delta$. The macro-induced variance is the convex quadratic

$$\sigma_{macro}^2 = \left[(1 + \theta_\pi)^2 \sigma_z^2 + (\theta_\pi + \varphi_u)^2 Y_u^2 \sigma_u^2\right] u^2 + (Bu - (1 - \beta))^2 \sigma_\delta^2,$$

minimized at $u^* = \frac{(1 - \beta)B\sigma_\delta^2}{(1 + \theta_\pi)^2 \sigma_z^2 + (\theta_\pi + \varphi_u)^2 Y_u^2 \sigma_u^2 + B^2 \sigma_\delta^2}$. The condition $\beta(1 + \theta_\pi + \varphi_\delta) > 1$ implies $(1 - \beta)(1 + \theta_\pi) < B$ and hence $u^* < \frac{1}{1 + \theta_\pi}$, so $\bar{\psi} > 0$. The comparative statics then follow exactly as in the proof of Proposition 2.

Nominal rate. Substituting $E_t[\pi_{t+1}] = \Pi_u \varphi_u u_t$ into (A.40) gives the expression in the proposition. The variance and the covariance are constants because inflation and return surprises are linear in the shocks with constant coefficients. $\square$

Proof of Proposition 6. From the proof above, $w_t = \frac{X_t - \psi\xi_t}{1 + \theta_\pi + \psi}$ and $r_t^f - E_{t-1}[r_t^f] = X_t - w_t = \frac{(\theta_\pi + \psi)X_t + \psi\xi_t}{1 + \theta_\pi + \psi}$. Substituting into $\hat{G}_\pi$, treating σ^2 as given, and differentiating with respect to ψ , we obtain

$$\frac{\partial \hat{G}_\pi(\psi, \sigma^2)}{\partial \psi} = \frac{2\psi}{(1 - \beta)\theta(1 + \theta_\pi + \psi)^3} \left[(1 + \theta_\pi)^2 \sigma_z^2 + (\theta_\pi + \beta\varphi_\delta)^2 \sigma_\delta^2 + (\theta_\pi + \varphi_u)^2 Y_u^2 \sigma_u^2 + \left(\frac{\sigma^2}{\alpha}\right)^2 \sigma_\mu^2 \right],$$

the analogue of the expression in the proof of Proposition 3. It is positive for $\psi > 0$ and zero at $\psi = 0$. $\hat{G}_\pi$ increases with σ^2 since σ^2 enters only through the noise term. Finally, Proposition 5 shows that $\frac{d\sigma^2}{d\psi} < 0$ for $\psi \in [0, \bar{\psi})$, and differentiating (A.48) gives the result. $\square$

A.6. FCI targeting with time-varying beliefs

This section analyzes the extension in Section 4.4.2, in which the aggregate asset price fluctuates with arbitrageurs' beliefs about future productivity in addition to noise. We show that belief shocks enter the equilibrium like noise shocks with a unit price impact, so that Propositions 2 and 3 extend directly.

Model with time-varying beliefs. Consider the baseline model in Section 3.1 with two changes. First, we modify Eq. (6) slightly so that

$$y_{t+1}^* = y_t^* + \varepsilon_{z,t+1}, \text{ where } \varepsilon_{z,t+1} \sim N(0, \bar{\sigma}_z^2). \quad (\text{A.49})$$

We assume $\varepsilon_{z,t}$ is i.i.d. Normally distributed with variance $\bar{\sigma}_z^2$. We reserve the notation σ_z^2 for the variance of $\varepsilon_{z,t+1}$ conditional on news. Second, arbitrageurs receive a signal (news) that *they believe* is informative about future productivity,

$$n_{z,t} = \varepsilon_{z,t+1} + e_{z,t}, \text{ where } e_{z,t} \sim N(0, \tilde{\sigma}_z^2), \quad (\text{A.50})$$

with $e_{z,t}$ i.i.d. Normally distributed. Under this belief, the posterior for future productivity is

$$\varepsilon_{z,t+1} \sim N(b_t, \sigma_z^2), \text{ where } b_t = \frac{1/\tilde{\sigma}_z^2}{1/\tilde{\sigma}_z^2 + 1/\bar{\sigma}_z^2} n_{z,t} \text{ and } \sigma_z^2 = \frac{1}{1/\tilde{\sigma}_z^2 + 1/\bar{\sigma}_z^2}. \quad (\text{A.51})$$

Ex ante (before observing the news), the belief shock is distributed as

$$b_t \sim N(0, \sigma_b^2), \text{ where } \sigma_b^2 = \bar{\sigma}_z^2 - \sigma_z^2. \quad (\text{A.52})$$

The ex-ante mean of beliefs is zero as the news is unpredictable, and the ex-ante variance of beliefs is equal to the variance reduction induced by (perceived) news.

We remain agnostic about whether these beliefs are correct. For concreteness, suppose $n_{z,t}$ is actually drawn according to $n_{z,t} = \eta_t + \varepsilon_{z,t+1} + e_{z,t}$ with $\eta_t \sim N(\bar{\eta}_t, \sigma_\eta^2)$. When $\bar{\eta}_t = \sigma_\eta^2 = 0$, beliefs are correct. When $\bar{\eta}_t = 0$ but $\sigma_\eta^2 > 0$, beliefs are unbiased but overconfident. When $\bar{\eta}_t \neq 0$, beliefs are also biased. The parameters $\bar{\eta}_t$ and σ_η^2 do not affect the equilibrium, so our analysis accommodates these possibilities. We assume the central bank shares arbitrageurs' beliefs, and we write E_t for expectations under this common belief.¹⁵ The rest of the model is the same as in Section 3.1. Arbitrageurs choose their portfolios to maximize expected log wealth under this belief, so Eqs. (9), (13), and (14) hold with expectations driven by the news.

Benchmark without gradualism. Suppose the central bank adjusts the interest rate freely. It closes the output gap in every period and state, so $p_t = p_t^*$ and $y_t = y_t^*$. Since $E_t[p_{t+1}^*] = y_t^* + b_t - m - \varphi_\delta \delta_t$, the asset pricing relation (14) implies that the interest rate that implements p^* is

$$r_t^{f*} = \rho - \frac{1}{2}\sigma^2 + (1 - \beta\varphi_\delta)\delta_t + b_t + \frac{\sigma^2}{\alpha}\mu_t. \quad (\text{A.53})$$

¹⁵Households' beliefs do not affect the equilibrium, since they do not make a portfolio choice and their consumption is driven by asset prices and demand shocks.

Relative to r_{star} in (15), the interest rate rises one-for-one with the belief shock to keep the asset price at p_{star} . The return (9) is $r_{t+1} = E_t[r_{t+1}] + (\varepsilon_{z,t+1} - b_t) - \beta\varepsilon_{\delta,t+1}$, so the return variance is $\sigma^2 = \sigma_z^2 + \beta^2\sigma_\delta^2$. Beliefs about future productivity are absorbed by the interest rate and do not add to return variance.

Equilibrium with gradualism and FCI targeting. With gradualism, the central bank solves (24) and announces expected p_{star} , $E_t[p_{t+1}^*] = y_t^* + b_t - m - \varphi_\delta\delta_t$, as its target. Discretion is the special case $\psi = 0$. Let $\tilde{\varepsilon}_{z,t} \equiv \varepsilon_{z,t} - b_{t-1}$ denote the productivity surprise relative to the previous period's belief, with $var_{t-1}(\tilde{\varepsilon}_{z,t}) = \sigma_z^2$, and let

$$X_t \equiv (1 - \beta\varphi_\delta)\varepsilon_{\delta,t} + b_t + \frac{\sigma^2}{\alpha}\varepsilon_{\mu,t} \quad \text{and} \quad \xi_t \equiv \tilde{\varepsilon}_{z,t} - \varepsilon_{\delta,t}. \quad (\text{A.54})$$

X_t is the surprise in the benchmark rate (A.53), and $\xi_t = p_t^* - E_{t-1}[p_t^*]$ is the surprise in the target.

Proposition 7 (Equilibrium with FCI Targeting and Time-Varying Beliefs). *Suppose the central bank follows the FCI targeting policy in (24) with $\psi \geq 0$, and the parameters satisfy $(1 + \theta)^2\alpha^2 > 4\sigma_\mu^2\sigma_{macro}^2(0)$ and $\beta(1 + \theta + \varphi_\delta) > 1$, where $\sigma_{macro}^2(\psi)$ is defined in (A.57). Then there is a locally stable equilibrium in which*

$$p_t = E_{t-1}[p_t^*] + \xi_t + \tilde{y}_t, \quad \tilde{y}_t = \frac{X_t - \psi\xi_t}{1 + \theta + \psi}, \quad r_t^f = r_t^{f*} - \tilde{y}_t, \quad (\text{A.55})$$

where r_t^{f*} is the benchmark rate in (A.53), evaluated at the equilibrium return variance. The return is

$$r_{t+1} = E_t[r_{t+1}] + \frac{(1 + \theta)\tilde{\varepsilon}_{z,t+1} + b_{t+1} + \frac{\sigma^2}{\alpha}\varepsilon_{\mu,t+1}}{1 + \theta + \psi} - \left(\beta - \frac{1 + \psi - \beta\varphi_\delta}{1 + \theta + \psi}\right)\varepsilon_{\delta,t+1}, \quad (\text{A.56})$$

and its variance σ^2 is the smaller positive solution to

$$\begin{aligned} \sigma^2 &= \sigma_{macro}^2(\psi) + \frac{1}{(1 + \theta + \psi)^2} \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2, \\ \text{where } \sigma_{macro}^2(\psi) &= \left(\frac{1 + \theta}{1 + \theta + \psi}\right)^2 \sigma_z^2 + \left(\beta - \frac{1 + \psi - \beta\varphi_\delta}{1 + \theta + \psi}\right)^2 \sigma_\delta^2 + \frac{\sigma_b^2}{(1 + \theta + \psi)^2}. \end{aligned} \quad (\text{A.57})$$

Let $\bar{\psi} = \arg \min_{\psi \geq 0} \sigma_{macro}^2(\psi) > 0$. For $\psi \in [0, \bar{\psi})$, stronger FCI targeting strictly reduces σ^2 and both of its components.

The equilibrium is that of Proposition 2 with the belief shock added to the shocks. The belief shock enters X_t with a unit coefficient: at the anticipated interest rate, it raises the expected future asset price and hence the current asset price and output one-for-one, while p_{star} is unchanged. It therefore behaves like a noise shock, except that its price impact does not depend on return variance, since beliefs are common across investors and do not induce trade. Under discretion, belief shocks add $\frac{\sigma_b^2}{(1 + \theta)^2}$ to the macro-induced return variance, which raises the price impact of noise. FCI targeting dampens the asset price response to belief shocks along with all other shocks, which reduces return variance and the price impact of noise.

Macro-stabilization effects of FCI targeting. The expected loss (31) can be written as $G^e(\psi) = \hat{G}(\psi, \sigma^2(\psi))$ as in (32), and the argument in the proof of Proposition 3 applies verbatim.

Substituting (A.55) and differentiating with respect to ψ for a given σ^2 gives

$$\frac{\partial \hat{G}(\psi, \sigma^2)}{\partial \psi} = \frac{2\psi}{(1-\beta)\theta(1+\theta+\psi)^3} \left[(1+\theta)^2 \sigma_z^2 + (\theta + \beta\varphi_\delta)^2 \sigma_\delta^2 + \sigma_b^2 + \left(\frac{\sigma^2}{\alpha} \right)^2 \sigma_\mu^2 \right],$$

which is positive for $\psi > 0$ and zero at $\psi = 0$, while $\hat{G}$ increases with σ^2 . Since $\frac{d\sigma^2}{d\psi} < 0$ on $[0, \bar{\psi})$ by Proposition 7, a small degree of FCI targeting reduces expected loss and $\psi^* > 0$. Belief shocks strengthen this conclusion: they add to the return variance that FCI targeting reduces, while the cost of the faster response to them is second order, as for noise.

Proof of Proposition 7. The proof follows that of Proposition 2 in Appendix A.3, with two changes. First, the expected future asset price is $E_t[p_{t+1}] = E_t[p_{t+1}^*] = y_t^* + b_t - m - \varphi_\delta \delta_t$, which does not depend on r_t^f , so $\frac{d\tilde{y}_t}{dr_t^f} = \frac{dp_t}{dr_t^f} = -1$ and the rate that closes the output gap is (A.53). Its surprise relative to $t-1$ is X_t in (A.54), since $E_{t-1}[b_t] = 0$. Second, the surprise in pstar relative to the previous period's belief is $p_t^* - E_{t-1}[p_t^*] = \xi_t$, with $\tilde{\varepsilon}_{z,t}$ in place of $\varepsilon_{z,t}$. Substituting $r_t^f - E_{t-1}[r_t^f] = X_t - \tilde{y}_t$ and $p_t - E_{t-1}[p_t^*] = \tilde{y}_t + \xi_t$ into the rule (25) gives $\tilde{y}_t$ in (A.55), and the asset price and the interest rate follow. The return (9) gives $r_{t+1} - E_t[r_{t+1}] = \xi_{t+1} + \tilde{y}_{t+1} + (1-\beta)\varepsilon_{\delta,t+1}$, which yields (A.56) and (A.57). With $u \equiv \frac{1}{1+\theta+\psi}$ and $B \equiv \theta + \beta\varphi_\delta$, the macro-induced variance is the convex quadratic $\left[(1+\theta)^2 \sigma_z^2 + \sigma_b^2 \right] u^2 + (Bu - (1-\beta))^2 \sigma_\delta^2$, minimized at $u^* = \frac{(1-\beta)B\sigma_\delta^2}{(1+\theta)^2 \sigma_z^2 + \sigma_b^2 + B^2 \sigma_\delta^2} < \frac{1}{1+\theta}$ under $\beta(1+\theta+\varphi_\delta) > 1$. The comparative statics then follow exactly as in the proof of Proposition 2. $\square$

A.7. FCI targeting with endogenous arbitrageur supply

In the main text, we took the fraction of arbitrageurs α that trade the market portfolio as an exogenous constant. Could endogenizing arbitrageur entry overturn the beneficial effects of FCI targeting? In this appendix, we address this question by developing a version of the model in which α is endogenous and driven by arbitrageurs' expected excess returns. We show that while FCI targeting reduces these returns (by reducing volatility) and induces some arbitrageur exit, our main results qualitatively still hold.

Model with endogenous arbitrageur entry. Consider the baseline model in Section 3.1 with one change: the fraction η of portfolio managers are noise traders as before whereas the remaining fraction $1 - \eta$ are *potential* arbitrageurs indexed by subscript $j \in [0, 1 - \eta]$. Potential arbitrageurs make a one-time decision (for simplicity) before time 0 whether to become an arbitrageur or to become an inelastic fund. We assume each fund manager is compensated by a fraction of the expected log returns they generate for households. Therefore, the expected lifetime benefit to becoming an arbitrageur is equal to the present discounted sum of expected *excess* log returns:

$$u^A = E \left[\sum_{t=0}^{\infty} \beta^t (E_t [\log (R_{t+1}^A)] - E_t [\log (R_{t+1})]) \right]. \quad (\text{A.58})$$

Here, R_{t+1}^A denotes the arbitrageurs' return and R_{t+1} is the inelastic fund's return—which is simply the return on the market portfolio. The entry cost is heterogeneous across agents and given by $\frac{1}{2} \frac{1}{1-\beta} c(j)$ for potential arbitrageur j (normalizing terms will simplify the expressions). We assume $c(j)$ is a strictly increasing function with the boundary condition $\lim_{j \rightarrow 1-\eta} c(j) = \infty$. The equilibrium fraction of arbitrageurs $\alpha \in (0, 1 - \eta)$ is then the solution to

$$\frac{1}{2} \frac{1}{1-\beta} c(\alpha) = u^A. \quad (\text{A.59})$$

In equilibrium, agents $j \in [0, \alpha]$ become arbitrageurs as their entry costs are sufficiently low, whereas agents $j \in (\alpha, 1 - \eta]$ become inelastic funds. We assume the benefits and costs are both infinitesimal (as arbitrageurs are infinitesimal) so they do not affect the resource constraints. The rest of the model is unchanged. For notational simplicity, we assume noise shocks are i.i.d. ($\varphi_\mu = 0$).

To characterize the equilibrium, first consider the arbitrageurs' return, $E_t [\log R_{t+1}^A]$. Up to a log-Normal approximation, we have

$$\begin{aligned} E_t [\log R_{t+1}^A] &= \max_{\omega_t^A} r_t^f + \omega_t^A \left(E_t [r_{t+1}] + \frac{(\sigma_{t,r_{t+1}})^2}{2} - r_t^f \right) - \frac{1}{2} (\omega_t^A)^2 (\sigma_{t,r_{t+1}})^2 \\ &= r_t^f + \frac{1}{2} \left(\frac{E_t [r_{t+1}] + \frac{(\sigma_{t,r_{t+1}})^2}{2} - r_t^f}{\sigma_{t,r_{t+1}}} \right)^2. \end{aligned}$$

Here, the second line substitutes the optimal portfolio weight in (12). Arbitrageurs' *optimal* expected return is increasing in the Sharpe ratio. Next note that in equilibrium the Sharpe ratio satisfies (see (11))

and (12))

$$\frac{E_t[r_{t+1}] + \frac{(\sigma_{t,r_{t+1}})^2}{2} - r_t^f}{\sigma_{t,r_{t+1}}} = \underbrace{\left(1 - \frac{\mu_t}{\alpha}\right) \sigma_{t,r_{t+1}}}_{\text{arbitrageur risk}}.$$

Combining these expressions, we calculate

$$E_t[\log R_{t+1}^A] = r_t^f + \frac{1}{2} \left(1 - \frac{\mu_t}{\alpha}\right)^2 \sigma_{t,r_{t+1}}^2.$$

Next consider the inelastic funds' return, which can be written as

$$\begin{aligned} E_t[\log R_{t+1}] &= r_t^f + \left(E_t[r_{t+1}] + \frac{(\sigma_{t,r_{t+1}})^2}{2} - r_t^f \right) - \frac{1}{2} (\sigma_{t,r_{t+1}})^2 \\ &= r_t^f + \left(1 - \frac{\mu_t}{\alpha}\right) \sigma_{t,r_{t+1}}^2 - \frac{1}{2} (\sigma_{t,r_{t+1}})^2. \end{aligned}$$

Here, the second line substitutes for the equilibrium Sharpe ratio as before.

Combining these observations, we calculate the per-period excess log return as

$$E_t[\log R_{t+1}^A] - E_t[\log R_{t+1}] = \frac{1}{2} \left(\frac{\mu_t}{\alpha}\right)^2 \sigma_{t,r_{t+1}}^2. \quad (\text{A.60})$$

The excess return is increasing in the amount of noise μ_t and in *the return variance*. With higher variance, arbitrageurs require a greater Sharpe ratio to absorb noisy flows. A greater Sharpe ratio in turn induces arbitrageurs to earn a greater expected excess return.

Note also that when the variance is constant over time, $\sigma_{t,r_{t+1}}^2 = \sigma^2$, we can combine (A.60) with (A.58) and (A.59) and use the simplifying assumption that μ_t is i.i.d. to obtain the equilibrium version of the entry condition

$$\frac{1}{2} \frac{1}{1-\beta} c(\alpha) = \frac{1}{2} \frac{1}{1-\beta} \frac{\sigma_\mu^2}{\alpha^2} \sigma^2 \implies c(\alpha) = \frac{\sigma_\mu^2}{\alpha^2} \sigma^2. \quad (\text{A.61})$$

All else equal, there is greater entry when the return variance is larger and the noise variance is larger—as this creates more frequent trading opportunities.

Equilibrium with discretion and FCI targeting. For a given α , Proposition 2 applies, with discretion as the special case $\psi = 0$. The return variance is the smaller root of the fixed point problem (30),

$$\sigma^2 = \sigma_{macro}^2(\psi) + \frac{1}{(1+\theta+\psi)^2} \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2.$$

The entry condition (A.61) implies $\frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2 = c(\alpha) \sigma^2$. Substituting and rearranging, the equilibrium fraction of arbitrageurs and the return variance satisfy

$$\alpha^2 c(\alpha) \left[1 - \frac{c(\alpha)}{(1+\theta+\psi)^2} \right] = \sigma_\mu^2 \sigma_{macro}^2(\psi) \quad \text{and} \quad \sigma^2 = \frac{\sigma_{macro}^2(\psi)}{1 - \frac{c(\alpha)}{(1+\theta+\psi)^2}}. \quad (\text{A.62})$$

The first equation determines α , and the second then determines σ^2 . The locally stable equilibrium corresponds to the smaller root of the fixed point problem, which requires $c(\alpha) < \frac{1}{2} (1+\theta+\psi)^2$. On this branch, the left side of the first equation is strictly increasing in α .

Consider first discretion, $\psi = 0$. Greater exogenous variance, $\sigma_{macro}^2(0)$ or σ_μ^2 , raises the right side of the first equation and hence the equilibrium α : more trading opportunities induce entry. It also raises σ^2 despite the entry, since $c(\alpha)$ rises with α . Entry mitigates but does not overturn the endogenous increase in variance.

Next consider FCI targeting. Over the range $\psi \in [0, \bar{\psi})$, FCI targeting lowers $\sigma_{macro}^2(\psi)$ and raises $(1 + \theta + \psi)^2$. Both changes lower the right side of the first equation in (A.62) and raise its left side for a given α , so the equilibrium α falls: the decline in return variance and the faster policy response to noise reduce arbitrageurs' excess returns and induce exit. Both changes also lower $\frac{c(\alpha)}{(1+\theta+\psi)^2}$, so the second equation implies that σ^2 falls. FCI targeting therefore still reduces the equilibrium return variance, albeit by less than with exogenous α . Since the exit is caused by the decline in variance, it mitigates but does not overturn that decline.

The limiting case of a perfectly elastic supply of arbitrage capital, $c(\alpha) = \bar{c}$, makes the logic transparent. Free entry pins down the ratio $\frac{\sigma^2}{\alpha^2}$, since the required excess return $\bar{c} = \frac{\sigma_\mu^2}{\alpha^2} \sigma^2$ depends on the noisy flow absorbed by each arbitrageur as well as on return variance. When variance falls, arbitrageurs exit until each remaining arbitrageur absorbs enough noisy flow to earn the required return, and this happens at a lower level of variance. The price impact of noise, $\frac{\sigma^2}{\alpha} = \sigma \sqrt{\bar{c}/\sigma_\mu^2}$, is then proportional to return volatility, so exit cannot undo the decline in noise-induced variance. Total variance remains anchored by its macro-induced component, $\sigma^2 = \sigma_{macro}^2(\psi) / \left[1 - \frac{\bar{c}}{(1+\theta+\psi)^2}\right]$, which FCI targeting reduces.

Finally, consider the effects of FCI targeting on the central bank's objective. The output gap and the interest rate surprise are still given by (A.38), with the price impact of noise

$$\frac{\sigma^2}{\alpha} = \frac{\alpha c(\alpha)}{\sigma_\mu^2},$$

which follows from the entry condition (A.61). Since α decreases with ψ and c is increasing, the price impact of noise is strictly decreasing in ψ . The argument in the proof of Proposition 3 then applies with the price impact in place of σ^2 : for a given price impact, discretion minimizes expected loss, so a small degree of FCI targeting generates a second-order cost, while the reduction in the price impact generates a first-order gain. It follows that Proposition 3 still holds with endogenous arbitrageur entry.

A.8. Discretionary target announcements and the full commitment solution

This appendix analyzes two departures from the FCI targeting framework in Section 4.1. First, we let the central bank choose the announced target freely in each period, rather than fixing it at the expected “pstar” (see Remark 4). Second, we characterize the full (linear) commitment solution (see Remark 5). Both exercises show that our main results are robust and clarify what FCI targeting with a fixed target does and does not capture. Throughout, we use the notation

$$X_t \equiv (1 - \beta\varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t}, \quad \xi_t \equiv \varepsilon_{z,t} - \varepsilon_{\delta,t}, \quad \Delta r_t \equiv r_t^f - E_{t-1} [r_t^f],$$

for the surprise in the benchmark interest rate r_t^{f*} , the surprise in “pstar,” and the interest rate surprise, respectively.

FCI targeting with discretionary target announcements. Suppose the central bank solves problem (24) with an announced target $\bar{p}_t$ in place of $E_{t-1} [p_t^*]$, and chooses both r_t^f and the target $\bar{p}_{t+1}$ in each period. We write the target as $\bar{p}_{t+1} = E_t [p_{t+1}^*] + c_t$, where c_t denotes the deviation of the target from the expected “pstar.” We conjecture (and verify) a linear equilibrium in which σ^2 is constant, the asset price satisfies $p_t = p_t^* + S_t$, and the deviation S_t has a component that is anticipated in the previous period and a surprise component, $S_t = m_t + s_t$ with $m_t = E_{t-1} [S_t]$. The output gap is $\tilde{y}_t = S_t$.

Substituting the conjecture into the asset price equation (14), and using the fact that (p_t^*, r_t^{f*}) satisfy the same equation, we obtain

$$r_t^f = r_t^{f*} + m_{t+1} - S_t. \quad (\text{A.63})$$

The announcement affects the current interest rate through m_{t+1} : an anticipated deviation of next period’s asset price raises the current asset price for a given interest rate. Since m_{t+1} is determined by c_t , which we will show is a function of the current interest rate surprise, we have $E_{t-1} [m_{t+1}] = 0$ and thus

$$\Delta r_t = X_t + m_{t+1} - s_t. \quad (\text{A.64})$$

Next consider the central bank’s optimality conditions. The condition for r_t^f is the interest rate rule (25), $\Delta r_t = \theta \tilde{y}_t + \psi (p_t - \bar{p}_t)$, where $p_t - \bar{p}_t = \xi_t + S_t - c_{t-1}$. Taking the expectation of this condition as of period $t - 1$, and using $E_{t-1} [\Delta r_t] = E_{t-1} [\xi_t] = 0$, we obtain the anticipated component of next period’s output gap as a function of the current deviation:

$$m_{t+1} = \varkappa c_t, \quad \text{where } \varkappa \equiv \frac{\psi}{\theta + \psi}. \quad (\text{A.65})$$

The central bank partially honors a target that deviates from the expected “pstar,” which creates an anticipated output gap in the same direction as the deviation.

The condition for c_t balances the effect of the deviation on the current loss against its effect on the expected future loss. The deviation affects the current loss through m_{t+1} , which raises p_t and $\tilde{y}_t$ one-for-one for a given interest rate [see (A.63)]. Using the interest rate rule, the marginal effect on the current loss is

$$\frac{\partial}{\partial c_t} \left[\tilde{y}_t^2 + \frac{1}{\theta} (\Delta r_t)^2 + \frac{\psi}{\theta} (p_t - \bar{p}_t)^2 \right] = \varkappa \left[2\tilde{y}_t + \frac{2\psi}{\theta} (p_t - \bar{p}_t) \right] = \frac{2\varkappa}{\theta} \Delta r_t.$$

The deviation affects the expected future loss only through the anticipated components m_{t+1} and $m_{t+1} - c_t$ of next period’s output gap and FCI gap, respectively [it does not affect next period’s in-

terest rate surprise, see (A.64), or the central bank's choices in later periods]. Using (A.65), this effect is $\beta \frac{\partial}{\partial c_t} \left[m_{t+1}^2 + \frac{\psi}{\theta} (m_{t+1} - c_t)^2 \right] = 2\beta \varkappa c_t$. Combining the two effects, the optimal deviation is

$$c_t = -\frac{1}{\beta\theta} \Delta r_t. \quad (\text{A.66})$$

The central bank announces a target below the expected “pstar” after a surprise interest rate increase, and vice versa. Intuitively, the anticipated future FCI gap substitutes for the costly current interest rate adjustment.

To complete the characterization, substitute (A.65 – A.66) into (A.64) to obtain $\Delta r_t = (X_t - s_t) / \lambda$, where

$$\lambda \equiv 1 + \frac{\varkappa}{\beta\theta} > 1.$$

Substituting this into the surprise component of the interest rate rule, $\Delta r_t = (\theta + \psi) s_t + \psi \xi_t$, and solving for s_t gives the following result.

Proposition 8 (FCI Targeting with Discretionary Target Announcements). *Suppose the central bank solves problem (24) with an announced target $\bar{p}_t$ in place of the expected “pstar,” choosing both r_t^f and $\bar{p}_{t+1}$, and the parameters satisfy $D^2 \alpha^2 > 4\sigma_\mu^2 \sigma_{macro}^2$, where D is defined below and σ_{macro}^2 denotes the macro-induced variance in (A.68). Then, there is a (locally stable) equilibrium in which the announced target, the interest rate surprise, the asset price, and output are given by*

$$\begin{aligned} \bar{p}_{t+1} &= E_t [p_{t+1}^*] - \frac{1}{\beta\theta} \Delta r_t, \\ \Delta r_t &= \frac{1}{\lambda} (X_t - s_t), \\ p_t &= E_{t-1} [p_t] + \frac{1}{D} \left[(1 + \lambda\theta) \varepsilon_{z,t} - (\lambda\theta + \beta\varphi_\delta) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right], \\ y_t &= \underbrace{y_t^* - \frac{\varkappa}{\beta\theta} \Delta r_{t-1}}_{m_t} + \underbrace{\frac{1}{D} \left[-\lambda\psi \varepsilon_{z,t} + (1 - \beta\varphi_\delta + \lambda\psi) \varepsilon_{\delta,t} + \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t} \right]}_{s_t}, \end{aligned} \quad (\text{A.67})$$

where $D \equiv 1 + \lambda(\theta + \psi)$. The return variance σ^2 is the smaller positive solution to

$$\sigma^2 = \underbrace{\left(\frac{1 + \lambda\theta}{D} \right)^2 \sigma_z^2 + \left(\beta - \frac{1 - \beta\varphi_\delta + \lambda\psi}{D} \right)^2 \sigma_\delta^2}_{\sigma_{macro}^2} + \frac{1}{D^2} \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2. \quad (\text{A.68})$$

Setting $\lambda = 1$ (equivalently, $c_t = 0$) recovers the equilibrium in Proposition 2. The discretionary announcement changes the equilibrium in three ways. First, the surprise components of the asset price and output are scaled by $D = 1 + \lambda(\theta + \psi) > 1 + \theta + \psi$: in particular, the pass-through of noise shocks, $1/D$, is lower than with a fixed target. The central bank stabilizes the FCI against current shocks not only with the interest rate but also with the announcement. Second, the interest rate surprise is smaller by the factor $1/\lambda$, since the announcement substitutes for the interest rate. Third, the output gap acquires an anticipated component m_t that depends on the previous period's interest rate surprise and has the opposite sign: the central bank pays for today's stabilization with a predictable output gap tomorrow. The equilibrium is thus history dependent even though the central bank acts with discretion

in each period. The announcement stabilizes the current FCI through its effect on the expected future FCI, a first-moment channel that operates for a given return variance. Return variance is therefore not a sufficient statistic for the gains from a discretionary target. The anticipated component m_t does not affect the variance since it is not a surprise. Nonetheless, in our calibration return variance is lower than with a fixed target at every ψ . The recruitment benefits of FCI targeting in the main text are therefore preserved, and the directional channel adds to them.

Does the discretionary announcement improve the central bank's true objective (16)? The expected per-period true loss is $E[s_t^2] + E[m_t^2] + \frac{1}{\theta} E[(\Delta r_t)^2]$, where $E[m_t^2] = \left(\frac{\kappa}{\beta\theta}\right)^2 E[(\Delta r_t)^2]$ captures the cost of history dependence. In numerical examples, the loss is lower than with a fixed target for every ψ .¹⁶ This is consistent with the general principle that history-dependent policy helps when the private sector is forward looking and interest rate adjustments are costly (Woodford, 2003). It is orthogonal to the recruitment effect: it would operate even if σ^2 were exogenous. We fix the target in the main text to isolate the recruitment effect and because a history-dependent target is operationally more demanding: it requires the central bank to commit to state-contingent deviations from “pstar” rather than to a level that market participants can compute from fundamentals.

Full commitment solution. We next characterize the policy that minimizes the central bank's true objective (16) under full commitment, restricting attention to stationary linear policies. Specifically, suppose the central bank commits to output gaps of the form

$$\tilde{y}_t = \sum_{k \geq 0} \left[a_k \varepsilon_{z,t-k} + b_k \varepsilon_{\delta,t-k} + n_k \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t-k} \right], \quad (\text{A.69})$$

and chooses the coefficients to minimize the unconditional expected per-period loss $E[\tilde{y}_t^2 + \frac{1}{\theta} (\Delta r_t)^2]$ (we ignore the initial periods, so the discount factor drops out). Since $p_t = p_t^* + \tilde{y}_t$, the asset price equation implies $r_t^f = r_t^{f*} + E_t[\tilde{y}_{t+1}] - \tilde{y}_t$, and thus the interest rate surprise depends only on the contemporaneous and one-period-ahead loadings:

$$\Delta r_t = (a_1 - a_0) \varepsilon_{z,t} + (1 - \beta\varphi_\delta + b_1 - b_0) \varepsilon_{\delta,t} + (1 + n_1 - n_0) \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t}.$$

The return surprise depends only on the contemporaneous loadings, $r_{t+1} - E_t[r_{t+1}] = (1 + a_0) \varepsilon_{z,t+1} - (\beta - b_0) \varepsilon_{\delta,t+1} + n_0 \frac{\sigma^2}{\alpha} \varepsilon_{\mu,t+1}$, so the variance is the smaller solution to

$$\sigma^2 = (1 + a_0)^2 \sigma_z^2 + (\beta - b_0)^2 \sigma_\delta^2 + n_0^2 \frac{(\sigma^2)^2}{\alpha^2} \sigma_\mu^2. \quad (\text{A.70})$$

Loadings at lags $k \geq 2$ raise the output gap loss without affecting either the interest rate surprise or the variance, so they are optimally zero. The loadings at lag one do not affect the variance and are chosen to trade off future output gaps against current interest rate surprises. For noise shocks, minimizing $n_1^2 + \frac{1}{\theta} (1 + n_1 - n_0)^2$ yields $n_1 = -\frac{1-n_0}{1+\theta}$, and similarly $a_1 = \frac{a_0}{1+\theta}$ and $b_1 = -\frac{(1-\beta\varphi_\delta)-b_0}{1+\theta}$. Substituting

¹⁶With the parameters in Appendix A.1, the loss-minimizing FCI targeting reduces the true loss relative to discretion by 65% with a fixed target and by 70% with a discretionary target. The full commitment solution described below reduces the loss by 72%.

these into the loss, the commitment problem reduces to

$$\min_{a_0, b_0, n_0} \frac{2 + \theta}{1 + \theta} a_0^2 \sigma_z^2 + \left[b_0^2 + \frac{((1 - \beta \varphi_\delta) - b_0)^2}{1 + \theta} \right] \sigma_\delta^2 + \left[n_0^2 + \frac{(1 - n_0)^2}{1 + \theta} \right] \left(\frac{\sigma^2}{\alpha} \right)^2 \sigma_\mu^2, \quad (\text{A.71})$$

subject to (A.70).

Comparing (A.71) with the discretionary loss, in which the rate surprise is weighted by $\frac{1}{\theta}$ rather than $\frac{1}{1+\theta}$, shows that the commitment to lag-one output gaps acts as if the central bank's speed of adjustment were $1 + \theta$ rather than θ : by spreading the response to a shock over two periods, the central bank reduces the interest rate surprise needed to achieve a given contemporaneous stabilization. This is the history-dependence channel. If σ^2 were exogenous, the optimal contemporaneous loadings would be $a_0 = 0$, $b_0 = \frac{1 - \beta \varphi_\delta}{2 + \theta}$, and $n_0 = \frac{1}{2 + \theta}$: noise pass-through would fall from $\frac{1}{1 + \theta}$ under discretion to $\frac{1}{2 + \theta}$, as under FCI targeting with $\psi = 1$, but supply shocks would still induce no output gap. With endogenous σ^2 , the optimality conditions for (a_0, b_0, n_0) acquire additional terms proportional to $\frac{\partial \sigma^2}{\partial a_0}, \frac{\partial \sigma^2}{\partial b_0}, \frac{\partial \sigma^2}{\partial n_0}$. These terms push the central bank to dampen the asset price response to every shock beyond what the history-dependence channel alone implies: $a_0 < 0$, so that supply shocks induce output gaps, $b_0 > \frac{1 - \beta \varphi_\delta}{2 + \theta}$, and $n_0 < \frac{1}{2 + \theta}$. This is the recruitment channel, and it is the channel that FCI targeting with a fixed target isolates.¹⁷

In summary, the full commitment solution combines two channels. It makes policy history dependent to economize on interest rate surprises, and it dampens the asset price response to shocks to recruit arbitrageurs. FCI targeting with a fixed target captures the second channel, which is our focus. FCI targeting with a discretionary target additionally captures part of the first channel.

B. Empirical Appendix

B.1. Data

B.1.1. Macroeconomic data

We download the following data from FRED (FRED series names in parentheses): nominal potential GDP (NGDPPOT), nominal GDP (GDP), nominal investment (GPDJ), nominal personal consumption expenditures (PCEC), GDP deflator (GDPDEF), PCE price index (PCEPI), PCE price index excluding food and energy (PCEPILFE), the Chicago Fed National Financial Conditions Index (NFCI), the 3-month yield (TB3), labor productivity (OPHNFB), labor share (PRS85006173), weekly hours (PRS85006023), employment (CE16OV), population (CNP16OV) and consumer sentiment (UMCSENT). We obtain the updated series for the Excess Bond Premium from Favara et al. (2016). In order to compute real variables, we divide the nominal variables by the GDP deflator. Furthermore, real investment and consumption are included in per-capita terms.¹⁸ For the new FCI index from Ajello et al. (2023), we use the baseline construction, which allows shocks to have effects up to 3 years. Using the 1-year version does not alter the results. We compute hours per capita as weekly hours times employment divided by population. Inflation is computed as 400 times the log-difference in the PCE price index. Real variables are computed

¹⁷With the parameters in Appendix A.1, the full commitment solution has $a_0 = -0.25$, $b_0 = 0.26$, and $n_0 = 0.19$, compared with $a_0 = 0$, $b_0 = 0.015$, and $n_0 = 0.25$ under history dependence alone. History dependence alone reduces the true loss relative to discretion by 40%. The full commitment solution reduces it by 72%.

¹⁸We do not normalize GDP by population, so that a linear combination of GDP and output gap directly recovers potential output. Results do not change if GDP is included in per-capita terms instead.

as nominal variables over the GDP deflator. For example, real GDP is computed as nominal GDP over the GDP deflator, real potential output is computed as nominal potential GDP over the GDP deflator, and so on. The output gap is computed as the log difference between real GDP and real potential GDP. Except for inflation, the FCI index, the EBP, and the nominal interest rate, all variables are transformed by taking logarithms and multiplying by 100.

Since the FCI index is available from 1990 onwards, we use the Chicago Fed FCI (NFCI) for the sample period 1973–1990 when computing the IRFs of monetary policy shocks. Ajello et al. (2023) show that their index is similar in sample to the Chicago Fed FCI, and estimated IRFs are similar if we use that FCI for the full sample. We use the utilization adjusted TFP series from Fernald (2014), available on his website. For the local projections in Appendix B.3.4, we use additional financial variables. For earnings and stock prices, we use the real values reported in the Shiller database. We convert to quarterly by taking the value at the end of the quarter.

B.1.2. Construction of the financial noise shock

In order to construct the shock, we follow Gabaix and Koijen (2021) closely. We use quarterly data from the Flow of Funds.¹⁹ We use the same sample as the VAR specification. We use unadjusted flows (FU), and for the levels we use unadjusted market values when available (LM), and otherwise the estimated level. We collect data on flows for the following sectors: 15 (households), 21 (state and local governments), 22 (state and local retirement funds), 26 (rest of the world), 34 (federal retirement funds), 51 (property and casualty insurance), 54 (life insurance companies), 55 (closed end funds), 56 (ETFs), 57 (private pension funds), 65 (mutual funds), 66 (securities brokers and dealers), and 76 (U.S.-chartered depository institutions). As in Gabaix and Koijen (2021), we use data on three asset classes: 30611 (Treasury securities), 30630 (corporate and foreign bonds), and 30641 (corporate equities). For returns data, we use ex-dividend returns on the CRSP value-weighted market portfolio. For GDP growth, we use the log difference of real GDP obtained from FRED. We adjust the data on flows for foreign holdings following Appendix C.1.3 in Gabaix and Koijen (2021).

We follow the same notation and conventions as Appendix C.1.2 in Gabaix and Koijen (2021). We construct a measure of the proportional change of the quantity of equity in sector i between $t - 1$ and t ($\Delta q_{it}^{\mathcal{E}}$) as follows: In the FoF, equity flows are defined by $\Delta F_{it}^{\mathcal{E}} = W_{it}^{\mathcal{E}} - W_{i,t-1}^{\mathcal{E}} R_t^X$. We assume the securities are adjusted at the end of the period, so $\Delta F_{it}^{\mathcal{E}} = (\Delta Q_{it}^{\mathcal{E}}) P_t^{\mathcal{E}}$, where Q_{it} is the amount of equities held by sector i at time t , and $P_t^{\mathcal{E}}$ is the price of each share. The relative flow in equities is $\Delta f_{it}^{\mathcal{E}} = \frac{\Delta F_{it}^{\mathcal{E}}}{W_{i,t-1}^{\mathcal{E}}}$.

The proportional change in quantity of equity is $\Delta q_{it}^{\mathcal{E}} = \Delta f_{it}^{\mathcal{E}} (R_t^X)^{-1} = \frac{\Delta Q_{it}^{\mathcal{E}}}{Q_{i,t-1}^{\mathcal{E}}}$. We assume that total wealth is given by the sum of holdings of the three asset classes we consider.

With our measure of $\Delta q_{it}^{\mathcal{E}}$ in hand, we proceed exactly as in Appendix B of Gabaix and Koijen (2021) in order to construct the financial flow shock series. We briefly expand on each of the steps below:

1. Construct precision weights $\tilde{E}_i = \min\{k \frac{\sigma_i^{-2}}{\sum_{j=1}^N \sigma_j^{-2}}, 1.5/N\}$ where k is chosen so that the weights add up to 1.
2. Run the panel regression:

$$\Delta q_{it} = \alpha_i + \beta_t + \gamma_i \Delta y_t + \delta_i t + \Delta \tilde{q}_{it} \quad (\text{B.1})$$

¹⁹Raw data is downloaded from <https://www.federalreserve.gov/datadownload/Build.aspx?rel=Z1>.

using $\tilde{E}_i$ as weights. Here Δy_t is quarter-on-quarter real GDP growth. In an alternative specification, we also control for changes in the consumer sentiment index ($Sent_t$), i.e., we add $\Delta Sent_t$ as an additional control. We implement the weighting scheme by multiplying each observation by $\tilde{E}_i^{1/2}$ and then running OLS, using dummies to capture the fixed effects. Denote the residuals of this transformed regression as $\tilde{E}_i^{1/2} \Delta \tilde{q}_{it}$

3. We run PCA on $\tilde{E}_i^{1/2} \Delta \tilde{q}_{it}$. In our baseline specification, we control for aggregate factors by removing the first N principal components (ordered in terms of share of variance explained) from $\tilde{E}_i^{1/2} \Delta \tilde{q}_{it}$. That is, we construct:

$$\Delta \tilde{q}_{it} = \tilde{E}_i^{1/2} \Delta \tilde{q}_{it} - \sum_{n=1}^N \lambda_{i,n} \eta_{t,n}^{PC} \quad (\text{B.2})$$

where $\eta_{t,n}^{PC}$ is principal component n at time t , and $\lambda_{i,n}$ is the loading of sector i on that principal component. Our baseline uses $N = 2$. Our results are essentially unchanged if we use $N = 3$ or $N = 4$ instead.

4. Finally, we construct the financial flow shock as:

$$Z_t^\mu = \sum_{i=1}^I S_{i,t-1} \Delta \tilde{q}_{it}$$

where $S_{i,t-1} = \frac{W_{i,t-1}^\varepsilon}{\sum_{j=1}^I W_{j,t-1}^\varepsilon}$ is the share of total equity held by sector i at time $t - 1$.

B.2. Bayesian VAR implementation details

For a VAR with M lags, let $\tilde{y} = [Y_{M+1}, \dots, Y_{T+M}]'$ be a matrix where each row is a date and each column is a macro variable. Let $\tilde{x} = [x_{M+1}, \dots, x_{T+M}]$ be a matrix that stacks controls for all equations: each x_t contains a constant and lagged macro variables. In this context, we can write the system to be estimated as:

$$\tilde{y} = \tilde{x}B + u \quad (\text{B.3})$$

where $u \sim N(\mathbf{0}, \Sigma \otimes I_T)$. Letting $\beta = \text{vec}(B)$, assume the following priors:

$$\beta | \Sigma \sim N(\text{vec}(b), \Sigma \otimes \Omega(\xi)) \quad (\text{B.4})$$

$$\Sigma \sim IW(\Psi(\xi), d) \quad (\text{B.5})$$

where ξ is a vector of hyperparameters. It is well known that the posterior distribution of β and Σ is conjugate with:

$$\beta | \Sigma, y \sim N(\hat{\beta}, \hat{V}), \quad (\text{B.6})$$

$$\Sigma | Y \sim IW\left(\Psi + \hat{u}'\hat{u} + (\hat{B} - b)' \Omega^{-1}(\hat{B} - b), T + d\right) \quad (\text{B.7})$$

where:

$$\begin{aligned}\hat{\beta} &= \text{vec}(\hat{B}), \\ \hat{B} &= (x'x + (\Omega)^{-1})^{-1} (x'y + (\Omega)^{-1}b), \\ \hat{V} &\equiv \Sigma \otimes (x'x + (\Omega)^{-1})^{-1} \\ \hat{u} &= \tilde{y} - \tilde{x}\hat{B}.\end{aligned}$$

We parametrize the hyperpriors exactly as in Giannone et al. (2015): we set b according to the Minnesota prior (all variables but the proxy are assumed random walks a priori), $d = n + 2$ where n is the number of variables in $\tilde{y}$, $\Psi = \text{diag}(\psi)$ where ψ is a vector of hyperpriors, and Ω is a diagonal matrix with i, i -th entry equal to $\lambda^2 / (s_j^2 \psi_j)$, where s_j is the lag order of the control and ψ_j is the prior variance of the shock in the equation of the variable that is being lagged.²⁰ The constant is assumed to have a diffuse prior. Thus, $\xi = (\lambda, \psi)'$. The key hyperparameter is λ , which controls the tightness of the prior in β , and thus controls the degree of shrinkage used in estimation. We select the hyperparameters to maximize the marginal likelihood. Given those hyperparameters, we can then obtain the posterior distribution of β and Σ in closed form using (B.6) and (B.7). We generate draws directly from these, and for each draw we can compute any objects of interest (impulse responses, Wold decompositions, Forecast variance ratios, forecasts, etc.)

In Section 2, we identify the shock using two different procedures: i) add the proxy first in the VAR and use recursive identification, ii) an SVAR-IV procedure. These two differ only in what the matrix $\tilde{x}$ includes, and what rotation of Σ is utilized. In the first case, as mentioned in the main text, we let $X_t = [Z_t^\mu, Y_t']'$ and estimate a standard BVAR in X_t . Thus, $\tilde{y}$ contains all macro variables plus the proxy, and $\tilde{x}$ includes a constant, two lags of each macro variable and two lags of the proxy.

For the SVAR-IV case, we estimate the following restricted system:

$$\begin{pmatrix} Z_t^\mu \\ Y_t \end{pmatrix} = \begin{pmatrix} c^z \\ c^y \end{pmatrix} + \sum_{\ell=1}^M \begin{pmatrix} 0 & B_\ell^{zy, \ell} \\ 0 & B_\ell^{yy, \ell} \end{pmatrix} \begin{pmatrix} Z_{t-\ell}^\mu \\ Y_{t-\ell} \end{pmatrix} + \begin{pmatrix} u_t^z \\ u_t^y \end{pmatrix} \quad (\text{B.8})$$

that is, we impose that the coefficients of lagged Z_t^μ are 0 in all equations. Thus, $\tilde{y}$ is the same as above, but $\tilde{x}$ does not include any lags of Z_t^μ .

For the Y_t equations, this is required by the assumption of invertibility: indeed, as shown in Plagborg-Møller and Wolf (2022), the shock is invertible if and only if it does not Granger cause macro outcomes. For the Z_t^μ , this is an assumption just made out of convenience: this way, all equations have the same regressors and thus we can use exactly the same formulas as above. We note that the constructed Z_t^μ is serially uncorrelated and that the marginal likelihoods of the restricted and unrestricted BVARs are quite close, so this restriction does not seem to be relevant.

Finally, for any given draw of β, Σ , we can estimate impulse responses to the noise shock as follows: for the unrestricted case, we just use a Cholesky decomposition. For the restricted case we can create a C matrix such that $\varepsilon = Cu$ using the fact that $\varepsilon_t^\mu = \frac{\text{Cov}(u_t^z, u_t^y)' [\text{Var}(u_t^y)]^{-1}}{\text{Cov}(u_t^z, u_t^y)' [\text{Var}(u_t^y)]^{-1} \text{Cov}(u_t^z, u_t^y)} \hat{u}_t^y$ for the first row, and then filling out the rest by orthogonalizing recursively. We then recover the rotation of interest by simply inverting C .

²⁰In this part only ψ denotes the hyperprior to be consistent with the original notation of Giannone et al. (2015).

	Unrestricted	Restricted (SVAR-IV)
λ	0.2182	0.2165
Log Marg. Lik.	-558.0599	-559.0485

Table 2: Estimated prior tightness (λ) and marginal likelihood. The restricted specification sets the coefficients of lagged values of the proxy to zero.

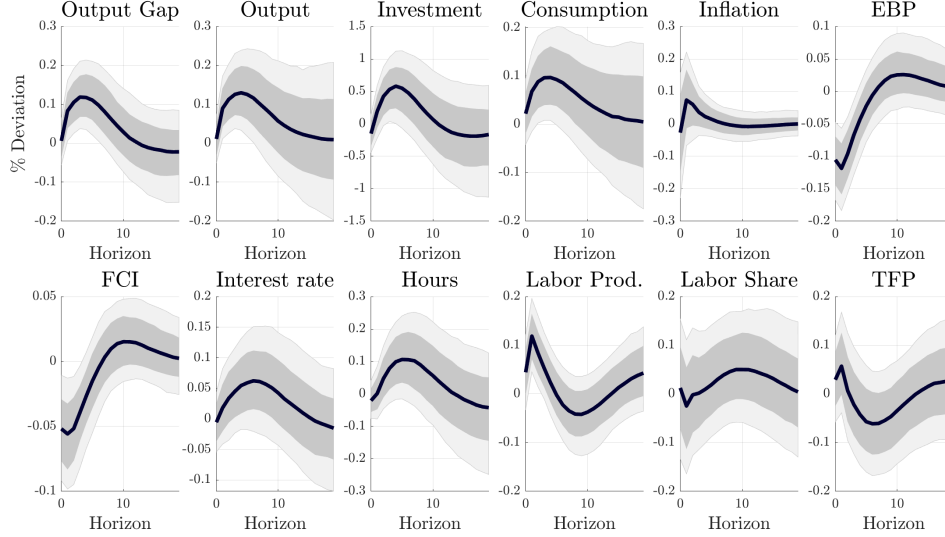


Figure 7: Impulse response to a financial noise shock for all variables. Unrestricted VAR with recursive identification. Shaded and light shaded grey bands indicate 68% and 90% credible sets, respectively.

B.3. Additional empirical results

B.3.1. Additional estimation results

Table 2 shows the estimated prior tightness (λ) as well as the log marginal likelihood. The degree of tightness is similar to that in Giannone et al. (2015). Both specifications yield very similar results in terms of fit, indicating that lagged Z_t^μ do not meaningfully improve fit and thus have their coefficients shrunk to 0. Figures 7 and 8 show impulse responses and variance decompositions for all macro variables included in the VARs.

B.3.2. Robustness: extending the sample to include COVID

In order to assess the stability of our estimates to including the recent COVID period, we extend the sample to 2024Q2. Figure 9 shows the estimated impulse response. The overall patterns are unchanged: upon the shock, there is a delayed response in the output gap and policy rates, and a faster response in inflation and FCI. Figure 10 depicts the forecast variance ratios. Results in this subsample are quite similar.



Figure 8: Identified Forecast Variance Ratios for the noise shock. Blue: SVAR-IV, assuming invertibility. Red: lower bound, assumes perfect measurement of the shocks. Black: recoverability-based FVR. The three lines are the medians, horizon by horizon, of each object. Light grey areas are 90% credible regions of the identified set, (Plagborg-Møller and Wolf, 2022)

B.3.3. Robustness for impulse response estimation

This subsection addresses two concerns about the construction of the shock and two about its estimation. The first concern is that the sectoral flows may still load on aggregate factors after removing two principal components, so that the shock partly reflects fundamentals. Figures 11 and 12 show the estimated IRFs when we instead remove 3 and 4 principal components in equation (B.2). The results are virtually identical, which indicates that the procedure adequately controls for aggregate factors in this setting (Gabaix and Koijen, 2024). The second concern is that households may move into equities at the same time as their sentiment, and hence their spending, improves. In that case the shock would capture the direct effect of sentiment on demand rather than the effect of flows through stock prices. Figure 13 shows the results when we also residualize the shock by changes in consumer sentiment. The responses are essentially unchanged. Turning to estimation, Figure 14 depicts the IRF estimated with the SVAR-IV procedure of Section 2.3 instead of the recursive scheme, and Figure 15 estimates the IRFs with local projections instead of the VAR. In both cases, results are similar to the baseline.

B.3.4. Robustness: drivers of the financial noise shock

Figure 16 shows the local projections version of Figure 5. As we can see, the overall patterns are unchanged. Figure 17 shows the IRFs for the extended VAR where we swap real dividends for real earnings. As we can see, earnings increase for the first few quarters, but this effect is short lived.

B.3.5. Monetary policy shock IRFs

Figure 18 contains the impulse response to the monetary policy shock identified by Aruoba and Drechsel (2025), which we use as an input in the next subsection. The responses are standard. The shock generates

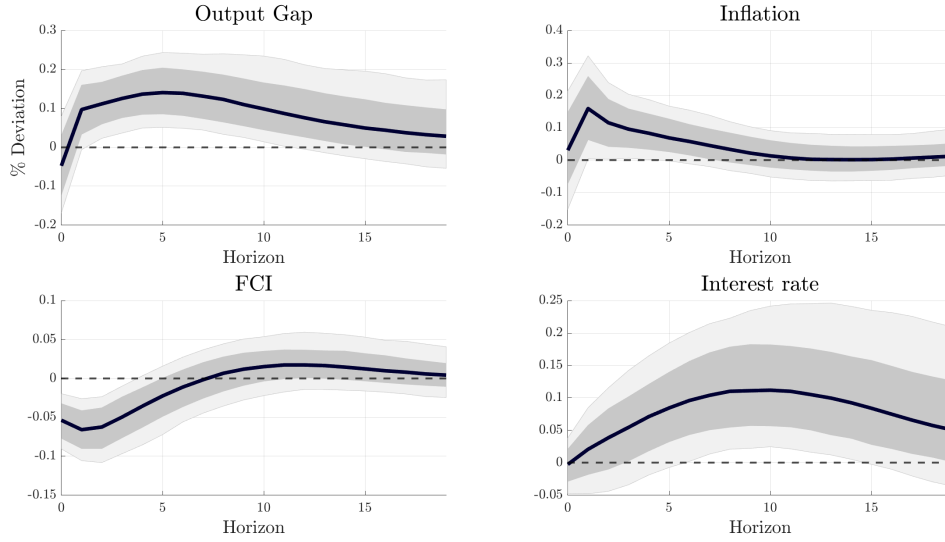


Figure 9: Impulse response to a financial noise shock, sample: 1990Q1–2024Q2. Shaded and light shaded grey bands indicate 68 and 90 credible sets respectively.

a small, marginally significant inflation response.

B.3.6. Counterfactual propagation of monetary policy shocks

One of the key observations of risk-centric models (Caballero and Simsek, 2020) is that monetary policy affects the economy via asset prices. Given our setting, we can test this claim empirically using the tools developed in McKay and Wolf (2023).

Specifically, we examine the following counterfactual question: how would a monetary policy shock propagate if financial conditions were unresponsive to monetary policy? To answer that question, we take the impulse-response of the Aruoba and Drechsel (2025) monetary policy shock, and use the identified response to a financial flow shock to approximately enforce $FCI_t = 0$ on impact and in expectation for the first year. Importantly, although the methodology is the same as in McKay and Wolf (2023), this is not a *policy* counterfactual. Instead, we are asking how a given policy shock would have propagated under a different mapping between monetary policy and financial conditions.²¹

Figure 19 shows the results. As we can see, the approximation is good for the first quarters, but we still get some delayed response of financial conditions at longer horizons. Crucially, the path of interest rates is basically unchanged. Turning to the output gap, the response is essentially zero at all horizons: if anything, the counterfactual rate increase initially *stimulates* output. Overall, the result is consistent with the key tenet of the risk-centric view of monetary policy: monetary policy affects the economy via financial conditions. Our results indicate that in a counterfactual economy where short-term interest rates and broader Financial Conditions are disconnected, monetary policy shocks would have essentially no impact on output gaps or inflation.

²¹The assumptions required for this to yield the correct counterfactual are analogous to the ones in McKay and Wolf (2023): we need that financial conditions enter in the rest of the private sector equations and in the monetary policy rule only through their expected values.

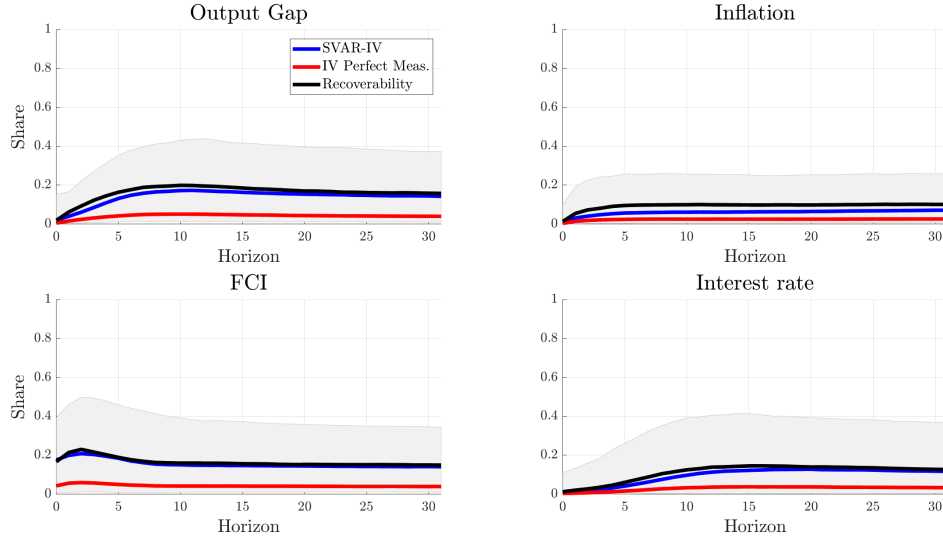


Figure 10: Identified Forecast Variance Ratios for the noise shock, sample: 1990Q1–2024Q2. Blue: SVAR-IV, assuming invertibility. Red: lower bound, assumes perfect measurement of the shocks. Black: recoverability-based FVR. The three lines are the medians, horizon by horizon, of each object. Light grey areas are 90% credible regions of the identified set, (Plagborg-Møller and Wolf, 2022)

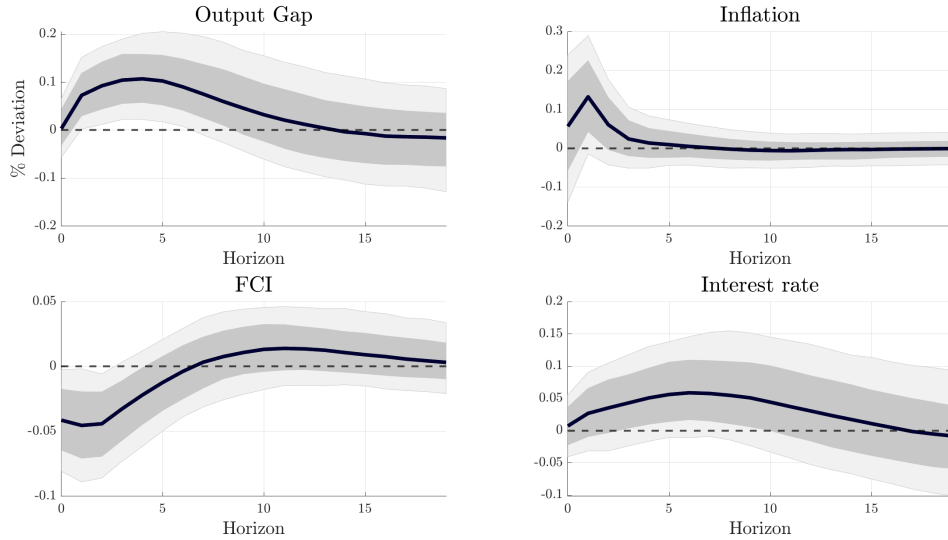


Figure 11: Impulse response to a financial noise shock, where the shock is identified controlling for 3 Principal Components in (B.2). Shaded and light shaded grey bands indicate 68% and 90% credible sets, respectively.

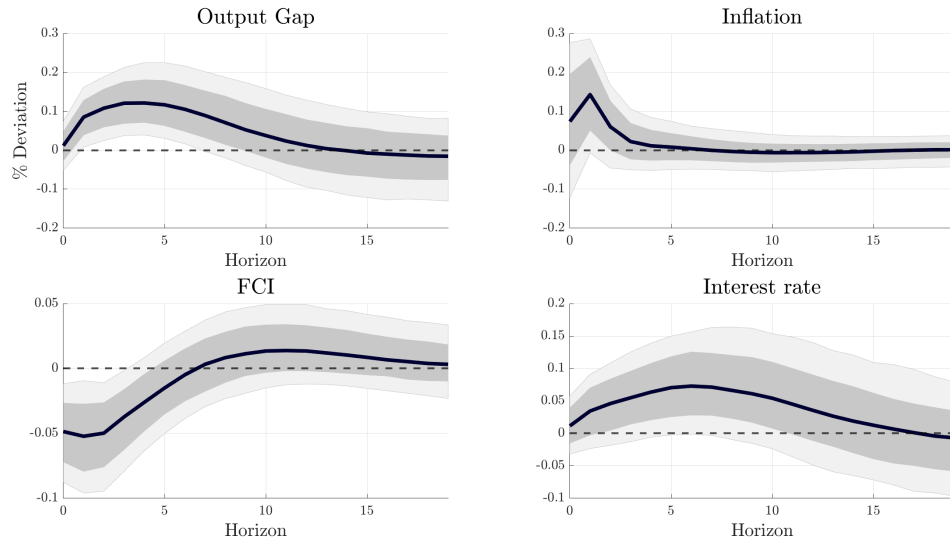


Figure 12: Impulse response to a financial noise shock, where the shock is identified controlling for 4 Principal Components in (B.2). Shaded and light shaded grey bands indicate 68% and 90% credible sets respectively.

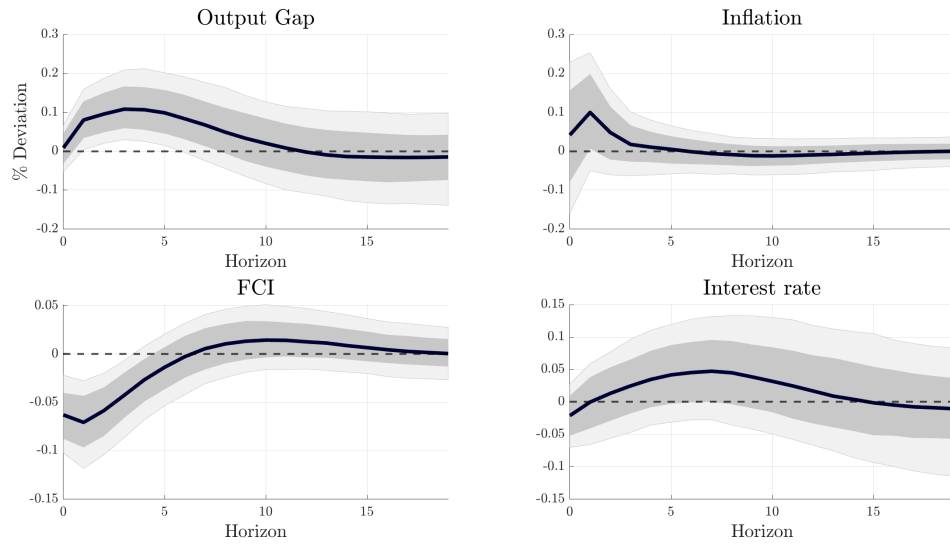


Figure 13: Impulse response to a financial noise shock, where we also residualize by changes in consumer sentiment when constructing the shock. Shaded and light shaded grey bands indicate 68% and 90% credible sets respectively.

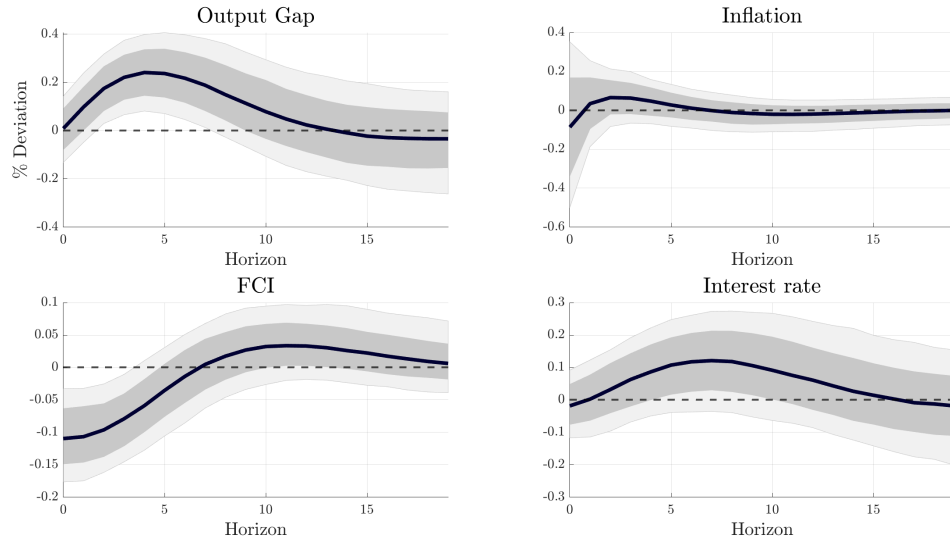


Figure 14: Impulse response to a financial noise shock, where the shock is identified using an SVAR-IV procedure. Shaded and light shaded grey bands indicate 68% and 90% credible sets respectively.

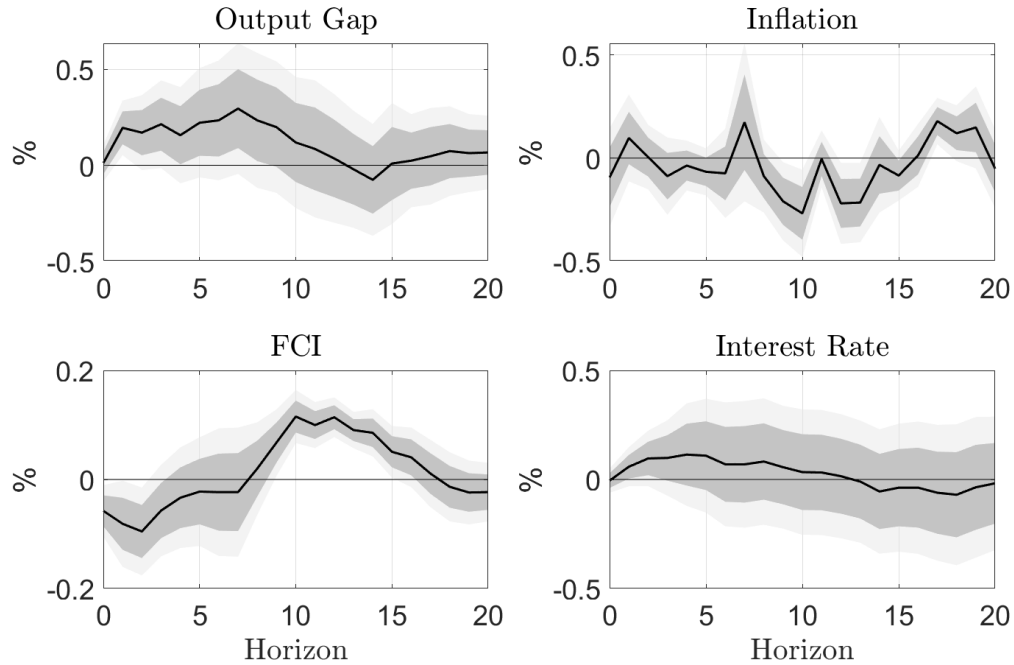


Figure 15: Impulse response to a financial noise shock, estimated via local projections. The regressions include two lags of the outcome variable as well as FCI. Shaded and light shaded grey bands indicate 68 and 90 percent heteroskedasticity-robust confidence intervals respectively.

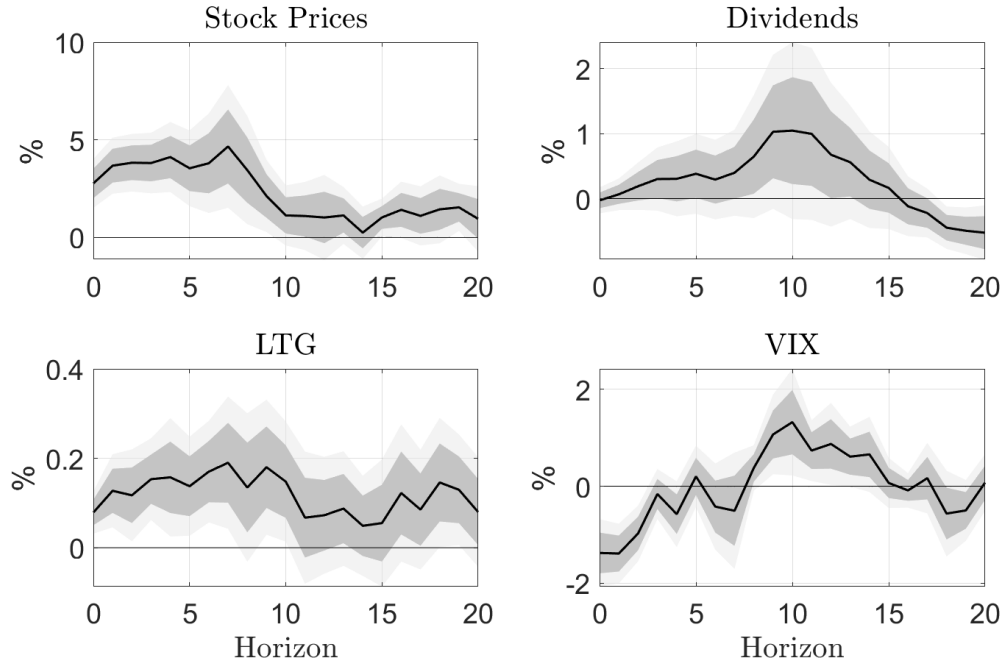


Figure 16: Impulse response to a financial noise shock for stock prices (S&P 500), real dividends, long-term earnings growth forecasts (Bordalo et al., 2024), and the VIX.

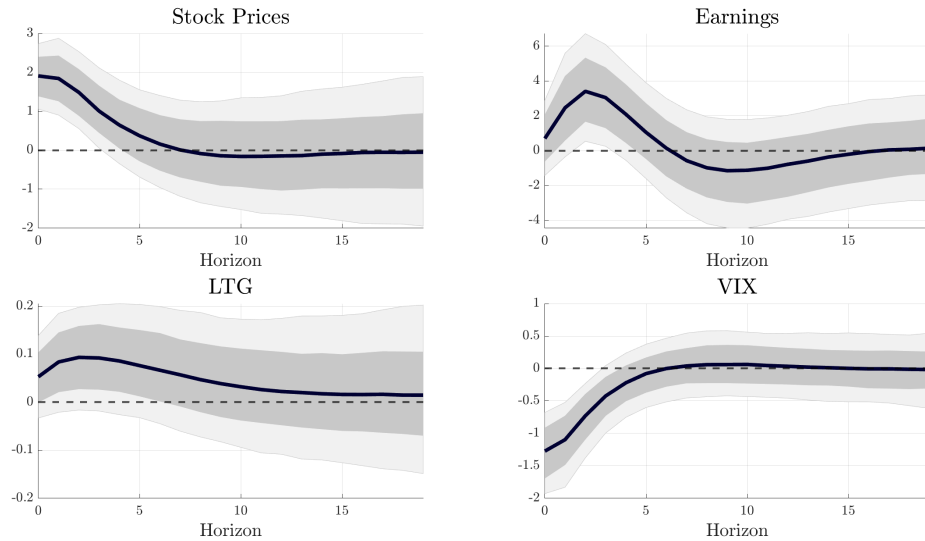


Figure 17: Impulse responses of selected financial variables to a financial noise shock. Shaded and light shaded grey bands indicate 68% and 90% credible sets, respectively.

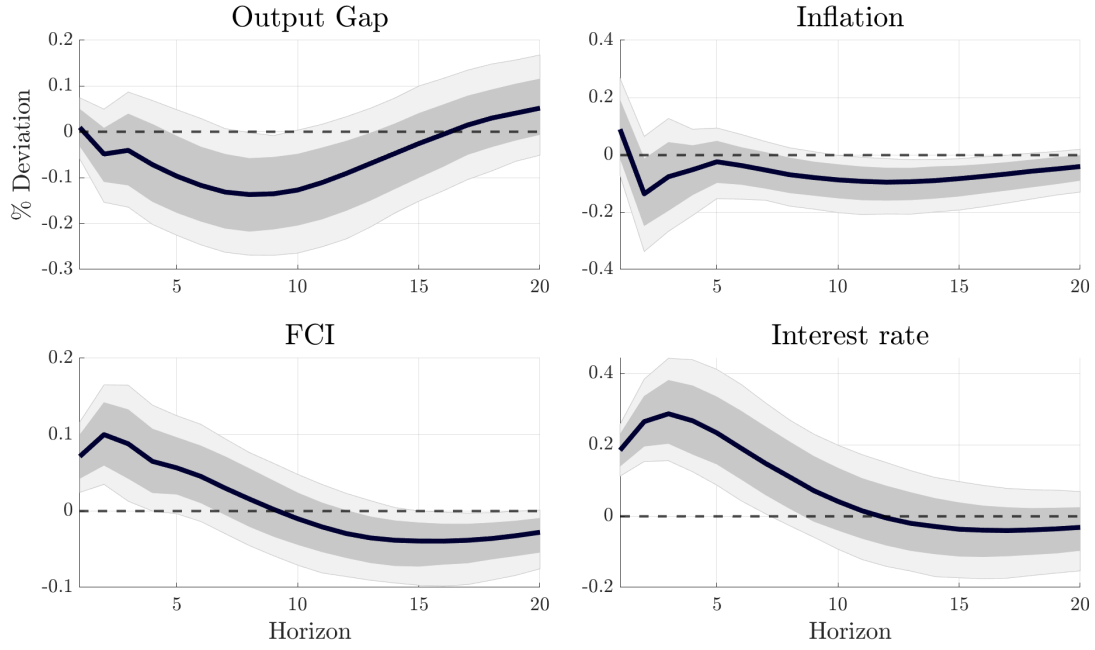


Figure 18: Impulse response to the Aruoba and Drechsel (2025) monetary policy shock. Shaded and light shaded grey bands indicate 68% and 90% credible sets respectively.

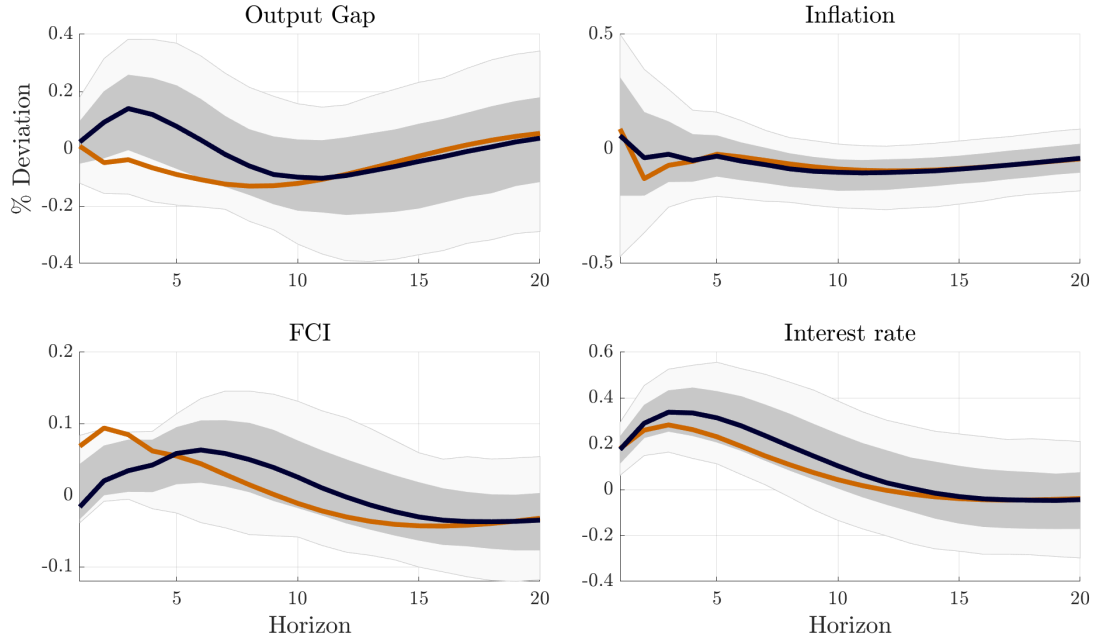


Figure 19: Counterfactual impulse response to a monetary policy shock, identified following Aruoba and Drechsel (2025). The orange line is the original point estimate, and the black line is the best approximation to the counterfactual impulse response. Shaded and light shaded grey bands indicate 68 and 90 confidence sets respectively.